

# Dilutive Financing\*

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## Abstract

This paper presents a dynamic model of firm financing where firms use financial slack to reduce rent extraction by financiers with bargaining power. Financing is lumpy because it is optimal to bargain infrequently. Moreover, firms may finance ‘early’ before exhausting internal funds to bargain when their outside options are better. Financing rents are thus endogenous to firms’ dynamic financing strategy. Firms with good financing alternatives raise financing early to reduce rents, whereas firms lacking such alternatives raise financing after exhausting funds to avoid frequently paying endogenously large rents. Investment irreversibility increases financing rents, and disproportionately so for less productive firms.

*JEL Classification:* E22, E44, G32

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# 1 Introduction

A pervasive feature of firm financing is that firms raise external financing infrequently in chunks rather than frequently in small increments. The canonical explanation in economics is based on the idea of fixed transaction costs, which dates back to [Baumol \(1952\)](#) and [Tobin \(1956\)](#).<sup>1</sup> In this paper, I present a new theory of dynamic firm financing that explains the infrequent “lumpy” financing as due to financier bargaining power. The theory also provides a natural explanation for another prevalent pattern which has received less attention in the literature: many firms raise external financing “early,” that is, before running out of cheaper internal funds. These two key aspects of ‘financial slack’ jointly arise in my model as part of firms’ optimal dynamic financing strategy to reduce rents that financiers are able to extract.

I consider an environment where a firm regularly needs financing, due to negative shocks or investment needs. While it may maintain internal funds in anticipation, these involve a carry cost.<sup>2</sup> The firm can raise funds from external financiers at any point, and there is no fixed transaction cost of financing. It is thus feasible, and first-best, to raise external financing incrementally whenever needed and keep zero internal funds.

The key element giving rise to the two dynamic patterns is a bargaining friction. A firm can bargain with financiers at any point to raise funds, but if that bargaining were to fail, the firm cannot immediately access *another* financier to bargain with again. This reduced accessibility upon a bargaining failure captures the idea that it can prove challenging in practice to find alternative sources of financing after having failed to raise funds from the primary one. For example, startups rely on venture capital funds with expertise for a specific industry and stage of the venture. Similarly, large firms resort to a handful of investment banks that address these firms’ sizable funding needs with syndicated financing. In either case, it plausibly takes some time to switch to another financier if the firm were to fail in bargaining with the current financier.<sup>3</sup>

This implies that, even if bargaining always succeeds in equilibrium, the surplus from it, a portion of which financiers extract as rents, includes not only the value that its success creates relative to firm value ex-ante but also the value of not causing such delay as the consequence of its failure. Therefore, financing dilutes firm value.

In response, firms choose to raise financing in excess of immediate funding needs, and maintain internal funds, in order to avoid needing to bargain with financiers and get diluted too often. Thus, financing is optimally lumpy. Moreover, firms may choose to raise financing early, before exhausting internal funds, in order to bargain when

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<sup>1</sup>[Baumol and Tobin \(1989\)](#) note that [Allais \(1947\)](#) first proposed the same basic mechanism.

<sup>2</sup>‘Internal funds’ may refer simply to cash reserves, but may also include, more broadly, any spare capacity for borrowing at a low interest rate, such as lines of credit.

<sup>3</sup>Appendix [A](#) shows how a bargaining failure may create conditions that hinder timely access.

their outside options are better. With financial slack retained at bargaining, firms may avoid having their “back against the wall”: even if financing were to fail such that firms cannot raise external financing right away, they could temporarily rely on these internal funds while seeking alternative sources of financing. By raising funds before financing is really needed, firms can thus improve their outside options and make a failure of bargaining less consequential, thereby making financing less dilutive.

In sum, financial slack reduces dilution of firm value by improving firms’ bargaining with financiers. Lumpy financing reduces the frequency of dilution, and early financing reduces its size. This, I argue, is what determines the dynamics of firm financing.

The point of departure is that financing costs – that is, financiers’ rents – not only determine firms’ optimal financing strategy, but also are determined by it. On one hand, the size of financing costs determines the optimal frequency of financing. On the other hand, the optimal *timing* of financing determines the size of financing costs. The first channel, when viewed in isolation, microfoundations fixed transaction costs based on financier bargaining power. Both channels combined, the theory generates novel implications on firm financing, as explained presently.

A key distinction of the model is that firms that raise financing early effectively offset financiers’ bargaining power: by keeping sufficient financial slack when bargaining for funds, they mitigate the consequence of a bargaining failure to the point where equilibrium financing costs do not increase in how much bargaining power financiers possess. It may thus be incorrect to observe that some firms incur small financing costs and conclude that their financiers must have little bargaining power. To the contrary, the small costs may reflect that these firms optimally keep large financial slack when raising financing precisely because their financiers’ bargaining power is substantial.

As a direct implication, the model predicts that firms with better access to financing may have more financial slack. Firms that are slow at accessing alternative sources of financing delay financing until running out of funds, because they cannot improve their outside options and reduce financing costs much by keeping financial slack when they bargain with financiers. In contrast, firms that can quickly access financing alternatives can greatly improve their outside options and reduce financing costs by raising financing when they have financial slack. In short, the incentive to raise financing early before exhausting financial slack *increases* in the quality of firms’ access to financing.<sup>4</sup>

Additionally, numerical exercise suggests important cross-sectional implications when investment is irreversible, such as for intangible capital or during downturns.<sup>5</sup> Consider productive firms investing heavily. If bargaining for financing were to fail, they are able to save some funds by reducing their investment expenses. Due to this

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<sup>4</sup>In standard views, early financing is due to firms’ exposure to fluctuations in financing conditions – and hence the incentive for it should decrease in the quality of access to financing.

<sup>5</sup>These conditions tend to coincide with difficulty of accessing alternative financiers.

ability to underinvest in response to a bargaining failure, these firms have good outside options vis-à-vis financiers when bargaining early with financial slack. In contrast, unproductive firms that invest little are endogenously constrained in their ability to underinvest: if bargaining were to fail, they have little expenses to cut down on – and existing capital cannot be divested. Since these firms cannot improve their outside options as much, they optimally bargain after running out of funds. As such, investment irreversibility may disproportionately increase financing costs for less productive firms.

In terms of applicability, this theory naturally speaks to startup firms relying on venture capital funds, in the sense that entrepreneurs in practice do worry about bargaining and dilution that financing involves. Regarding large and established firms, it predicts that these firms have, in equilibrium, little reason to worry about bargaining with a financial institution such as an investment bank because they keep large financial slack. Across both cases, the paper makes a single claim: how much bargaining ‘matters’ in equilibrium depends endogenously on firms’ optimal financial slack.

Methodologically, this paper’s dynamic bargaining framework is tractable. I obtain a general optimality condition for choosing to bargain when bargaining is dilutive, characterize the unique Markov-perfect equilibrium as simple  $(s, S)$  bounds, and establish comparative statics in bargaining power as well as access to alternative financing. As an aside, the model allows for a standard numerical solution given essentially as single-agent optimization, with an added binary state variable to track outside options.

The rest of the paper is organized as follows. Section 2 describes related literature. Section 3 presents the core mechanism with a two-period model. Section 4 builds the main model with full dynamics in infinite horizon, and Section 5 characterizes optimal financing strategy. Section 6 introduces the investment choice. Section 7 concludes.

## 2 Related Literature

This paper makes contributions to four branches of the literature.

**New framework of firm financial slack based on bargaining.** As mentioned, the classic theory by Baumol (1952) and Tobin (1956) posits fixed transaction costs to explain households’ cash-holdings. The theory has been applied to the firm context by Décamps, Mariotti, Rochet and Villeneuve (2011) and Bolton, Chen and Wang (2011) to rationalize lumpy firm financing. Bolton, Chen and Wang (2013) introduce fluctuations in fixed transaction costs to explain early financing with precautionary incentives.<sup>6</sup> In contrast, this paper suggests that both lumpy financing and early financing may arise from financier bargaining power, and generates a distinct prediction

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<sup>6</sup>The same mechanism is used by Alvarez and Lippi (2009) to study household money demand.

that firms with greater access to financing have greater incentives for early financing.

In [Hugonnier, Malamud and Morellec \(2014\)](#), firms face search frictions in financing. As such, when firms’ matching rate increases, financing becomes, on average, both less lumpy and less ‘early’ – that is, less financial slack upon financing. In this paper, firms can always access financing but have limited alternatives. When firms’ access to financing alternatives improves, financing may become less lumpy yet ‘earlier.’<sup>7</sup>

Empirically, this paper explains why large, established and creditworthy firms still maintain sizable financial slack, as reported by [Graham \(2022\)](#) and [Albertus, Glover and Levine \(2025\)](#). It is also consistent with [Opler, Pinkowitz, Stulz and Williamson \(1999\)](#), [Bates, Kahle and Stulz \(2009\)](#), [Graham and Leary \(2018\)](#) and [Begenau and Palazzo \(2021\)](#), who document that firms with steep growth have large financial slack.

**Endogenous timing of trades and endogenous dynamic outside options.** The seminal work by [Rubinstein and Wolinsky \(1985\)](#) extends the foundational bargaining game of [Rubinstein \(1982\)](#) to endogenize agents’ outside options through search frictions. In this class of models, the timing of trades is exogenous and driven by the random search process. Recently, [McClellan \(2024\)](#) endogenizes the timing of trades when the agent’s outside option evolves exogenously. In contrast, both the timing of trades and the evolution of outside options are endogenous in my model.

This paper also relates to the literature on holdup such as [Hart and Moore \(1990\)](#) in that a first-best choice worsens agents’ subsequent bargaining position. The contribution is that it studies optimal dynamic strategy given recurrence of holdup problems.

[Kiyotaki, Moore and Zhang \(2024\)](#) show that repeated rounds of bargaining dilute agents’ payoffs and induce underinvestment. On the other hand, [Lagos and Zhang \(2022\)](#) study how outside options off-path may limit the counterparty’s bargaining power on-path. These two mechanisms are parsimoniously unified in my model.

**Complementing the literature on dynamic firm financing.** This paper expands the large literature on debt dilution, such as [Myers \(1977\)](#), [Brunnermeier and Oehmke \(2013\)](#), [Admati, DeMarzo, Hellwig and Pfleiderer \(2018\)](#), [DeMarzo and He \(2021\)](#) and [Donaldson, Koont, Piacentino and Vanasco \(2024\)](#), by showing that dilution may drive the dynamics of firms’ financing and investment even in the absence of capital structure.

The paper builds on risk management literature such as [Froot, Scharfstein and Stein \(1993\)](#) by rationalizing financial slack in excess of investment needs. It also suggests that ease of liquidation may induce early refinancing of debt, instead of renegotiation

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<sup>7</sup>On a theoretical note, they use a standard continuous-time model in which financing is identically accessible – by the same matching rate – regardless of bargaining outcome. Therefore, financing is never dilutive and there is no incentive to strategically choose when to bargain and when not. My theory suggests that this ‘dominance of bargaining’ may not be robust. It holds in such continuous-time limits because instantaneous matching probability is zero, and fails, for example, under the same search frictions but in discrete time (see Section 5.1).

upon default as in [Bolton and Scharfstein \(1996\)](#) and [Hart and Moore \(1998\)](#).

**Capital liquidity, productivity and financing.** The financial frictions proposed in this paper become severer when investment becomes more irreversible. This holds despite the absence of collateralized debt in the model, and arises because the ability to divest capital to obtain funds improves firms’ outside options vis-à-vis financiers. Thus, this paper extends the results from the vast literature emphasizing how the liquidity of capital affects borrowing constraints – including [Kiyotaki and Moore \(1997\)](#), [Lorenzoni \(2008\)](#), etc. – to general forms of firm financing including equity.

At the same time, when investment is irreversible, financing costs may be lower for productive firms investing heavily. In [Rampini and Viswanathan \(2010\)](#) with collateral constraints, productive firms with small capital are heavily constrained in financing investment. The difference suggests that the cross-section of firm growth in intangible-intensive sectors with high irreversibility may be qualitatively different depending on whether financing mainly involves collateral constraints or bargaining frictions.

### 3 Core Mechanism

This section presents a stylized model that captures the core economic insights that due to financiers’ bargaining power, firms may optimally raise financing infrequently and hence in a lumpy fashion, and also early – that is, before internal funds are exhausted.

**Setup.** There are two periods with three dates  $t \in \{0, 1, 2\}$ . There is no discounting. There is a project that lasts for the two periods. In each period, the project spends a unit of funds; with any less funds, it fails. The project yields a payoff  $v > 2$  on terminal date  $t = 2$ . An agent called entrepreneur owns a firm that runs the project.<sup>8</sup>

The firm cannot self-finance the project. There are two financiers, one visiting the firm on date 0 and the other on date 1, who can provide it. Each of them has bargaining power and, if asked by the firm to provide funds, extracts a fraction  $1 - \theta$ ,  $\theta \in (0, 1)$ , of the surplus from bargaining in addition to being compensated for the cost of funds normalized to unity. A financier cannot bargain with the firm or provide funds except on the date of his visit. Funds can be kept internally, but it requires a marginal carry cost  $\varphi > 0$  to keep funds for one period.<sup>9</sup>

Denote  $h_t$  as the firm’s holding of internal funds on date  $t$ ; ‘financial slack’ is when  $h_t$  is positive. By backward induction, the firm raises from the second financier  $\max\{0, 1 - h_1\}$  units of funds on date 1. If  $h_1 \geq 1$ , then there is no need to bargain

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<sup>8</sup>The present setup gives a minimal environment that allows for multiple rounds of bargaining so that the core mechanism can be captured. The main model in Section 4 will entertain generic business profiles that allow for operating profits subject to occasional losses due to volatility.

<sup>9</sup>This can represent either the entrepreneur’s utility cost or decay by a factor of  $\frac{\varphi}{1+\varphi} \in (0, 1)$ .

with the second financier. If  $h_1 < 1$ , the project fails without more funds. The firm thus bargains with the second financier and pays him the cost of funds  $1 - h_1$  plus his rent, which is a fraction  $1 - \theta$  of the surplus from bargaining  $v - (1 - h_1)$ . Due to the carry cost  $\varphi > 0$ ,  $h_1 \in \{0, 1\}$  in equilibrium as long as  $h_0 \leq 1$ . Firm value on date 1 is, then,  $v$  if  $h_1 = 1$ , and  $v - 1 - (1 - \theta)(v - 1) = \theta(v - 1) \equiv \hat{v}$  if  $h_1 = 0$ .

Relative to firm value on date 1 given  $h_1 = 0$ , the above financing creates  $v - 1 - \hat{v}$  in surplus. But the second financier extracts a fraction  $1 - \theta$  of not just this, but also the value,  $\hat{v} - 0$ , of not losing the project as the consequence of failing the bargaining.<sup>10</sup>

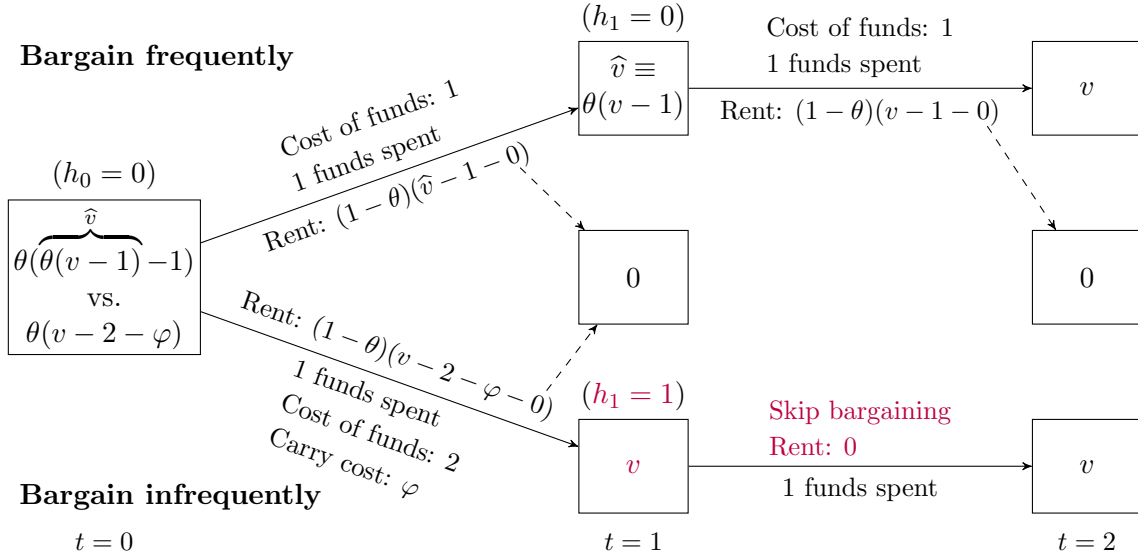


Figure 1: Frequency of bargaining

Rectangles show firm value at each node. The amount of internal funds on date  $t$  is  $h_t$ . Dashed arrows represent the firm's outside options – i.e., what happens if bargaining fails. Firm values and rents are determined as:

$$\text{Firm value}_t = \underbrace{\text{Firm value}_{t+1} - \text{Costs}_t}_{=\text{Continuation value}_t} - \text{Rent}_t, \quad \text{Rent}_t = (1 - \theta) \underbrace{(\text{Continuation value}_t - \text{Outside option}_t)}_{=\text{Bargaining surplus}_t}.$$

**I. Frequency of bargaining and lumpy financing.** Assume zero initial funds,  $h_0 = 0$ . The entrepreneur chooses whether to bargain twice on both dates  $t \in \{0, 1\}$  or once on date  $t = 0$ . If she bargains frequently (that is, twice), the entrepreneur raise a unit of funds from the first financier on date 0. No funds are kept internally,  $h_1 = 0$ , so that firm value on date 1 is  $\hat{v}$  due to the second bargaining. If she instead bargains infrequently (that is, once), she raises two units from the first financier. One is spent immediately, and the other is kept,  $h_1 = 1$ , at the carry cost  $\varphi$ . Since there is no need to bargain with the second financier, firm value on date 1 is  $v$ .

<sup>10</sup>Section 5.1 defines dilution precisely as the second component of financiers' rent:  $(1-\theta)(\hat{v}-0)$  in this case. While the definition is crucial in the formal analysis throughout Section 5, here I do not treat it explicitly in order to preserve the benefits of the game-tree representation.

Either way, the first financier extracts as his rent a fraction  $1 - \theta$  of firm value on date 1 minus the cost of funds and, if applicable, the carry cost; this amount comprises not just the surplus that the bargaining creates relative to firm value on date 0 but also the value of not losing the project as the consequence of a failure of the bargaining.

The total payoffs for all three agents – the entrepreneur and the two financiers – are maximized if and only if bargaining is frequent, since it avoids the carry cost,  $\varphi$ . But the entrepreneur's payoff is firm value on date 0. As Figure 1 describes, she optimally bargains infrequently if and only if  $\theta(v - 2 - \varphi) \geq \theta(\hat{v} - 1)$ , or, equivalently,

$$(1 - \theta)(v - 1) \geq \varphi; \quad (1)$$

that is, if the rent that the second financier would extract upon frequent bargaining,  $(1 - \theta)(v - 1)$ , exceeds the carry cost upon infrequent bargaining,  $\varphi$ .<sup>11</sup>

Despite the carry cost, the entrepreneur may optimally avoid bargaining with the second financier by raising funds on date 0 in excess of the immediate funding needs.

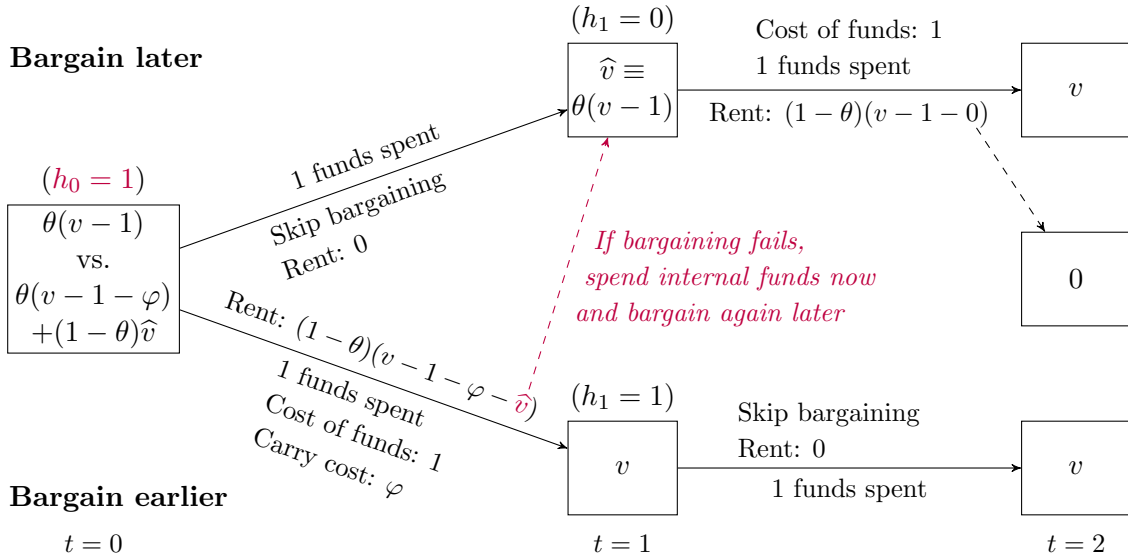


Figure 2: Timing of bargaining

**II. Timing of bargaining and outside options.** Next, suppose instead that the firm initially has one unit of funds,  $h_0 = 1$ . This means that the entrepreneur only needs to raise one unit for the second period, and thus can bargain just once in any case. But she must now choose when to bargain. If she bargains on date 1 with the second financier, then her payoff is  $\hat{v}$ . Note that during this bargaining, the entrepreneur has no financial slack,  $h_1 = 0$ , and is thus at the brink of losing the project. If instead she bargains earlier on date 0 with the first financier, she is no longer at the brink; even

<sup>11</sup>The entrepreneur may also optimally abandon the project if  $\theta(v - 1) \leq 1$  and  $v - 2 - \varphi \leq 0$ .

if bargaining were to fail on date 0, the entrepreneur can spend the funds  $h_0 = 1$  now and bargain later with the second financier. Thus, her outside option value is  $\widehat{v}$ , not 0.

How do better outside options affect the optimality of bargaining on date 0 versus on date 1? On either date when bargaining takes place, the firm gets the continuation value – that is, the difference between tomorrow’s firm value and today’s costs of funds and carry – minus the financier’s rent. The financier extracts as his rent a fraction  $1 - \theta$  of the value that his financing creates relative *not* to firm value today, but to the firm’s outside option value if that bargaining were to fail. That is,

$$\begin{aligned} \text{Financing payoff}_t &= \text{Continuation value}_t - \text{Rent}_t \\ &= \text{Continuation value}_t - (1 - \theta)(\text{Continuation value}_t - \text{Outside option}_t) \\ &= \theta \text{ Continuation value}_t + (1 - \theta) \text{ Outside option}_t. \end{aligned}$$

Thus, better outside options make financing less dilutive to firm value by a factor of  $1 - \theta$ . On the other hand, bargaining on date 0 involves the carry cost,  $\varphi$ , since one unit of funds is kept until date 1. This lowers the continuation value. But the entrepreneur only bears a fraction  $\theta$  of it since the financier bears the rest as a reduction in his rent.

As Figure 2 outlines, the entrepreneur thus optimally bargains earlier if and only if  $\theta(v - 1 - \varphi) + (1 - \theta)\widehat{v} \geq \theta(v - 1)$ , or, equivalently,

$$(1 - \theta)\widehat{v} \geq \theta\varphi; \tag{2}$$

that is, if the reduction in dilution due to better outside options,  $(1 - \theta)\widehat{v}$ , outweighs her loss from the carry cost,  $\theta\varphi$ .

**Discussion.** Financial slack may endogenously arise on date 1, despite the carry cost, if it is optimal to bargain infrequently rather than frequently or earlier rather than later. These results obtain because a failure of bargaining makes financing less accessible: The firm can choose to bargain on any date, but if bargaining were to fail, then it must wait one period before being able to bargain again.<sup>12</sup> Because a failure of bargaining is thus consequential, firm value ex-ante is always strictly greater than the firm’s outside option value upon a failure of bargaining on the same date. Thus, when the firm chooses to bargain, it simultaneously captures a fraction  $\theta$  of the surplus that bargaining creates relative to its value ex-ante, and incurs a fraction  $1 - \theta$  of the loss that a failure of the bargaining would entail relative to its value ex-ante. The second component thus makes financing dilutive.

The entrepreneur, in response, bargains infrequently by raising enough at a time to sustain multiple periods without needing to bargain and pay rents, thereby reducing

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<sup>12</sup>Supplemental Appendix SA.1 discusses setups where multiple financiers visit each period.

the frequency of dilution. She may also optimally raise funds earlier to bargain when her outside options are better, thereby reducing the size of dilution.

The second result of early bargaining leads to a distinct comparative statics. Again given the initial endowment  $h_0 = 1$ , now assume that if bargaining were to fail on date 0, the second financier cancels his visit on date 1.<sup>13</sup> Essentially, the firm’s access to financing is now ‘weaker’ because it gets lost for good upon a bargaining failure. Then, the firm has its “back against the wall” vis-à-vis either financier, even if it bargains early with financial slack: the project fails if bargaining fails because the firm can no longer raise additional funds. Since the firm’s outside option value is now invariably zero on both dates  $t \in \{0, 1\}$ , the left-hand side of Inequality (2) is zero, such that early bargaining is suboptimal for any parameters. In short, it is only firms with *better* access to financing that may optimally bargain early with financial slack.

The above analysis captures this paper’s core mechanism, but it is incomplete. The initial funds  $h_0 \in \{0, 1\}$  reflect the firm’s preceding choice, and so must be endogenized. In particular, although I separately discussed infrequent bargaining and early bargaining for clarity of each insight, they should interact in principle.<sup>14</sup> Therefore, I transition to a fully dynamic setup in infinite horizon for more complete analysis.

## 4 Model

This section sets up the main model of firm financing subject to bargaining frictions.

**Environment.** Time is continuous and infinite  $t \in [0, \infty)$ . All agents are risk neutral and have a common time discount rate  $\rho > 0$ . Agents called insiders own a firm to run a business. The business has an exogenous cash flow profile, given in Section 4.1. The firm may hold internal funds (or ‘funds’)  $h_t \geq 0$ , to which cash flow accrues. Internal funds yield a return  $r < \rho$ .<sup>15</sup> The spread  $\varphi \equiv \rho - r > 0$  is the associated carry cost.

Insiders may frictionlessly receive a nonnegative dividend. If internal funds are depleted without immediate financing, the business fails with zero salvage value.

**Financing bargaining.** To avoid business failure, the firm needs regular financing. Insiders are penniless and hence must raise funds for the firm from homogeneous agents called financiers. Insiders can choose when to bargain à la Nash with financiers for funds.<sup>16</sup> Bargaining weights for insiders and financiers are  $\theta \in (0, 1)$  and  $1 - \theta$ ,

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<sup>13</sup>This may reflect adverse selection problem – see Appendix A.1 in connection to Section 4.

<sup>14</sup>For example, if early bargaining at  $t = 0$  is not optimal, the firm at  $t = -1$  with no funds  $h_{-1} = 0$  may find it optimal to raise three units instead of two from the financier on that date.

<sup>15</sup>Internal funds can, via a simple reformulation of cash flow in Section 4.1, represent slack against borrowing constraint in a risk-free instantaneous lending market with interest rate  $r < \rho$ .

<sup>16</sup>Other axiomatic bargaining solutions, such as Kalai and Smorodinsky (1975) and Kalai (1977), are equivalent to Nash (1950) (under suitable generalizations) in this setup due to the

respectively. Financiers provide funds and receive a proportional ownership stake.

To focus on the bargaining-driven dynamics of financing, I do not separate the timing of bargaining from the timing of financing: at any  $t \geq 0$ , no agents can commit to a future transaction at  $t' > t$ . For further simplicity, I assume that financiers are deep-pocketed but upon financing, they become penniless insiders.

**Outside options.** If insiders can walk away from bargaining and immediately find alternative financiers, bargaining is trivial due to perfect competition among financiers. I therefore assume that if bargaining with one group of financiers were to fail, it takes time to find another group of financiers to bargain with for funds. I call this time lag ‘exclusion.’ Excluded insiders are ‘re-included’ at a Poisson rate  $\gamma \geq 0$  that parametrizes the accessibility of alternative financing. Thus, insiders face a stochastic time lag in finding another financier upon a failed bargaining. Re-inclusion means restoring the ‘access,’ i.e., the ability to freely choose when to bargain with financiers.

**Asymmetry.** The model’s core premise is that insiders can bargain again at anytime after bargaining succeeds, but if it were to fail, they must wait some time to be able to bargain again. This asymmetry with respect to bargaining outcome is, arguably, rather plausible.<sup>17</sup> Corporate managers can forecast whether funds will be needed soon and start engaging with financial investors in advance. If, however, they fail to reach an agreement during bargaining, the managers should start looking for other options for financing. Because a failure of bargaining is, as Appendix A shows, prone to creating conditions that trigger or intensify other frictions, accessing such alternatives after a bargaining failure is likely more difficult than accessing the primary options without such a preceding failure. Even if bargaining does not fail in equilibrium such that this asymmetry does not manifest itself, what would happen if it did determines what does happen in terms of bargaining payoff – and hence, when it is optimal to bargain.

**Minor comments.** The assumption that financiers become penniless upon financing represents that each financier is limited in funding capacity but can syndicate with others; it is also without loss to assume that insiders bargain with an intermediary that syndicates competitive financiers (see Appendix A.2). Also, the restriction to equity financing rather than generic contracts can be microfounded through inability to prevent renegotiation given each financier’s bargaining power. Lastly, unobservability of internal funds  $h_t$  can rationalize why only insiders get to choose when to bargain.

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assumptions of deep-pocketed financiers and risk-neutral preferences.

<sup>17</sup>In comparison, standard models of continuous-time search frictions with bargaining are symmetric in that the matching rate is identical regardless of bargaining outcome. This symmetry fails in standard discrete-time models of search frictions with bargaining: if bargaining were to fail, the firm must wait one period before regaining the chance to be matched. Thus, the present asymmetry with  $\gamma < \infty$  can also be thought of as a continuous-time analogue of what is natural in discrete-time models; one of the footnotes in Section 5.1 discusses it in detail.

## 4.1 Cash flow

The business has an exogenous cash flow profile. Consider two concrete examples:

1. The business is a ‘startup’ that incurs a fixed flow expense  $\kappa dt$  with  $\kappa > 0$  (‘burn rate’), until succeeding upon a stopping time  $\tau$  at a Poisson rate  $\lambda > 0$ . Upon success, the business earns a terminal payoff  $\Pi > \frac{\kappa}{\lambda}$  and ends.
2. The business is an ‘operating firm’ that indefinitely earns an average flow profit  $\pi dt$ , where  $\pi > 0$ , but subject to occasional losses due to volatility  $\sigma dB_t$ , where  $\sigma > 0$  and  $B_t$  is a standard Brownian motion.

Generically, time- $t$  cash inflow is  $\mu dt + \sigma dB_t$ , where  $\mu \in \mathbb{R}$ ,  $\sigma \geq 0$ ; for startups,  $\mu \equiv -\kappa < 0$  and for operating firms,  $\mu \equiv \pi > 0$ . The business may ‘succeed’ upon a stopping time  $\tau$  at a Poisson rate  $\lambda \geq 0$  with a terminal dividend  $\Pi + h_\tau$ ,  $\Pi \in \mathbb{R}$ .

**Assumption 1.**  $\mu + \lambda\Pi > 0$ . If  $\sigma = 0$ , then  $\mu < 0$  and either  $r \leq 0$  or  $\frac{\mu + \lambda\Pi}{\rho + \lambda} < \frac{|\mu|}{r}$ .

The first part gives a positive net present value of the business  $\frac{\mu + \lambda\Pi}{\rho + \lambda} > 0$  in the absence of frictions. The second part ensures that external financing is regularly needed.<sup>18</sup>

## 4.2 Value functions and bargaining for financing

I consider Markov-perfect equilibrium. By linear preference, optimal dividend policy is  $\max\{0, h_t - \bar{h}\}$  above a threshold  $\bar{h} \geq 0$ . Letting  $V(h)$  denote equilibrium value of insiders (including ones that previously were financiers) given funds  $h$ ,  $\bar{h}$  solves dividend optimality  $V'(\bar{h}) = 1$ , i.e., smooth pasting, and super contact  $\sigma > 0 \implies V''(\bar{h}) = 0$ .

Suppose that at a given point, firm insiders have chosen to bargain with financiers for funds. Let  $V_o(h)$  denote insiders’ reservation value (‘outside option if bargaining fails’) given funds  $h$ , and  $x \in [0, 1]$  their retained ownership. Nash bargaining solves

$$\max_{x \in [0, 1], \bar{h} \geq 0} \left( xV(\bar{h}) - V_o(h) \right)^\theta \left( (1 - x)V(\bar{h}) - (\bar{h} - h) \right)^{1 - \theta}.$$

The solution is straightforward:  $\bar{h} \in \arg \max_{\tilde{h}} V(\tilde{h}) - (\tilde{h} - h)$ , and

$$x(h)V(\bar{h}) = \left[ V(\bar{h}) - (\bar{h} - h) \right] - (1 - \theta) \left[ V(\bar{h}) - (\bar{h} - h) - V_o(h) \right]. \quad (3)$$

The funding target  $\bar{h}$  maximizes continuation value,  $V(\bar{h}) - (\bar{h} - h)$ .<sup>19</sup> Ownership is split such that insiders’ value from bargaining  $x(h)V(\bar{h})$  is continuation value,  $V(\bar{h}) - (\bar{h} - h)$ ,

<sup>18</sup>Even without volatility  $\sigma = 0$ , loss occurs with positive probability  $\mu < 0$  (i.e., certainty), and fully covering losses with returns on internal funds  $h_t \geq |\mu|/r$  is suboptimal.

<sup>19</sup>The funding target, taken as the infimum of the  $\arg \max$ , coincides with the dividend payout threshold because both equalize marginal value of funds with marginal cost of funds. This identity may fail given transaction costs of dividend payout, but this paper’s results are robust.

minus what financiers extract in rents. Crucially, they extract a fraction  $1-\theta$  of not just the value that bargaining creates relative to firm value ex-ante,  $V(\bar{h}) - (\bar{h} - h) - V(h)$ , but also the value of not letting the bargaining fail and thus making financing less accessible,  $V(h) - V_o(h)$ . This second component is the focus of analysis in Section 5.

### 4.3 Dynamic programming

The simplifications in the setup make the financing history payoff-irrelevant, allowing for a straightforward dynamic programming for the pair of value functions  $(V, V_o)$  that depend only on the amount of internal funds  $h$ .

**Boundary conditions.** First, dividend optimality gives  $V'(\bar{h}) = 1$  by smooth pasting and, if  $\sigma > 0$ ,  $V''(\bar{h}) = 0$  by super contact.<sup>20</sup> Second, the business fails if funds are depleted without access to financing, that is,  $V_o(0) = 0$ . Third, insiders have access to financing, that is,  $V(h) \geq x(h)V(\bar{h})$ , where equality means optimal financing; since financing must be optimal at  $h = 0$ ,  $V(0) = x(0)V(\bar{h})$ .

**HJB equations.** When insiders are neither receiving dividends nor raising financing, their equilibrium value function  $V$  and reservation value function  $V_o$  solve

$$\begin{aligned}\rho V(h) - rhV'(h) &= \mu V'(h) + \frac{1}{2}\sigma^2 V''(h) + \lambda(\Pi + h - V(h)), \\ \rho V_o(h) - rhV_o'(h) &= \mu V_o'(h) + \frac{1}{2}\sigma^2 V_o''(h) + \lambda(\Pi + h - V_o(h)) + \gamma(V(h) - V_o(h)).\end{aligned}\quad (4)$$

The last term in Equation (4) reflects that if financing were to fail such that insiders' value function becomes  $V_o$ , they can regain access to financing – that is, revert to  $V$  – at a Poisson rate  $\gamma$ . Since financing does not fail in equilibrium, this Poisson arrival at rate  $\gamma$  never actually occurs on path.

## 5 Analysis

The model's equilibrium concept is Markov perfection in that strategies are restricted to depend only on  $h$  and whether insiders have access to financing. This section presents the analysis of the equilibrium – which exists and is unique.

**Lemma 1** (Existence and uniqueness). *There exist unique  $V$  and  $V_o$ , the equilibrium value function of insiders and their reservation value function.*<sup>21</sup>

<sup>20</sup>The same holds for  $V_o$  with respect to the corresponding dividend threshold  $\bar{h}_o$ .

<sup>21</sup>The proof in Appendix B.1 leverages the two-step recursion shown in Section 4.3: equilibrium value function  $V$  depends on itself because (i)  $V$  depends on  $V_o$  through bargaining  $\theta$ , and (ii)  $V_o$  depends on  $V$  through re-inclusion  $\gamma$ . The expected discounting at rate  $\rho > 0$  due to the

Given the assurance, Sections 5.1 and 5.2 discuss lumpy financing and early financing. Section 5.3 illustrates how financial slack reduces dilution, and Section 5.4 characterizes equilibrium financing costs. Section 5.5 presents comparative statics.

## 5.1 Dilution and lumpy financing

Let  $\bar{h} = \inf\{\arg \max_h V(h) - h\}$ .<sup>22</sup> Financiers' rents in Equation (3) can be decomposed into

$$(1 - \theta) \left[ \underbrace{\left( V(\bar{h}) - (\bar{h} - h) - V(h) \right)}_{\equiv N(h)} + \underbrace{\left( V(h) - V_o(h) \right)}_{\equiv E(h)} \right].$$

Above,  $N(h)$  is the 'on-path' surplus from bargaining when insiders choose to enter into bargaining with financiers, and  $E(h)$  is insiders' loss due to exclusion: if bargaining fails once they have entered into it, insiders get  $V_o(h)$  instead of  $V(h)$ . As shown, financiers receive as their rents a fraction  $1 - \theta > 0$  of not just  $N(h)$  but also  $E(h)$ .

Substitute the above decomposition into Equation (3) to obtain

$$\begin{aligned} x(h)V(\bar{h}) &= V(\bar{h}) - (\bar{h} - h) - (1 - \theta)[N(h) + E(h)] \\ &= V(h) + \theta N(h) - (1 - \theta)E(h). \end{aligned}$$

It is thus optimal to bargain at  $h \leq \bar{h}$ , that is,  $V(h) \leq x(h)V(\bar{h})$ , if and only if  $(1 - \theta)E(h) \leq \theta N(h)$ . Since insiders in equilibrium can bargain at any time they choose to,  $V(h) \geq x(h)V(\bar{h})$  holds for any  $h$ . Therefore, this model's optimality condition for choosing to bargain is

$$(1 - \theta)E(h) = \theta N(h). \quad (5)$$

Equation (5) demonstrates why it may sometimes be suboptimal to bargain. If a failure of bargaining were not consequential, that is,  $\gamma \rightarrow \infty$ , then the loss from exclusion,  $E(h)$ , would vanish (given  $h \neq 0$ ) and financiers' rents would be proportional to the on-path surplus,  $N(h)$ , relative to equilibrium value,  $V(h)$ . Therefore, financing would not dilute the firm value, and it would always be optimal to bargain in order to capture a portion of that on-path surplus. Because a failure of bargaining is consequential, that is,  $\gamma < \infty$ , financiers' rents are proportional to more than the on-path surplus,  $N(h) + E(h)$ . Thus, financing dilutes firm value,  $V(h)$ , by  $(1 - \theta)E(h) > 0$ . As a result, it is not optimal to bargain when the portion of the on-path surplus that insiders can retain by doing so,  $\theta N(h)$ , is smaller than the dilution,  $(1 - \theta)E(h)$ .<sup>23</sup>

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time delay in re-inclusion  $\gamma < \infty$  makes it contractionary. Accordingly,  $V$  exists as a unique fixed point by contraction mapping theorem. Step (ii) then determines  $V_o$ .

<sup>22</sup>When  $h_t > \bar{h}$ , the firm immediately pays out dividends  $h_t - \bar{h}$  to insiders. Thus,  $h \mapsto V(h) - h$  is constant above  $\bar{h}$ , and therefore choosing the funding target  $\bar{h}$  as the infimum is without loss.

<sup>23</sup>This is why in standard models of continuous-time search frictions with the same matching

In short, despite the ability to finance anytime without fixed transaction costs, financing is dilutive and thus optimally infrequent and ‘lumpy’ if it involves bargaining  $\theta < 1$  and its failure is consequential  $\gamma < \infty$ .<sup>24</sup> Defining  $B \equiv \{h \in [0, \bar{h}] \mid V(h) = x(h)V(\bar{h})\}$  as the set of  $h$  at which bargaining is optimal, this is formalized below.<sup>25</sup>

**Proposition 1.** *Financing is lumpy and intermittent, i.e.,  $\sup B < \bar{h}$ .*

*Proof.* First, both  $V$  and  $V_o$  are continuous: see Appendix B.1. Next,  $\theta N(\bar{h}) = 0$  since  $N(\bar{h}) = 0$  trivially. Therefore, it suffices to show that dilution is positive at  $h = \bar{h}$ , that is,  $(1 - \theta)E(\bar{h}) > 0$ . By Assumption 1, there is a finite time interval over which, without financing, internal funds  $\bar{h}$  get depleted with nonzero probability. Also, over any finite time interval, re-inclusion fails to occur with nonzero probability since  $\gamma < \infty$ . In sum, there is nonzero probability that today’s exclusion causes business failure within a finite time interval. Without exclusion, positive surplus is retained at depletion since  $\theta > 0$ . Thus,  $V_o(\bar{h}) < V(\bar{h})$ , and  $(1 - \theta)E(\bar{h}) = (1 - \theta)(V(\bar{h}) - V_o(\bar{h})) > 0$  since  $\theta < 1$ .  $\square$

## 5.2 Outside options and early financing

So far, two facts have been established about insiders’ optimal financing strategy,  $B$ :  $0 \in B$  and  $\sup B < \bar{h}$ . The following lemma establishes that it is actually always an interval from zero: that is, optimal financing strategy is monotone in internal funds  $h$ . While this is a technical result, I include a partial proof in the article because the derivation points at the core of how this model departs from the canonical paradigm based on fixed transaction costs of financing.

**Lemma 2** (Monotone financing strategy). *If  $h \in B$ , then  $[0, h] \subset B$ .*

*Proof.* Suppose  $B \setminus \{0\}$  is nonempty. Define  $F$  as the payoff function when insiders immediately raise financing given any funds  $h$  and subsequently follow optimal strategy:

$$F(h) \equiv x(h)V(\bar{h}) = \theta(V(\bar{h}) - (\bar{h} - h)) + (1 - \theta)V_o(h) \leq V(h). \quad (6)$$

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rate regardless of bargaining outcome, financing is never dilutive: the left-hand side of Equation (5) is zero since  $E(h) = 0$ , i.e.,  $V_o(h) = V(h)$  (and the right-hand side can be strictly greater due to search frictions). Note that this does not hold under standard search frictions in discrete time. In those models, if a firm is matched on a date for a chance to bargain but that bargaining were to fail, it must wait until the next date for a chance to be matched again. As such, a failure of bargaining makes financing less accessible by wasting one chance for a match. Therefore,  $E(h) > 0$ , i.e.,  $V_o(h) < V(h)$ , when it is strictly optimal to bargain, and thus financing is dilutive (when it is optimal not to bargain,  $E(h) = 0$  since waiting one period involves no loss, which implies that  $N(h) \leq 0$ , but then bargaining must be trivial since  $N(h) \geq 0$  always); Section 3 illustrates it with matching probability equal to unity. In continuous-time limits,  $E(h) = 0$  always, since the waiting is instantaneous. Therefore,  $\gamma < \infty$  is, in a sense, a continuous-time analogue of what is natural in standard discrete-time settings.

<sup>24</sup>The converse also holds, i.e., financing is not lumpy if  $\theta \rightarrow 1$  or  $\gamma \rightarrow \infty$ : see Proposition 3.

<sup>25</sup>Obviously,  $B$  is nonempty, since  $0 \in B$ .

Immediate financing is optimal on the set  $B$ , and hence is preferred to an instantaneously delayed financing: for  $h \in B \setminus \{0\}$ ,

$$\rho F(h) - rhF'(h) \geq \mathcal{H}(F)(h) \equiv \mu F'(h) + \frac{1}{2}\sigma^2 F''(h) + \lambda(\Pi + h - F(h)). \quad (7)$$

Also, dividend optimality conditions for  $V$  imply that

$$\rho V(\bar{h}) - r\bar{h} = \mathcal{H}(V)(\bar{h}) = \mu + \lambda(\Pi + \bar{h} - V(\bar{h})). \quad (8)$$

Equation (6) implies that  $F'(h) = \theta + (1 - \theta)V'_o(h)$  and  $F''(h) = (1 - \theta)V''_o(h)$ . Thus,

$$\mathcal{H}(F)(h) = \theta \mathcal{H}(V)(\bar{h}) + (1 - \theta)\mathcal{H}(V_o)(h). \quad (9)$$

Substituting (4), (6), (8), (9) into (7) cancels out  $\mathcal{H}(F)(h)$  and gives: for  $h \in B \setminus \{0\}$ ,

$$(1 - \theta)\gamma(V(h) - V_o(h)) \geq \theta\varphi(\bar{h} - h), \quad (10)$$

which is equivalent to Inequality (7). For the rest of the proof, see Appendix B.2.  $\square$

Inequality (10) captures the core trade-off associated with early financing. Suppose that financing is optimal at  $h_t = h > 0$ . This implies that insiders prefer bargaining earlier now to delaying bargaining by an instant  $dt$  as a one-shot deviation.<sup>26</sup> Two factors are relevant in this comparison. On one hand, there is a carry cost when funds  $\bar{h} - h_t$  come in at  $t$  instead of at  $t + dt$ . This reduces the on-path surplus,  $N(h_t) \equiv V(\bar{h}) - (\bar{h} - h_t) - V(h)$ , by  $\varphi(\bar{h} - h_t) dt$ . When insiders bargain earlier at  $t$ , they incur a fraction  $\theta$  of this loss since financiers bear the rest  $1 - \theta$  as a reduction in rents. This is the right-hand side. On the other hand, there is a chance of finding alternative financiers during  $(t, t + dt]$ . This raises insiders' reservation value,  $V_o(h_t)$ , by  $\gamma(V(h_t) - V_o(h_t)) dt$ , and hence reduces the loss from exclusion,  $E(h_t) \equiv V(h_t) - V_o(h_t)$ , by as much. Thus, when insiders bargain earlier at  $t$ , dilution,  $(1 - \theta)E(h_t)$ , is reduced by a fraction  $1 - \theta$  of the amount. This is the left-hand side.

Despite the on-path loss from the carry cost involved, earlier financing mitigates the off-path consequence of a bargaining failure. Even though the internal funds kept at bargaining are not deployed on the equilibrium path where financing is readily accessible, they *are* deployed *off* the equilibrium path where financing is less accessible, in order to sustain the business until other financiers are found. By improving outside options and thus making a failure of bargaining less consequential, therefore, earlier financing makes financing less dilutive.

More technically, Lemma 2 allows the equilibrium to be characterized by  $(s, S)$

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<sup>26</sup>This comparison is marginal in time ( $t$  versus  $t + dt$ ), rather than in funds ( $h$  versus  $h + dh$ ).

bounds  $(\underline{h}, \bar{h})$ , where  $\underline{h} \equiv \sup B$ . When  $\lim_{s \rightarrow t^-} h_s = \underline{h}$ , insiders raise financing in the amount  $\Delta h \equiv \bar{h} - \underline{h} > 0$  so that  $h_t = \bar{h}$ . Henceforth, I refer to  $\underline{h}$  both as ‘financing threshold’ – to address the dynamics of financing – and ‘funding reserve’ – to focus on the amount of financial slack when financing is raised. I say that insiders engage in ‘early financing’ if  $\underline{h} > 0$ , that is, if insiders raise financing before running out of funds. I also refer to  $\bar{h}$  and  $\Delta h$  as ‘funding target’ and ‘financing amount,’ respectively.

**Proposition 2** (Monotonicity of early financing). *Given the other parameters, there exists  $\underline{\gamma} \in (0, \infty)$  such that early financing is optimal, i.e.,  $\underline{h} > 0$ , if and only if  $\gamma > \underline{\gamma}$ . Early financing is thus never optimal, i.e.,  $\underline{h} = 0$ , if  $\gamma = 0$ .*

Proposition 2 formalizes the logical implication from the discussion of Lemma 2: The quicker insiders can find alternative financiers upon a failure of financing, the better they are at mitigating its consequence – and thus at reducing the size of dilution – by raising external financing with some internal funds. Therefore, early financing, as a binary choice  $\mathbb{1}(\underline{h} > 0)$ , is weakly *increasing* in firms’ access to alternative financing  $\gamma$ .

### 5.3 Illustration

I now graphically explain how financial slack reduces dilution. Consider a startup that incurs a fixed expense  $\kappa \, dt$  until success at Poisson rate  $\lambda$  with a terminal payoff  $\Pi$ . Baseline parameters are:  $(\rho, r) = (0.05, 0)$ ,  $(\theta, \gamma) = (0.5, 1)$  and  $(\kappa, \lambda, \Pi) = (2, 0.1, 50)$ .

Here, I use the term ‘financing cost’ to denote financiers’ rents. Also, this part does not explicitly decompose it into dilution,  $(1 - \theta)E(h)$ , and the rest,  $(1 - \theta)N(h)$ , in order to keep the graphs simple – similar to Section 3.<sup>27</sup>

**Size-frequency tradeoff.** Figure 3 illustrates the relationship between financial slack and financing cost. In the main plot that shows optimal financing strategy, insiders raise financing once every  $\frac{\Delta h}{\kappa} \approx 1.8$  periods (until success) and ‘very early’ with a large funding reserve  $\underline{h} \approx 9.1$ . At each financing, insiders incur a small financing cost: 0.18 in value. If this cost were ‘fixed,’ the strategy in the main plot would be strictly dominated by further delaying financing with lower thresholds  $\tilde{h} < \underline{h}$ , as the cost would be less frequently incurred. The bottom two subplots, Figures 3a and 3b, show that these deviations are indeed not optimal.<sup>28</sup> Although frequency decreases, the size of the cost endogenously increases, to 1.5 with  $\tilde{h} = 0.5\underline{h}$  and even up to 7.88 with  $\tilde{h} = 0$ .

<sup>27</sup>Section 5.4 shows that there is a tight relationship between dilution,  $(1 - \theta)E(h)$ , and the total financing cost,  $(1 - \theta)[N(h) + E(h)]$ , when financing is optimal,  $h \leq \underline{h}$ .

<sup>28</sup>For ease of graphical illustration, I assume that these deviations – although appearing as recurrent in the two subplots – are taken by the agents as ‘one-shot’ until each time of financing. Anticipated deviations post financing would affect firm value ex-post, endogenously altering the ‘schedule’ of financing rents ex-ante as shown in Figure 4.



Figure 3: Financial slack and financing cost

The plots describe a startup with parameters  $(\rho, r) = (0.05, 0)$ ,  $(\theta, \gamma) = (0.5, 1)$ ,  $(\kappa, \lambda, \Pi) = (2, 0.1, 50)$ . The graphs in solid black tracks the history of internal funds  $h_t$  until business success – at  $t = 10$  in this sample. Financing (i.e.,  $h_t$  jumping from financing threshold  $\underline{h}$  to funding target  $\bar{h}$ ) involves financiers' rent extraction given in magenta bar. The bottom subplots depict suboptimally lower financing thresholds. An operating firm  $\pi dt + \sigma dB_t$  with  $(\pi, \sigma) = (1, 2)$  yields the same patterns, including how counterfactual financing thresholds affect the size of financing cost.

Why does financing cost go up when insiders raise financing with smaller financial slack? The answer lies in insiders' outside options. In Figure 4, I decompose valuation on the vertical axis, assuming a one-shot strategy of immediate financing at each  $h$  across  $[0, \bar{h}]$  on the horizontal axis.<sup>29</sup> The rightmost edge in blue is the funding target,  $\bar{h}$ , and the red vertical line the financing threshold,  $\underline{h}$ .

The solid black line at the top of Figure 4 is firm value upon financing,  $V(\bar{h})$ . To attain this 'post-money' value, financiers must provide financing  $\bar{h} - h$ , represented as the height of the light gray area right below it; this amount decreases in  $h$  at a unit slope. Bargaining surplus, however, is not simply the difference between post-money value,  $V(\bar{h})$ , and 'money,'  $\bar{h} - h$ . Insiders possess outside options at bargaining,  $V_o(h)$ , represented as the height of the dark gray area. Even if financing were to fail, insiders with financial slack may still seek alternative funding sources at Poisson rate  $\gamma = 1$  to prevent business failure. But if financing were to fail without slack, the business

<sup>29</sup>This is what  $F$  in the proof of Lemma 2 represents, and is suboptimal on  $(\underline{h}, \bar{h}]$ . One distinction is that the present comparison is in terms of funds  $h$ , rather than time  $t$ .



Figure 4: Financial slack, outside options, and financing cost

This plot addresses the same startup example as in Figure 3. It shows value decomposition when insiders with funds  $h \in [0, \bar{h}]$  choose to raise financing – suboptimally on  $(\underline{h}, \bar{h}]$ . Continuation value from bargaining is  $V(\bar{h}) - (\bar{h} - h)$ , which is the height of the bottom of the light gray area, and outside option value is the height of the dark gray area. The bargaining surplus,  $V(\bar{h}) - (\bar{h} - h) - V_o(h) (= N(h) + E(h))$ , is in between the two, which is split, by ratio  $(1 - \theta : \theta)$ , to financiers’ rents in magenta and insiders’ retained surplus in light blue.

promptly fails. Thus, the outside option value,  $V_o$ , rises steeply in financial slack,  $h$ .

Therefore, the bargaining surplus, which is the two colored areas in the middle, decreases in financial slack. This is what gets divided, according to the ratio  $(1 - \theta : \theta)$ , into financiers’ rents in magenta and insiders’ surplus retention in light blue.

In short, financial slack monotonically reduces the size of financing cost by improving the firm’s outside options and thereby making financing less dilutive. But then, why is the optimal funding reserve,  $\underline{h}$ , not even higher? The answer is the frequency. Given funding target,  $\bar{h}$ , a higher financing threshold,  $\underline{h}$ , decreases financing amount,  $\Delta h$ , and thus increases the frequency of incurring the financing cost, while its size no longer decreases as steeply.<sup>30</sup>

**Access to alternative financing  $\gamma$ .** When is early financing not optimal? As Proposition 2 shows, it is not optimal when insiders cannot quickly access alternative financing,  $\gamma \leq \underline{\gamma}$ . Then, financial slack does not improve insiders’ outside options enough to justify the increases in carry cost and financing frequency.

Figure 5 illustrates it with  $\gamma = 0$  for an otherwise identical startup. Financial slack increases insiders’ outside options at a much lower slope, since insiders cannot access alternative financing if financing were to fail. Consequently, financial slack does not reduce the size of financing cost enough to compensate for its increased frequency. These insiders instead delay financing until funds are depleted and, when they do raise funds, raise a much larger amount,  $\Delta h \approx 18.27 > 3.64$ , to reduce the frequency instead.

<sup>30</sup>There is a gap even at the top  $V(\bar{h}) - V_o(\bar{h}) > 0$ , which is a mere 0.022 but certainly nonzero.

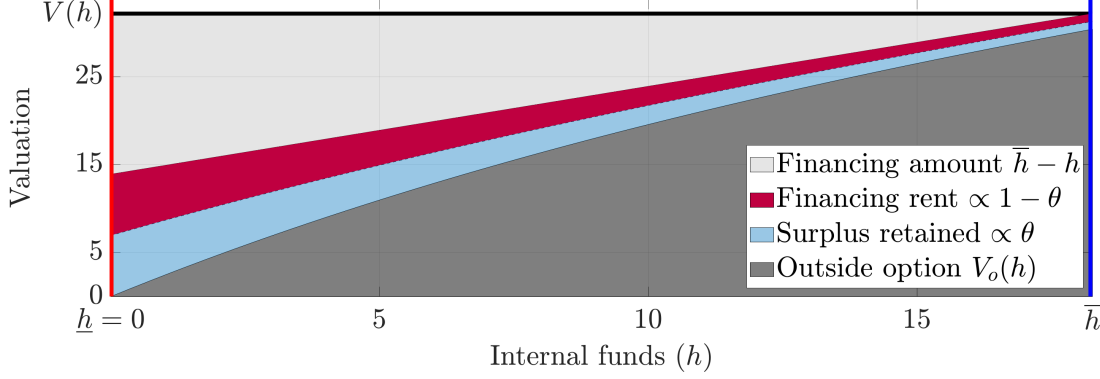


Figure 5: Poor access to alternative financing  $\gamma = 0$

The height of the plot is adjusted so that slopes are visually comparable to those in Figure 4.

## 5.4 Endogenous financing costs

As illustrated, there are two types of optimal financing strategies, depending on whether financing is early. Since financing costs are endogenous to financing strategy, their equilibrium expression looks qualitatively different in each case, as formalized below.

**Lemma 3** (Endogenous financing costs). *In equilibrium, financiers extract rents from insiders through financing in the amount of:  $(1-\theta)(V(\bar{h})-\bar{h})$  if insiders delay financing until funds are depleted, i.e.,  $\underline{h} = 0$ ; and, if insiders finance early, i.e.,  $\underline{h} > 0$ ,*

$$\frac{\varphi}{\gamma}\Delta h + \frac{1}{2}\frac{\sigma^2}{\theta\gamma}\left(V''^{(+)}(\underline{h}) - V''^{(-)}(\underline{h})\right) \geq \frac{\varphi}{\gamma}\Delta h. \quad (11)$$

For clarity, first consider startups with  $\sigma = 0$  such that the lower bound in (11) attains. Lemma 3 states that, in equilibrium, firms that raise financing after running out of funds pay financing costs proportional, by a factor of financiers' bargaining power,  $1 - \theta$ , to firm value net of financing,  $V(\bar{h}) - \bar{h}$ , whereas firms that raise financing early pay financing costs proportional, by the ratio of carry cost to the speed of accessing alternative financing,  $\frac{\varphi}{\gamma}$ , to financing amount,  $\Delta h$ . Thus, the model explains what in equilibrium resemble two distinct types of transaction costs – ‘fixed’ costs covarying with firm value and ‘variable’ costs proportional to financing amount – as in fact arising from firms' optimal choice of whether to bargain early with financiers for funds.

The reason why the expression  $\frac{\varphi}{\gamma}\Delta h$  in (11), given  $\sigma = 0$ , does not involve bargaining power,  $\theta$ , is because of early financing. From the discussion of Lemma 2, earlier financing at  $h_t > 0$  reduces dilution,  $(1 - \theta)E(h_t)$  by the amount  $(1 - \theta)\gamma E(h_t) dt$ , and hence at the rate  $\gamma dt$ . Note that by Equation (5),  $h_t \leq \underline{h}$  if and only if a fraction  $\theta$  of financiers' rents is due to dilution.<sup>31</sup> Thus, earlier financing at  $h_t \in (0, \underline{h}]$  reduces

<sup>31</sup>Substitute Equation (5) into  $(1 - \theta)[N(h_t) + E(h_t)] = \frac{1-\theta}{\theta}N(h_t) + (1 - \theta)E(h_t)$ .

financiers' rents at the rate  $\theta\gamma dt$ . On the other hand, it reduces the on-path surplus by the amount  $\varphi(\bar{h} - h_t) dt$ , and insiders bear a fraction  $\theta$  of it due to bargaining. Therefore, at an optimal 'interior' financing threshold,  $\underline{h} > 0$ , that equalizes the marginal costs and benefits of earlier financing,  $\theta$  is canceled out, giving (11).

Economically, given  $h \leq \underline{h}$ , a higher financier bargaining power,  $1 - \theta$ , lowers firm value,  $V(h)$ , and hence raises the on-path surplus from bargaining,  $N(h)$ . Thus, while financiers' rents,  $(1 - \theta)[N(h) + E(h)]$ , become larger ceteris paribus, dilution,  $(1 - \theta)E(h)$ , becomes a smaller fraction,  $\theta$ , of the rents. Since earlier financing reduces dilution at the marginal rate of  $\gamma$ , its benefit does not scale up with financiers' total rents when their bargaining power increases. But at the same time, earlier financing becomes less costly to insiders because when the marginal carry cost involved,  $\varphi(\bar{h} - h)$ , reduces the on-path surplus,  $N(h)$ , financiers bear a larger fraction,  $1 - \theta$ , of it. Therefore, insiders optimally raise financing even earlier to reduce dilution further, such that the optimized financing rent,  $(1 - \theta)[N(\underline{h}) + E(\underline{h})]$ , ends up not increasing.<sup>32</sup>

In short, the cancellation of  $\theta$  in deriving (11) represents that by optimally financing early, insiders are in effect offsetting financier bargaining power,  $1 - \theta$ , with the financial slack that they bring to the bargaining table,  $\underline{h} > 0$ .

**Robustness to option effects.** Generally when  $\sigma > 0$ , there is a third effect from the option value of waiting. When  $h_t \rightarrow \underline{h}^+$ , delaying financing may avert dilution from financing if insiders subsequently receive positive cash flows,  $\mu dt + \sigma dB_t > 0$ ; its expected value is  $\frac{1}{2}\sigma^2(V''^{(+)}(\underline{h}) - V''^{(-)}(\underline{h}))$ . This pushes down financing threshold and, as (11) shows, adds an option component to financing costs on top of the 'variable-cost' component, because waiting for an option value worsens *outside* options at bargaining if negative cash flow shocks follow and thus endogenously increases dilution.

The comparative statics in Section 5.5 is, however, largely robust to this option effect precisely because of early financing. When firms bargain with financial slack, they are reducing the size of dilution and thereby lowering the option value of avoiding bargaining in the first place. Thus, the size of the effect mostly comoves with overall financing costs that are being optimized through early financing.

## 5.5 Comparative statics of financial slack

I now present comparative statics. While some claims are formally established only under the analytically tractable case of  $\sigma = 0$  without the option effect, numerical simulations suggest that the patterns are robust even when  $\sigma > 0$ , consistent with the

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<sup>32</sup>Earlier financing also reduces the portion of financiers' rents other than dilution,  $(1 - \theta)N(h)$ , because the carry cost involved reduces the on-path surplus,  $N(h)$ . To be clear, this is an indirect effect: insiders have no incentive to reduce the on-path surplus in order to reduce what financiers extract from it, since insiders retain the remainder,  $\theta N(h)$ , which accrues to firm value,  $V(h)$ .

discussion on it in Section 5.4. For clarity, let  $r \in [0, \rho)$ ; a negative yield (very) slightly complicates exposition. I will first explain intuitions and then discuss the significance.

**Proposition 3** (Comparative statics in  $\theta$  and  $\gamma$ ). *Equilibrium,  $(\underline{h}, \bar{h})$ , is invariant to  $\gamma \leq \underline{\gamma}$ . **Funding target**,  $\bar{h}$ , is decreasing in  $\theta$ , and decreasing in  $\gamma \geq \underline{\gamma}$ .<sup>33</sup> **Financing threshold**,  $\underline{h}$ , is zero for any  $\theta$  above some  $\underline{\theta} < 1$ , and positive and non-monotonic for  $\gamma \in (\underline{\gamma}, \infty)$ . **Financial slack vanishes**,  $\bar{h}, \underline{h}, \Delta h \rightarrow 0$ , as either  $\theta \rightarrow 1$  or  $\gamma \rightarrow \infty$ .*

*If  $\sigma = 0$ , the following also hold: **Financing threshold**,  $\underline{h}$ , is decreasing in  $\theta$  when  $\underline{h} > 0$ ; **Financing amount**,  $\Delta h$ , – in case financing is early (i.e.,  $\underline{h} > 0$  such that  $\Delta h < \bar{h}$ ) – is constant (increasing) in  $\theta$  given  $r = 0$  ( $r > 0$ ), and decreasing in  $\gamma$  given  $r = 0$ ; **Financing rent**,  $(1 - \theta)[N(\underline{h}) + E(\underline{h})]$ , is constant (increasing) in  $\theta$  if  $\underline{h} > 0$  given  $r = 0$  ( $r > 0$ ), decreasing in  $\theta$  if  $\underline{h} = 0$ , decreasing in  $\gamma \in [\underline{\gamma}, \infty)$  given  $r = 0$ , and vanishing as either  $\theta \rightarrow 1$  or  $\gamma \rightarrow \infty$ .*

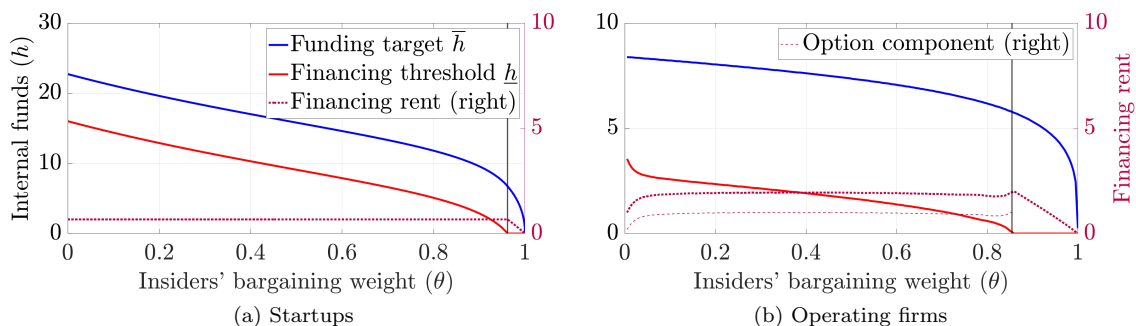


Figure 6: Comparative statics in bargaining power  $\theta$

Parameters are  $(\rho, r, \gamma) = (0.05, 0, 0.5)$  and, for startups,  $(\kappa, \lambda, \Pi) = (2, 0.1, 50)$  and, for operating firms,  $(\mu, \sigma) = (1, 2)$ ; see Section 4.1. Frictionless net present values are set at 20 for both cases. Vertical lines are early financing thresholds. The ‘option component’ in Figure 6b is the second term in (11) from Lemma 3, which is zero for startups in Figure 6a since  $\sigma = 0$ .

**Intuitions.** First, an increase in either bargaining weight,  $\theta$ , or access to alternative financing,  $\gamma (\geq \underline{\gamma})$ , lowers incentives to keep financial slack, and hence decreases funding target,  $\bar{h}$ , – down to zero in the limit along with  $\underline{h}$  and  $\Delta h$ .<sup>34</sup> Second, funding reserve,  $\underline{h}$ , is decreasing in  $\theta$  as well. As Lemma 3 shows, insiders reduce the size of dilution by financing early,  $\underline{h} > 0$ . With a higher  $\theta$ , financiers extract a smaller rent, and so insiders have less incentives to reduce the size of dilution.

Funding reserve,  $\underline{h}$ , is, however, non-monotonic in access to alternative financing,  $\gamma$ . With  $\gamma$  above but near  $\underline{\gamma}$ , insiders finance early to improve their outside options. It

<sup>33</sup>Within this proposition, ‘decreasing’/‘increasing’ indicates strict monotonicity only.

<sup>34</sup>If  $\theta = 0$ , any allocation is an equilibrium since insiders are indifferent. Thus, the game is not upper-hemicontinuous as  $\theta \rightarrow 0$ . Note that the surplus from bargaining is *fully* ‘externalized’ to financiers when  $\theta = 0$ . Thus, there exists a first-best equilibrium where insiders with choice adopt the best strategy for financiers with benefit. Still, due to zero surplus, insiders’ equilibrium value function is given by lack of access to any financing, i.e., coinciding with  $V_0$  given  $\gamma = 0$ .

still takes quite long to find another financier, so that large funding reserves are needed to make it sufficiently likely before the business fails. As alternative financiers become quicker to find, insiders need less funding reserves,  $\underline{h}$ , to make it at least as likely.

Lastly, financing amount,  $\Delta h$ , determines the frequency of financing, and hence of dilution. Lemma 3 shows that given  $\sigma = 0$  and early financing, the size of financing cost is  $\frac{\Delta h}{\gamma}$ , which is constant in  $\theta$  and decreasing in  $\gamma$ . Therefore, its optimal frequency, which is decreasing in financing amount,  $\Delta h$ , is constant in  $\theta$  and increasing in  $\gamma$ .<sup>35</sup> The claims on financing rent,  $(1 - \theta)[N(\underline{h}) + E(\underline{h})]$ , then follow from Lemma 3.

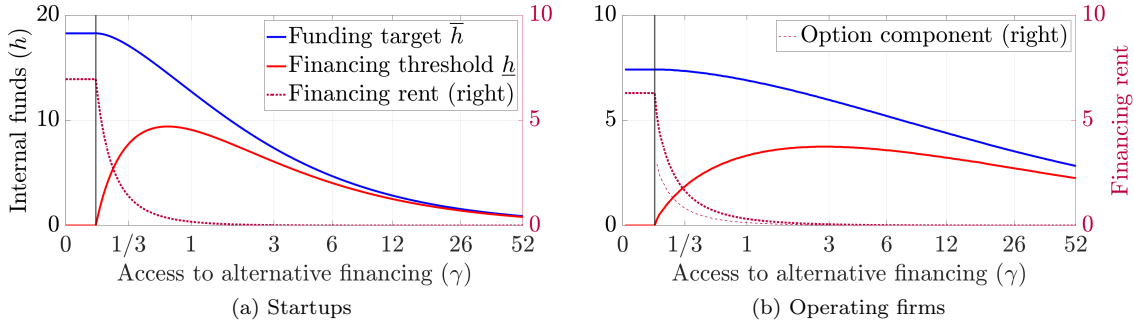


Figure 7: Comparative statics in quality of access to financing  $\gamma$

$\theta = 0.5$ . See Figure 6 for explanation of other parameters and graph components. Financing rent is never zero.

**Significance.** Proposition 3 is significant in two regards. First, as Figure 6a shows (and as Figure 6b closely aligns), financing rent is increasing in financiers' bargaining power,  $1 - \theta$ , but only up to  $\underline{h} = 0$ : on the part of the domain for  $\theta$  over which financing is early,  $\underline{h} > 0$ , financing rent is nonincreasing in financiers' bargaining power,  $1 - \theta$ , consistent with the discussion of Lemma 3. This suggests that it may be incorrect to observe small financing costs and then conclude that financiers must have small bargaining power. The small financing costs in equilibrium may in fact be the result of firms optimally choosing to bargain with financiers having large bargaining power,  $1 - \theta \gg 0$ , when they have *correspondingly* large financial slack,  $\underline{h} \gg 0$ .

Second, the non-monotonicity of financing threshold,  $\underline{h}$ , in access to alternative financing,  $\gamma$ , as shown in Figure 7, suggests broad applicability. Conceptually, varying  $\gamma$  over its domain  $[0, \infty)$  covers firms facing perfectly monopolistic financiers up to firms facing perfectly competitive financiers asymptotically. Excluding the right tail corresponding to an extremely competitive financial sector, this model states that there is a spectrum of firms along two distinct types of financing dynamics. On one hand, firms that cannot quickly access alternative sources of financing raise financing highly infrequently in large chunks and only when they have exhausted most or all of their

<sup>35</sup>If  $r > 0$ , a higher  $\bar{h}$ , holding financing amount,  $\Delta h$ , fixed, implies less frequent financing due to higher internal yields,  $rh_t dt$ ; since  $\frac{\partial \bar{h}}{\partial \theta} < 0$ ,  $\Delta h$  is thus increasing in  $\theta$  when  $\underline{h} > 0$ .

internal funds. On the other hand, firms that can quickly (but not extremely quickly) do so raise financing highly frequently in small increments even as they always keep a sizable amount of cheaper internal funds.<sup>36</sup> Many classes of firms – including young startup firms, financially distressed firms, and large, established and creditworthy firms – seem to fall onto different points along this spectrum.

And this broad applicability directly arises from this paper’s core insight that, because of the bargaining frictions, the incentive to raise financing early increases in the quality of firms’ access to financing. Young startup firms and distressed firms have poor access to alternative financing sources, and therefore raise financing highly infrequently in large chunks and only when funds are exhausted. Established and creditworthy firms as well as the so-called “hot” startups have great access to alternative financing sources, and *therefore* raise financing even when they have large financial slack. In both cases, financiers may possess sizable bargaining power. But how much of it translates into observed financing costs depends endogenously on how firms respond to it through their optimal financing strategy given their quality of access to financing.

## 6 Investment Extension

This section considers investment as firms’ explicit choice to analyze how it interacts with the bargaining frictions. Firms that can choose how much to invest have some control over the flow of funds. As such, investment choice should affect firms’ outside options when they bargain with financiers. Specifically, firms that can quickly obtain large extra funds by divesting capital or reducing their investment expenses have good outside options when they keep funds as they bargain: if external financing were to fail, such firms can do so in order to avoid running out of internal funds. Financiers, therefore, can only extract small rents from such firms when they raise financing early.

In this paper, divestment and reducing investment expenses are thus equivalent off-path responses to reduce on-path dilution. But they are distinct economic activities; the idea of investment irreversibility highlights that divesting existing capital is often difficult – such as for intangible capital or during downturns. Investment choice thus creates two separate channels – divestment and reducing investment expenses – through which this paper’s mechanism based on financier bargaining power may operate.

This section explores how firms with investment technology choose optimal financing strategy vis-à-vis financiers possessing bargaining power. Section 6.1 extends the setup with investment choice, and Section 6.2 explains how investment affects financing costs depending on its reversibility or lack thereof.

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<sup>36</sup>Consistent with the intuition, Supplemental Appendix SA.3 shows that such firms may have funding reserve  $\underline{h}$  that substantially exceeds their investment needs – see Figure SA.3b.

## 6.1 Extended setup

I modify the cash flow profile with a standard investment model à la [Hayashi \(1982\)](#).

**Production and investment.** A firm owns capital,  $K_t$ , and employs it to produce

$$(A dt + \sigma dB_t) K_t$$

in cash flow, with  $A, \sigma > 0$ . The firm can invest  $I_t \in \mathcal{I}$  into its capital stock, subject to adjustment cost;  $\mathcal{I} = \mathbb{R}$  means that capital can be divested into funds, whereas  $\mathcal{I} = \mathbb{R}_+$  means that it cannot, that is, investment is perfectly irreversible. Given flow investment  $I_t dt$ , the firm incurs an additional flow cost in funds  $\Psi(I_t/K_t)K_t dt$ ;  $\Psi$  satisfies  $\Psi(0) = \Psi'(0) = 0 < \Psi''$ . When the firm is neither financing nor paying dividends, internal funds,  $H_t \geq 0$ , evolve as: writing  $i_t \equiv I_t/K_t$ ,

$$dH_t = (A - i_t - \Psi(i_t))K_t dt + \sigma K_t dB_t.$$

For simplicity, assume a quadratic cost:  $\Psi(i) \equiv \psi \frac{i^2}{2}$  for some  $\psi > 0$ . Given depreciation rate  $\delta \geq 0$ , the firm's capital stock,  $K_t$ , evolves as  $dK_t/K_t = (i_t - \delta) dt$ . While there is no explicit capital trade,  $i_t < 0$  given  $\mathcal{I} = \mathbb{R}$  means divestment of capital into funds.

**Investment optimization.** Define  $W(K, H)$  as firm value given capital  $K > 0$  and funds  $H \geq 0$ . Letting  $h \equiv H/K$  and  $V(h) \equiv W(1, h)$ , homogeneity in  $(K, H)$  gives  $W(K, H) = KV(h)$ . Per-capital HJB equation is

$$\rho V(h) - rhV'(h) = \max_{i \in \mathcal{I}} \left\{ (A - i - \Psi(i))V'(h) + \frac{1}{2}\sigma^2 V''(h) + (i - \delta) \underbrace{(V(h) - hV'(h))}_{=W_K} \right\}.$$

**No alternative financiers.** To highlight how investment choice affects financing strategy, I set  $\gamma = 0$  in this extension. This is also sensible given the present focus on investment irreversibility: finding alternative financiers upon a failure of financing is likely more difficult when either the firm's capital is largely intangible and thus hard to liquidate or the economy is in a downturn such that capital is illiquid in general.

## 6.2 Early financing and investment irreversibility

Because investment choice limits tractability, here I focus on delivering the intuition. The key goal of the exposition is to show that the specific numerical patterns presented arise directly from the same economic mechanism analyzed in Sections [3](#) through [5](#).<sup>[37](#)</sup>

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<sup>37</sup>Supplemental Appendix [SA.4](#) provides suggestive analysis in support of the explanation.



Figure 8: Investment irreversibility, financing strategy, and financing costs

Each plot is comparative statics in firm productivity  $A \in [0.17, 0.18]$ , depending on reversibility of investment. Bargaining parameters are  $(\theta, \gamma) = (0.5, 0)$ , and other parameters (except for  $A$  being varied) are adopted from Bolton et al. (2011):  $(\rho, r) = (0.06, 0.05)$ ,  $(A, \sigma, \psi, \delta) = (0.18, 0.09, 1.5, 0.1007)$ . Both  $y$ -axes are identical across plots, and on a per-capital basis. In both plots, financing rent is never exactly zero.

Figure 8 depicts optimal financing strategy as a function of firm productivity, depending on whether investment is irreversible. First, consider the case of reversibility,  $i \in \mathbb{R}$  in Figure 8a. Despite the absence of alternative financiers upon a bargaining failure,  $\gamma = 0$ , firms over the domain  $A \in [0.17, 0.18]$  still raise financing early, and financing costs are small. But in the case of irreversible investment,  $i \geq 0$ , in Figure 8b, it is only more productive firms that raise financing early, whereas less productive ones delay financing until funds are depleted and incur sizable financing costs.

This stark difference in firms' cross-section of financing costs depending on irreversibility stems from this paper's core claim: financing costs are endogenous to firms' financing strategy, because their outside options at bargaining are endogenous to it. To make the point, I proceed in two steps: I first (i) illustrate how firms' ability to reduce investment incentivizes early financing, and then (ii) explain how irreversibility of investment differentially affects this incentive based on productivity.

**(i) Underinvestment as a strategic choice.** Suppose the firm has failed in bargaining with financiers. Since  $\gamma = 0$ , it can never raise external financing. But if it can, for example, sell off its existing capital stock, the firm can occasionally do so upon a negative shock to avoid business failure. Therefore, firms that can *strategically* underinvest in response to a bargaining failure have good outside options vis-à-vis financiers.

In this context, early financing improves firms' outside options if it takes time to obtain sufficient funds through underinvestment. The convex adjustment cost,  $\Psi$ , in the present setup implies that divestment must be optimally smoothed out over time. More generally, disposal of existing capital stock typically requires that the firm needs to find a willing buyer, which may be difficult for a large chunk of capital. Accordingly, firms that have lost access to external financing can divest with ease upon negative shocks if they are sufficiently away from the brink of depleting their funds.



Figure 9: Ease of divestment opportunities,  $\hat{\lambda}$ , given  $\gamma = 0$

Plots track internal funds, financing costs and divestment given stochastic arrivals of divestment opportunities (vertical dashed lines in gray). In the gray area above financing threshold in red, divestment is optimal on the path of equilibrium; the black marker indicates actual divestment. All axes are identical across plots.

To illustrate, I modify the main model from Section 4 with lumpy divestment opportunities. The business has ‘normalized’ running cash flow  $\pi dt + \sigma dB_t$ , similar to operating firms. At a Poisson arrival rate of  $\hat{\lambda} > 0$ , the business obtains opportunities to downsize future cash flow by a factor of  $\eta \in (0, 1)$  in return for receiving funds  $\xi > 0$ . I let  $(\rho, r) = (0.07, 0)$ ,  $(\pi, \sigma) = (1, 1)$  and  $(\eta, \xi) = (0.9, 0.7)$  so that divestment is not first-best. Lastly,  $\theta = 0.5$  and there are no alternative financiers,  $\gamma = 0$ .

Consistent with the above discussion, Figure 9 shows that firms that can divest more swiftly can improve their outside options vis-à-vis financiers when they raise financing with financial slack. This is because if bargaining were to fail, firms with a higher  $\hat{\lambda}$  have a better chance for timely divestment given that they have some funds to begin with. These firms thus choose to raise financing early because they are better at reducing the size of dilution by improving their outside options.

The case of  $\hat{\lambda} = 10$  in Figure 9d is particularly enlightening. Because financing costs are (endogenously) so low, these firms never divest in equilibrium – the gray area where divestment is optimal is below the financing threshold in red. Thus, their equilibrium dynamics is indistinguishable from operating firms in Sections 4 and 5. The fact that early financing still arises despite the absence of alternative external financing,  $\gamma = 0$ , (see Proposition 2) shows that this *off-path* ability to underinvest is, in a sense, a form of alternative *self*-financing – that is, a substitute for  $\gamma$ .

**(ii) Heterogeneous effects of investment irreversibility.** Now I return to the case with convex investment adjustment cost. Based on the above discussion, the part

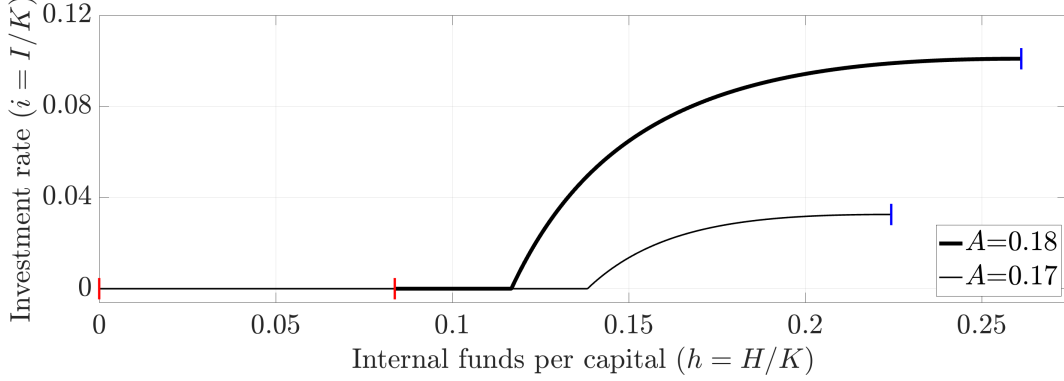


Figure 10: No alternative external financiers  $\gamma = 0$  and no divestment  $i \geq 0$

The black curves, in different thickness, represent optimal investment policy given internal funds on the equilibrium path, bounded by funding target  $\bar{h}$  in the blue line segments and financing threshold  $\underline{h}$  in the red ones.

of Figure 8b under investment irreversibility where financing is not early,  $\underline{h} = 0$ , makes sense: there are no alternative external financiers,  $\gamma = 0$ , and divestment is excluded – both on and off equilibrium path. Nevertheless, firms with higher productivity still raise financing early. To understand why productivity matters for the early financing incentive under irreversibility, I begin by discussing investment policy.

Figure 10 compares two cases of  $A \in \{0.17, 0.18\}$ , depicting optimal investment rate on the equilibrium path,  $i(h)$ , as a function of internal funds per capital,  $h$ , bounded by funding target,  $\bar{h}$ , and financing threshold,  $\underline{h}$ . As shown, one obvious effect of productivity,  $A$ , is that more productive firms invest more.

The reason why only productive firms raise financing early when investment is irreversible is exactly because this differential on-path investment has crucial off-path implications. Specifically, more productive firms that invest more on path have greater room to cut down on expenses to save funds *if* necessary. In contrast, less productive firms invest less on path, thereby endogenously constraining their ability to reduce investment off path, given that divestment is precluded. Thus, more productive firms have greater incentives to raise financing early in order to improve their outside options and reduce dilution when bargaining with financiers.

Figure 11 illustrates the mechanism. Each subplot contains the investment policy in Figure 10 for each productivity case and additionally depicts two objects. First, ‘fallback underinvestment’ is the difference between firms’ optimal investment depending on whether they have access to financing,  $i(h)$ , or do not,  $i_o(h)$ . Second, ‘financing rent’ is the amount that the firm must pay financiers if they choose to raise financing given funds  $h$  – which is thus suboptimal for  $h > \underline{h}$ .

The key observation is that due to investment irreversibility, productive firms can underinvest more if bargaining were to fail. Because on-path investment,  $i(h)$ , is higher

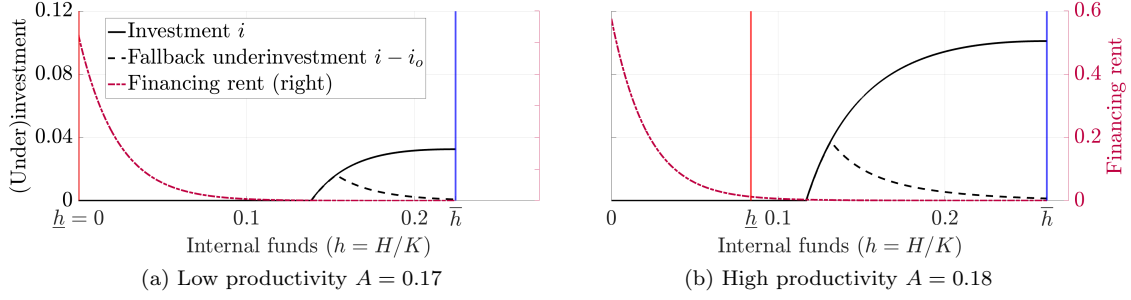


Figure 11: Investment and fallback underinvestment

Fallback underinvestment,  $i(h) - i_o(h)$ , is how much firms with funds  $h$  would reduce investment upon losing access to external financing, e.g., due to a failure of bargaining (off the equilibrium path); where the dashed line representing it is not visible, it coincides with the solid line representing investment,  $i(h)$ . Both y-axes are identical across plots. All values are on a per-capital basis. In both plots, financing rent is never exactly zero.

in Figure 11b, fallback underinvestment,  $i(h) - i_o(h)$ , is higher than in Figure 11a.<sup>38</sup> As such, productive firms raise financing early to improve their outside options with their endogenous ability to underinvest, whereas unproductive firms delay financing until funds are depleted and incur endogenously large financing costs.

In sum, investment irreversibility disproportionately increases financing costs for less productive firms, because more productive firms invest more and therefore can have better outside options if financing were to fail. Thus, this paper suggests that financing frictions are endogenously less severe for more productive firms in intangible-intensive sectors and/or during business downturns with low capital liquidity.

And the result directly arises from the model's core economic mechanism that because of financier bargaining power, firms with good financing alternatives – in the present case, the ability to reduce investment expenses – choose to raise financing early in order to improve their outside options and thereby make financing less dilutive.

## 7 Conclusion

I present a dynamic theory of firm financing based on financier bargaining power. Given that a failure of bargaining is consequential, financing dilutes firm value on the equilibrium path where bargaining always succeeds. In response, firms choose to raise financing in a lumpy fashion and maintain internal funds in order to bargain with financiers and get diluted infrequently. Moreover, firms may choose to raise financing early, before running out of internal funds, in order to bargain when their outside options are better and thereby make financing less dilutive. Thus, lumpy financing reduces the frequency of dilution and early financing reduces its size. In short, it is how firms use financial slack to improve their bargaining with financiers that determines

<sup>38</sup>With reversibility, unproductive firms may increase fallback underinvestment via  $i_o(h) < 0$ .

the dynamics of firm financing.

The paper provides a new theory of economic inaction due to the optimality of infrequent bargaining, as an alternative to the canonical paradigm based on fixed transaction costs dating to [Allais \(1947\)](#), [Baumol \(1952\)](#) and [Tobin \(1956\)](#). Furthermore, the model endogenizes costs of action such that they not only determine optimal dynamic strategy, but also are determined by it. This feature leads to novel implications as explored in the paper in the context of firm financing.

Looking forward, this paper suggests further research in three broad directions. First, it generates a cross-sectional prediction that financial slack is non-monotonic in how specialized firms’ financing relationships are – such as startups tied to stage-specific venture capital or borrowers tied to a single lead arranger in syndicated lending. This can be potentially tested using empirical proxies such as lender concentration, relationship tenure, or, simply, firms’ credit ratings. Second, the model’s restriction to equity financing raises in turn a theoretical question of optimal dynamic security design in the presence of financier bargaining power. Third, the implication from investment irreversibility suggests that financier bargaining power may contribute to widening cross-sectional dispersion in firm growth in intangible-intensive sectors or during business downturns.

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## Appendix A Microfoundations

This appendix provides two microfoundations for the core premise of the main model: Although firms can bargain again anytime after a success of bargaining, a failure of bargaining disables them from bargaining again for some period.

In Appendix A.1, a failure of bargaining precipitates an adverse selection problem that induces firms to ‘wait for news’ à la [Daley and Green \(2012\)](#) to bargain with other financiers. Appendix A.2 shows how reputation concerns may allow financiers to credibly commit to dissolving the relationship with a firm upon a failure of bargaining.<sup>39</sup>

For the sake of brevity, here I focus on delivering economic intuitions rather than conducting formal analysis. The key claim is that a failure of bargaining is prone to creating conditions that intensify underlying frictions – such as information or search. This naturally gives rise to asymmetric accessibility of financing depending on the outcome of bargaining, which is not manifested in equilibrium because bargaining always succeeds in it. Therefore, this model is potentially relevant in contexts in which those other frictions are expected to be present but difficult to observe explicitly.

### A.1 Model 1: due diligence and dynamic lemons’ market

**Setup.** Firms are ‘startups’ as specified in Section 4.1. A fraction  $\varepsilon \in (0, 1)$  of startups are ‘lemons,’ while the rest are ‘normal’; the type is private knowledge to the firm’s insiders. A lemon is different from normal startups in that its terminal payoff is  $\tilde{\Pi} < \frac{\kappa}{\lambda}$ , instead of  $\Pi > \frac{\kappa}{\lambda}$ , such that net present value is negative. Lemons can try to ‘cheat’ financiers by concealing their type until bargaining succeeds, and can also terminate the business anytime. The present exposition will involve letting  $\varepsilon \rightarrow 0^+$ .

**Public signals.** Let  $m \in \{-1, 1\}$  denote the startup’s type, with  $m = 1$  for lemons. Each firm generates public noisy signals  $ds_t = m dt + \sigma_s dZ_t$  where  $\sigma_s > 0$ ,  $Z_t$  is a standard Brownian motion. Thus, a positive flow signal  $ds_t > 0$  indicates a higher likelihood of being a lemon. But when the market’s ex-ante belief  $\zeta_0 \equiv \mathbb{P}(m = 1) = \varepsilon$  converges to zero, public signals alone have little effects on belief updating. To see this, note that

$$\mathbb{P}(m = 1 \mid \{s_{t'}\}_{t' \leq t}) = \frac{\zeta_0 \phi\left(\frac{s_t - t}{\sigma_s \sqrt{t}}\right)}{\zeta_0 \phi\left(\frac{s_t - t}{\sigma_s \sqrt{t}}\right) + (1 - \zeta_0) \phi\left(\frac{s_t + t}{\sigma_s \sqrt{t}}\right)}, \quad (\text{A.1})$$

where  $\phi : \mathbb{R} \rightarrow \mathbb{R}_{++}$  is a standard normal density. When  $\zeta_0 \rightarrow 0^+$ , the above belief based on public signals converges to zero for all  $t$  and regardless of the actual type.<sup>40</sup>

<sup>39</sup>This model is similar to [Hendershott, Li, Livdan and Schürhoff \(2020\)](#). Relative to their model, this model also endogenizes firms’ cost of breaking a relationship in terms of time delay.

<sup>40</sup>The convergence is not uniform in time conditional on  $m = 1$ , since for any  $\varepsilon > 0$ , public

**Due diligence.** When bargaining with a firm, financiers conduct due diligence at an infinitesimal cost to obtain an additional private signal about the startup's type. The signal features both false negatives and false positives: A lemon is falsely signaled as normal ('false negative') with probability  $\Delta \in (0, 1/2)$ , and a normal firm is falsely signaled as a lemon ('false positive') with probability  $\delta\varepsilon \in (0, 1/2)$ . Given a prior market belief  $\zeta_t^p \equiv \lim_{t' \rightarrow t^-} \zeta_{t'}$ , a negative (positive) result at  $t$  induces the financiers' private belief  $\hat{\zeta}_t = \hat{\zeta}_t^-$  ( $\hat{\zeta}_t = \hat{\zeta}_t^+$ ), where

$$\hat{\zeta}_t^- \equiv \frac{\zeta_t^p \Delta}{\zeta_t^p \Delta + (1 - \zeta_t^p)(1 - \delta\varepsilon)} < \zeta_t^p < \frac{\zeta_t^p(1 - \Delta)}{\zeta_t^p(1 - \Delta) + (1 - \zeta_t^p)\delta\varepsilon} \equiv \hat{\zeta}_t^+. \quad (\text{A.2})$$

The assumption that the probability of false positives is proportional to  $\varepsilon$  represents that when there are few lemons out there, it should be correspondingly easy for normal firms to pass due diligence.<sup>41</sup> As  $\varepsilon \rightarrow 0^+$ , this implies that even when lemons have incentives to continue their businesses in the hope of 'cheating' financiers by obtaining a false negative with a non-vanishing probability  $\Delta$ , financiers as well as normal firms have little reason to worry about it ex-ante since there are vanishingly few lemons in the first place; nevertheless, conditional on a positive due diligence result, the financiers assign a non-vanishing probability that they are dealing with a lemon trying to 'cheat' them.<sup>42</sup> Therefore, even though adverse selection is negligible ex-ante, it is not negligible conditional on a positive due diligence result such that financiers plausibly find it optimal to let the bargaining fail upon obtaining one.

Crucially, I assume that, while both success and failure of bargaining are publicly observable, whether the failure occurred due to a positive due diligence result is not. Since in equilibrium bargaining fails if and only if financiers assign a sufficient probability to the firm being a lemon,  $\zeta_t = \hat{\zeta}_t$  in equilibrium; that is, the market interprets – correctly in equilibrium – a success of bargaining as fully revealing a negative due diligence result and its failure as fully revealing a positive one. Thus, a failure of bargaining – regardless of the cause – discretely increases market belief on the startup

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signals are fully revealing asymptotically:  $\lim_{t \rightarrow \infty} \tilde{\zeta}_t = \mathbb{1}(m = 1)$  almost surely, where  $\tilde{\zeta}_t \equiv \mathbb{P}(m = 1 \mid \{s_{t'}\}_{t' \leq t})$ . But note that

$$d\tilde{\zeta}_t = \frac{2}{\sigma_s^2} \tilde{\zeta}_t(1 - \tilde{\zeta}_t) \left( (m - (2\tilde{\zeta}_t - 1)) dt + \sigma_s dZ_t \right)$$

by Itô's lemma. Thus, given  $\tilde{\zeta}_0 \rightarrow 0^+$  but conditional on  $m = 1$ , a stopping time for the drift and the diffusion in  $d\tilde{\zeta}_t$  to become sufficiently greater than zero almost surely diverges to infinity. Since a startup operates at most until the stopping time  $\tau_\lambda$  at a Poisson rate  $\lambda > 0$  which is almost surely finite and independent of  $\{\tilde{\zeta}_t\}_t$  conditional on  $m$ , a *curtailed* process  $\mathbb{1}(t \leq \tau_\lambda)\tilde{\zeta}_t$  almost surely converges to zero uniformly in time as  $\tilde{\zeta}_0 \rightarrow 0^+$ , even when conditioned on  $m = 1$ .

<sup>41</sup>Otherwise, due diligence would be nearly uninformative when  $\varepsilon \rightarrow 0^+$  since even conditional on a positive result, a firm would still be highly unlikely to be a lemon.

<sup>42</sup>Supposing that  $\zeta_t^p \approx \beta\varepsilon$  for some  $\beta > 0$ , (A.2) implies that  $\hat{\zeta}_t^+ \rightarrow \frac{\beta(1-\Delta)}{\beta(1-\Delta)+\delta} > 0$  as  $\varepsilon \rightarrow 0^+$ .

being a lemon  $\zeta_t = \hat{\zeta}_t^+ > \zeta_t^p$  by a non-vanishing size and away from zero.

**Dynamic signaling.** The above reasoning thus implies that a failure of bargaining makes subsequent public signals  $\{s_{t'} - s_t\}_{t' > t}$  non-vanishingly informative about the startup's hidden type. Thus, firms that have failed in bargaining then ‘wait for news’ à la [Daley and Green \(2012\)](#): When market belief goes up to an upper threshold due to unfavorable public signals  $ds_t > 0$ , lemons mix between termination and continuation while normal firms continue to wait, so that market belief does not go up further; When it goes down to a lower threshold due to favorable public signals  $ds_t < 0$ , all firms choose to receive due diligence again in order to bargain with financiers.<sup>43</sup> Lemons have incentives to wait for it due to the chance  $\Delta$  of ‘cheating’ financiers.

**Implications to bargaining.** Because the cause of a bargaining failure itself is not publicly observable, the market belief and the financiers’ private belief diverge on the off-equilibrium path on which the bargaining has failed despite a negative due diligence result; that is,  $\zeta_t = \hat{\zeta}_t^+ > \hat{\zeta}_t^- = \hat{\zeta}_t$ . Therefore, a normal firm that has obtained a negative due diligence result has its outside options vis-à-vis the financiers characterized by being forced by the market’s wrong belief into the above dynamic signaling stage. If public signals are more precise (that is,  $\sigma_s$  is lower), then the average waiting period is endogenously shorter. Thus, firms with superior public signals about their quality can bargain again more quickly following a failure of bargaining.<sup>44</sup>

## A.2 Model 2: intermediary reputation

**Setup.** There are a continuum of deep-pocketed and infinitely lived financiers. Each financier is risk-neutral and has a time preference represented by a discount rate  $\rho > 0$ . There are also a continuum of firms, each as described in Section 4.

There are  $N$  infinitely lived intermediaries,  $N \in \mathbb{N}$ . Each firm has a long-lasting relationship with one and only one intermediary, whereas each intermediary maintains relationships with a continuum of firms. When a firm needs financing, it must bargain

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<sup>43</sup>Relative to [Daley and Green \(2012\)](#), the direction of favorable signals is flipped so that pooled sales occur at lower threshold.

<sup>44</sup>This model, even asymptotically with  $\varepsilon \rightarrow 0^+$ , does not map cleanly into the details of Section 4 (the way that Appendix A.2 does with  $N \rightarrow \infty$ ) in two technical regards, although neither undermines this paper’s core insights. First, the duration of exclusion is not exponentially distributed. But most of the formal results in Section 5 – specifically, Lemma 1 and Proposition 1 as well as Proposition 2 with suitable modifications – hold under the core premise of time delay until regaining the ability to bargain. Second, the new value function upon re-inclusion must be lower than the original equilibrium value function before a bargaining failure. This is because the market belief before a bargaining failure is close to zero due to  $\varepsilon \rightarrow 0^+$  whereas the dynamic signaling may end before the market belief returns close to zero. But since the belief eventually converges to zero conditional on  $m = -1$  and the startup not succeeding, the theoretical framework in Section 5 still largely applies.

with the intermediary – direct access to financiers is disallowed.<sup>45</sup> The firm’s bargaining power is  $\theta \in (0, 1)$ . The relationship enables the firm to bargain with the intermediary at any time and however often. As part of the bargaining, the intermediary syndicates financing at zero cost from financiers who are perfectly competitive, and receives funds upfront as compensation for the syndication service.<sup>46</sup> Financiers at  $t$  cannot commit to providing funds at  $t' > t$ , and intermediaries at  $t$  cannot commit to syndicating financing at  $t' > t$ .<sup>47</sup> Given that financiers are perfectly competitive with respect to intermediaries, the assumption in Section 4 that financiers become penniless upon providing financing can be relaxed. Both the firm and the intermediary have the choice of unilaterally walking away from the bargaining, which leads to its failure.

Each intermediary is of a hidden type  $j \in \{R, T\}$ , with a common prior  $\mathbb{P}(j = T) = \nu \in (0, 1)$  i.i.d. across intermediaries. The type matters if bargaining between a firm and an intermediary were to fail. If the intermediary is ‘rational’  $j = R$ , it chooses between retaining the relationship with the firm or dissolving it. A tough intermediary  $j = T$ , on the other hand, must dissolve the relationship upon a bargaining failure.

If the relationship dissolves, the stray firm searches for another relationship. Each intermediary – including the one that the firm has left – gets matched to a stray firm at a Poisson rate  $\hat{\gamma} \geq 0$ . If a match occurs (at a Poisson rate  $N\hat{\gamma}$ ), the firm enters into the new – or restored – relationship through which it can bargain anytime for financing with the newly matched – or previous – intermediary. I assume that search frictions for stray firms do not vanish in the limit, that is,

$$\lim_{N \rightarrow \infty} N\hat{\gamma} = \gamma < \infty,$$

which implies that  $\lim_{N \rightarrow \infty} \hat{\gamma} = 0$ .

**Intermediary reputation.** I consider a perfect Bayesian equilibrium in which success or failure of any bargaining, along with counterparty identities, are publicly observable. Suppose that an intermediary fails bargaining with a firm at  $t$ . If belief at this node  $t$  induces tough intermediaries to occasionally let bargaining fail in order to signal their type, rational intermediaries can mimic the strategy because each relationship that a rational type chooses to dissolve following a bargaining failure is massless in its

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<sup>45</sup>This modified setup allows for a relationship to be entertained without altering the solution to the bargaining problem in Section 4.2. Without this modification, a relationship with a financier would mean that the financier is both a provider of funds to the firm and also one of the firm’s insiders, such that the financier’s ownership stake becomes another state variable; thus, the bargaining problem would need modifying along with the solution.

<sup>46</sup>The stark asymmetry in bargaining power between firms and financiers vis-à-vis intermediaries can be thought of as reflecting unique business characteristics of each firm as opposed to perfect substitutability of funding across financiers.

<sup>47</sup>Alternatively, financiers can commit to providing funds to a firm up to some amount, and any committed funds yield a reduced return  $r < \rho$  before drawdown.

total payoff, as an intermediary deals with a continuum of firms. Thus, a failure of bargaining is assigned zero probability from all preceding nodes. I assume no belief updating at this node, which trivially satisfies standard belief refinement criteria.

Now suppose that after a bargaining failure at  $t$ , the same intermediary bargains with the same firm at  $t + s$ . For simplicity of illustration, suppose that all firms are startups as specified in Section 4.1, such that a stray firm that survives (and does not succeed) until being matched raises financing immediately upon the match.<sup>48</sup> Letting  $\sigma_R \in [0, 1]$  be the rational intermediary's mixing probability for retaining the relationship upon a bargaining failure and  $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$  a density over the time interval until its next bargaining with the firm, the updated market belief at  $t + s$  is

$$\mathbb{P}_{t+s}(j = T) = \frac{\nu \hat{\gamma} \exp(-N \hat{\gamma} s)}{(\nu + (1 - \nu)(1 - \sigma_R)) \hat{\gamma} \exp(-N \hat{\gamma} s) + (1 - \nu) \sigma_R f(s)}.$$

As  $N \rightarrow \infty$ , the above does not converge to zero only if  $\sigma_R f(s)$  converges to zero. Otherwise, the odds that a stray firm is rematched to the same intermediary are negligible given a large  $N$ , so the event strongly signals prior retention – and hence, rationality.

Therefore, rational intermediaries have incentives to delay bargaining with the same firm after a bargaining failure in order to avoid damaging their reputation for being tough and worsening their bargaining position.<sup>49</sup> In the limit as  $N \rightarrow \infty$ , rational types can credibly commit to dissolving the relationship upon a failure of bargaining.

**Equivalence of bargaining.** Despite the modifications in the details of bargaining, the setup is exactly equivalent to the main model. Suppose a firm with funds  $h$  has chosen to bargain with the intermediary. By the previous discussion, the intermediary's reservation value in connection to each relationship is (asymptotically) zero and independent of its type. Letting  $z$  be the intermediary's upfront fee, bargaining solves

$$\begin{aligned} \max_{z \geq 0, \bar{h} \geq h} & \left( xV(\bar{h}) - V_o(h) \right)^\theta (z - 0)^{1-\theta} \\ \text{s.t.} & (1 - x)V(\bar{h}) = (\bar{h} - h) + z; \end{aligned}$$

the constraint ensures fair compensation to financiers for providing funds and also covering the fee  $z$  charged by the intermediary. Substituting it, the intermediary extracts  $z - 0 = (1 - x)V(\bar{h}) - (\bar{h} - h)$  in rents. Thus, the solution coincides with Section 4.2.

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<sup>48</sup>This does not hold in general because funds may have drifted above the financing threshold. While the expression then becomes complicated, the reasoning itself is unaffected.

<sup>49</sup>This holds even for an arbitrarily small  $\nu > 0$  because of the stark assumption that an intermediary maintains a continuum of massless relationships. If an intermediary only maintains finitely many atomistic relationships, the statement should fail for a sufficiently small  $\nu > 0$ .

## Appendix B Omitted Proofs

### B.1 Lemma 1 (Existence and uniqueness).

*Proof.* Let  $\mathcal{W}$  be the space of continuous nonnegative bounded functions on  $\mathbf{H} \equiv [0, H]$  with an arbitrary  $H \gg 0$ , and complete it with  $L^\infty$  supremum metric. Keep cash flow as in Section 4.1, and also the assumptions on dividends and business failure upon funding depletion without prompt financing.<sup>50</sup> Define  $T, T_o : \mathcal{W} \rightarrow \mathcal{W}$  as: for  $W \in \mathcal{W}$ ,

1.  $h \mapsto T_o W(h)$  is the value function of ‘excluded’ insiders lacking the ability to raise financing (so that the boundary condition is  $T_o W(0) = 0$ ), who, upon ‘re-inclusion’ that occurs at the stopping time  $\tau_\gamma$  given by a Poisson rate  $\gamma \in [0, \infty)$ , receive a terminal payoff of  $W(h_{\tau_\gamma})$ ; and
2.  $h \mapsto TW(h)$  is the value function of non-excluded insiders who can, at any  $t \geq 0$ , either (i) forgo financing, or (ii) raise financing through bargaining with financiers where insiders’ outside option from bargaining is given by  $W(h_t)$ :
  - if  $W(h_t)$  is sufficiently high, ‘bargaining’ involves zero financing and zero rents since (iii) insiders *take* the outside option such that  $TW(h_t) = W(h_t)$ .

By the Theorem of the Maximum, the above transformations are well-defined as self-maps given the continuity restriction in  $\mathcal{W}$ .

An equilibrium value function  $V$  is a fixed point of the *concatenated* map  $\hat{T} \equiv T \circ T_o$ , and vice versa; if  $V$  exists, then  $V_o \equiv T_o V$ . If  $\hat{T}$  is a contraction on  $\mathcal{W}$ , by contraction mapping theorem there exists a unique fixed point  $V \in \mathcal{W}$  such that  $V = \hat{T}V$ .

The claim on  $\hat{T}$  follows from Blackwell’s Lemma. First, obviously  $\hat{T}$  is nondecreasing:  $W' \geq W$  on  $\mathbf{H}$  implies  $\hat{T}W' \geq \hat{T}W$  on  $\mathbf{H}$ . Second, let us show that there exists some  $\zeta \in [0, 1)$  such that for any  $W \in \mathcal{W}$  and  $w \in \mathbb{R}_{++}$ ,  $\hat{T}(W + w) \leq \hat{T}W + \zeta w$ .

Recall that  $\tau_\gamma$  occurs at a Poisson rate  $\gamma \in [0, \infty)$ . Then,

$$T_o(W + w) \leq T_o W + \zeta_\gamma^\rho w, \quad (\text{B.1})$$

where  $\zeta_\gamma^\rho \equiv \mathbb{E}[e^{-\rho\tau_\gamma}] = \frac{\gamma}{\gamma + \rho} < 1$ .<sup>51</sup> As an aside,  $T_o$  is a contraction on  $\mathcal{W}$ , and its unique fixed point is invariant to  $\gamma \in [0, \infty)$ . Next, for any  $w \in \mathbb{R}_{++}$ ,

$$T(W + w) \leq TW + w. \quad (\text{B.2})$$

<sup>50</sup>Also enforce that whenever  $h_t > H$ , dividend payout of at least  $h_t - H$  is mandatory.

<sup>51</sup>The Poisson structure is not essential; one can, for example, assume that re-inclusion deterministically occurs at  $\tau_\gamma \equiv 1/\gamma > 0$ , in which case  $T_o W(h)$  is excluded insiders’ value at  $t = 0$  with  $h_0 = h$  given the otherwise identical environment. As long as re-inclusion takes some time with nonzero probability  $\mathbb{P}(\tau_\gamma > 0) > 0$ , the bonus dividend  $w \in \mathbb{R}_{++}$  that is received only upon re-inclusion is, from  $t = 0$ , discounted in expectation (at least) by a factor of  $\mathbb{E}[e^{-\rho\tau_\gamma}] < 1$ .

This is because marginal benefit from uniform global improvement in reservation value function does not exceed unity. The upper bound is attained wherever on  $h_t \in \mathbf{H}$  (iii) the outside option  $W(h_t)$  is so good that insiders take it,  $TW(h_t) = W(h_t)$ . Otherwise, marginal benefit is  $1 - \theta$  wherever on  $\mathbf{H}$  insiders optimally (ii) raise positive funds through bargaining, and zero wherever on  $\mathbf{H}$  insiders optimally (i) forgo financing.<sup>52</sup>

$T$  is not strictly a contraction, but it is not ‘expansionary’: its contraction coefficient is unity. Therefore,  $\hat{T}$  inherits the coefficient  $\zeta_\gamma^\rho \in [0, 1)$  from  $T_o$ : for any  $w \in \mathbb{R}_{++}$ ,

$$\hat{T}(W + w) = T(T_o(W + w)) \leq T(T_o W + \zeta_\gamma^\rho w) \leq T(T_o W) + \zeta_\gamma^\rho w = \hat{T}W + \zeta_\gamma^\rho w.$$

The first inequality is from Inequality (B.1) and monotonicity of  $T$ , and the second from Inequality (B.2). By Blackwell’s Lemma,  $\hat{T}$  is a contraction on  $\mathcal{W}$ , as desired.<sup>53</sup>  $\square$

## B.2 Lemma 2 (Monotone financing strategy)

*Proof.* (Continued from Section 5.2) Rewrite Inequality (10) as: for  $h \in B \setminus \{0\}$ ,  $G(h) \equiv (1 - \theta)\gamma(V(h) - V_o(h)) - \theta\varphi(\bar{h} - h) \geq 0$ . Define  $\tilde{G}$  as follows: for  $h \in [0, \bar{h}]$ ,

$$\tilde{G}(h) \equiv (1 - \theta)\gamma(F(h) - V_o(h)) - \theta\varphi(\bar{h} - h).$$

Obviously,  $\tilde{G} \leq G$ , and  $\tilde{G}(h) = G(h)$  if and only if  $h \in B$ . Note that

$$\tilde{G}'(h) = \theta\left(\varphi - (1 - \theta)\gamma(V_o'(h) - 1)\right),$$

which implies that (i)  $\tilde{G}'' \geq 0$  given weak concavity of  $V_o$  from standard dynamic programming arguments, and (ii)  $\tilde{G}'(h) < 0$  if and only if  $F'(h) > 1 + \frac{\varphi}{\gamma}$ . Also,

$$(1 - \theta)(F(h) - V_o(h)) = \theta(V(\bar{h}) - F(h) - (\bar{h} - h)), \quad (\text{B.3})$$

which is an identity that reflects standard properties of bargaining.

I establish an intermediate claim: if  $h \in B \setminus \{0\}$ , then  $\rho F(\tilde{h}) - r\tilde{h}F'(\tilde{h}) > \mathcal{H}(F)(\tilde{h})$  for any  $\tilde{h} < h$ . Since  $F(h) = V(h)$  and  $\tilde{G}(h) = G(h) \geq 0$  by  $h \in B \setminus \{0\}$ , it holds that  $\theta\gamma(V(\bar{h}) - V(h) - (\bar{h} - h)) \geq \theta\varphi(\bar{h} - h)$  by Equation (B.3), and thus,

$$\frac{V(\bar{h}) - V(h)}{\bar{h} - h} = \frac{1}{\bar{h} - h} \int_h^{\bar{h}} V'(y) dy \geq 1 + \frac{\varphi}{\gamma}.$$

Since  $V$  is weakly concave (by standard arguments) and  $V'(\bar{h}) = 1$ , it must be that

<sup>52</sup>These are pointwise effects of a rise in  $W(h)$  on  $TW(h)$ . When  $W$  globally shifts up uniformly, marginal effect on any  $TW(h)$  is a weighted average of 1,  $1 - \theta$  and 0, and hence unity at most.

<sup>53</sup>Supplemental Appendix SA.2 graphically illustrates the proof.

$V'(h) > 1 + \frac{\varphi}{\gamma}$ . Note that  $F'(h) = V'(h)$  for any  $h \in B \setminus \{0\}$ , either trivially if  $h$  is in the interior of the set or by smooth pasting otherwise, such that  $F'(h) > 1 + \frac{\varphi}{\gamma}$  and so  $\tilde{G}'(h) < 0$ . Since  $\tilde{G}'' \geq 0$  globally and  $\tilde{G}(h) \geq 0$  for  $h \in B \setminus \{0\}$ , it follows that  $\tilde{G} > 0$  on  $[0, h)$ . Thus,  $G \geq \tilde{G} > 0$  on  $[0, h)$ , since  $V \geq F$  globally. The intermediate claim then holds since  $G > 0$  is equivalent (by construction) to Inequality (10) being strict, which is equivalent to Inequality (7) being strict.

Now, suppose by way of contradiction that there exists some  $\hat{h} \in B \setminus \{0\}$  such that for any sufficiently small  $\varepsilon > 0$ ,  $\hat{h} - \varepsilon \notin B$ . Take  $h' < \hat{h}$  such that  $h' \in B$  but  $(h', \hat{h})$  is disjoint with  $B$ ; it exists since  $0 \in B$ . Define  $D \equiv V - F$ ; it satisfies  $D(h') = D(\hat{h}) = 0$  by value matching but  $D > 0$  on  $(h', \hat{h})$ . Let  $h^*$  be the maximizer of  $D$  on  $(h', \hat{h})$ , such that  $D'(h^*) = 0$  and  $D''(h^*) \leq 0$ . But then,

$$\begin{aligned} \rho V(h^*) - rh^*V'(h^*) &> \rho F(h^*) - rh^*F'(h^*) > \mathcal{H}(F)(h^*) \\ &= \mu F'(h^*) + \frac{1}{2}\sigma^2 F''(h^*) + \lambda(\Pi + h^* - F(h^*)) \\ &\geq \mu V'(h^*) + \frac{1}{2}\sigma^2 V''(h^*) + \lambda(\Pi + h^* - V(h^*)) = \mathcal{H}(V)(h^*) \end{aligned}$$

since  $V(h^*) > F(h^*)$ ,  $V'(h^*) = F'(h^*)$ , and  $V''(h^*) \leq F''(h^*)$ . This contradicts the HJB equation for  $V$  which should hold at  $h = h^*$ . This completes the proof; for subsequent reference, the present reasoning implies that if  $G(0) \leq 0$ , then  $\underline{h} = 0$ .  $\square$

### B.3 Proposition 2 (Monotonicity of early financing)

*Proof.* Since  $V(0) = F(0)$ ,

$$G(0) = \theta \left( (1 - \theta)\gamma(V(\bar{h}) - \bar{h}) - \varphi\bar{h} \right). \quad (\text{B.4})$$

Since Part 1 of Appendix B.5 shows that  $V(\bar{h}) - \bar{h} > 0$  is nondecreasing in  $\gamma$  while  $\bar{h}$  nonincreasing in  $\gamma$ , it follows that  $G(0)$  is negative when  $\gamma = 0$ , strictly increasing in  $\gamma$ , and strictly positive for sufficiently high  $\gamma$ .<sup>54</sup> Also, note that it is impossible that  $F'(0) > V'(0)$ ; if it holds, then the identity  $F(0) = V(0)$  – since  $0 \in B$  – implies that  $F > V$  on a neighborhood above  $h = 0$ , a contradiction.

Writing the objects as functions of  $\gamma \in [0, \infty)$ , it holds that  $V'(0; \gamma)$ , taken as a derivative of  $V$  with respect to  $h$  at  $h = 0$ , is constant in  $\gamma$  as long as  $\underline{h}(\gamma) = 0$ , because  $V_0(0; \gamma)$  is identically zero and thus equilibrium value function  $V$  is unaffected by  $\gamma$  when  $\underline{h}(\gamma) = 0$ . In addition,  $F'(0; \gamma) = \theta + (1 - \theta)V'_o(0; \gamma)$  is strictly increasing in  $\gamma$  and goes to infinity as  $\gamma \rightarrow \infty$ , because for any  $h > 0$ ,  $V_o(h; \gamma) > 0$  is strictly increasing in  $\gamma$  even as  $V_o(0; \gamma) = 0$  by the boundary condition. Therefore, the following object is

<sup>54</sup>Only *strict* monotonicity of  $V(\bar{h}) - \bar{h}$  for  $\gamma \geq \underline{\gamma}$  from Part 1 depends on Proposition 2.

well-defined, positive, and finite:

$$\underline{\gamma} \equiv \inf \{ \gamma \geq 0 \mid G(0; \gamma) > 0 \text{ and } F'(0; \gamma) = V'(0; \gamma) \} \in (0, \infty).$$

When  $\gamma < \underline{\gamma}$ , either  $G(0; \gamma) \leq 0$  or  $F'(0; \gamma) < V'(0; \gamma)$  such that  $\underline{h}(\gamma) = 0$ . When  $\gamma > \underline{\gamma}$ , then  $\underline{h}(\gamma) > 0$  since otherwise,  $F'(0; \gamma) > V'(0; \gamma)$ . Lastly, suppose that  $\gamma = \underline{\gamma}$ . If  $G(0; \underline{\gamma}) = 0$ , then obviously  $\underline{h}(\underline{\gamma}) = 0$ . Suppose that  $G(0; \underline{\gamma}) > 0$ . For  $\gamma < \underline{\gamma}$  sufficiently close,  $G(0; \gamma) > 0$  and  $\underline{h}(\gamma) = 0$ . Letting  $\gamma \rightarrow \underline{\gamma}^-$ , the invariant HJB equation must still hold for  $V(h; \underline{\gamma})$  at  $h = 0$ . Since  $F'(0; \underline{\gamma}) = V'(0; \underline{\gamma})$ , it follows from  $G(0; \underline{\gamma}) > 0$  that  $\frac{1}{2}\sigma^2 F''(0; \underline{\gamma}) < \frac{1}{2}\sigma^2 V''(0; \underline{\gamma})$ . If  $\sigma = 0$ , this is a contradiction, and so  $G(0; \underline{\gamma}) = 0$ . Otherwise,  $F''(0; \underline{\gamma}) < V''(0; \underline{\gamma})$  such that  $F(h; \underline{\gamma}) < V(h; \underline{\gamma})$  for small  $h > 0$ . In either case, therefore,  $\underline{h}(\underline{\gamma}) = 0$ .  $\square$

## B.4 Lemma 3 (Endogenous financing costs)

*Proof.* Financiers' rent when  $\underline{h} = 0$  is from (3). For (11), rearrange  $\rho V(\underline{h}) - r\underline{h}V'(\underline{h}) = \mathcal{H}(V)(\underline{h})$  (using the right second derivative when  $\sigma > 0$ ) following the steps in the proof of Lemma 2 given value matching  $V(\underline{h}) = F(\underline{h})$  and smooth pasting  $V'(\underline{h}) = F'(\underline{h})$ , apply Equation (B.3), and solve for  $V(\bar{h}) - F(\bar{h}) - \Delta h$  which is financiers' rents. The inequality follows from  $\rho F(\bar{h}) - r\bar{h}F'(\bar{h}) \geq \mathcal{H}(F)(\bar{h})$  since  $\bar{h} \in B \setminus \{0\}$ .  $\square$

## B.5 Proposition 3 (Comparative statics in $\theta$ and $\gamma$ )

*Proof.* Constancy of  $(\underline{h}, \bar{h})$  in  $\gamma \in [0, \underline{\gamma}]$  is because  $\gamma \leq \underline{\gamma}$  implies  $V_o(\underline{h}) = V_o(0) = 0$ .

**Part 1: Funding target  $\bar{h}$ .** Take  $\gamma_2 > \gamma_1 \geq \underline{\gamma}$  and consider the equilibrium with  $\gamma = \gamma_2$ . When insiders bargain with financiers, they choose  $\bar{h}_2$  to maximize  $V(\bar{h}_2; \gamma_2) - \bar{h}_2$ . Suppose that they agree, as a one-shot deviation, to choose  $\bar{h}_1$  instead and then mimic the optimal financing strategy under  $\gamma = \gamma_1$  (i.e., refinance at  $\underline{h}_1$ , pay out above  $\bar{h}_1$ ) until next financing. Denote the payoff function associated with this strategy as  $\tilde{V}$ . Note that  $\tilde{V}(\bar{h}_1; \gamma_2) > V(\bar{h}_1; \gamma_1)$ , as the reservation value at  $h_t = \underline{h}_1 > 0$  is strictly higher with  $\gamma = \gamma_2 > \gamma_1$ . Since  $\bar{h}_2$  without the one-shot deviation is optimal,  $V(\bar{h}_2; \gamma_2) - \bar{h}_2 \geq \tilde{V}(\bar{h}_1; \gamma_2) - \bar{h}_1 > V(\bar{h}_1; \gamma_1) - \bar{h}_1$ . Finally, since

$$V(\bar{h}; \gamma) - \bar{h} = \frac{1}{\rho + \lambda} (\mu + \lambda \Pi - \varphi \bar{h}), \quad (\text{B.5})$$

from evaluating Equation (8) given smooth pasting  $V'(\bar{h}) = 1$  and super contact  $\sigma > 0 \implies V''(\bar{h}) = 0$ , we have  $\bar{h}_2 < \bar{h}_1$ . Strict monotonicity in  $\theta$  is proven similarly.

**Part 2: Asymptotics.** Since  $\bar{h}$  decreases in  $\theta$ , it converges as  $\theta \rightarrow 1^-$ . Note that

$$V(\bar{h}) - V(\underline{h}) - \Delta h = \int_{\bar{h}-\Delta h}^{\bar{h}} V'(h) - 1 \, dh = (1 - \theta)(V(\bar{h}) - \Delta h - V_o(\underline{h})). \quad (\text{B.6})$$

Since  $V(\bar{h}) - \Delta h - V_o(\underline{h}) = V(\bar{h}) - \bar{h} - (V_o(\underline{h}) - \underline{h})$  is bounded above in  $\theta$  by the frictionless value  $\frac{\mu+\lambda\Pi}{\rho+\lambda} > 0$ , dividend optimality condition  $V'(\bar{h}) = 1$  and concavity of  $V$  require that  $\Delta h \rightarrow 0$  as  $\theta \rightarrow 1^-$ . Next, let  $\tilde{h} \equiv \lim_{\theta \rightarrow 1} \bar{h}$ , which exists by monotone convergence. Since  $V(\bar{h}) - \bar{h}$  is also bounded above,  $G(0)$  as defined in Equation (B.4) converges to  $-\varphi\tilde{h} \leq 0$  as  $\theta \rightarrow 1$ . Since  $G(0) \leq 0$  implies  $\underline{h} = 0$  by Appendix B.2 and  $\Delta h \rightarrow 0$  as  $\theta \rightarrow 1$ , it follows that  $\tilde{h} = \lim_{\theta \rightarrow 1} (\underline{h} + \Delta h) = 0$ .

Next,  $\bar{h}$  also converges as  $\gamma \rightarrow \infty$ , again by monotone convergence. Note that  $V(h) - V_o(h) \rightarrow 0$  as  $\gamma \rightarrow \infty$  pointwise for  $h > 0$ , even as  $V(0) - V_o(0) = V(0)$  does not converge to zero. Suppose by way of contradiction that  $\bar{h} \rightarrow \tilde{h} > 0$  as  $\gamma \rightarrow \infty$ . Then, either (i)  $\Delta h \rightarrow \tilde{\Delta h} > 0$ , or (ii)  $\underline{h} \rightarrow \tilde{\underline{h}} > 0$ .<sup>55</sup>

In case (i), financiers' rents converge to

$$V(\bar{h}) - V(\underline{h}) - \Delta h \rightarrow \int_{\tilde{h}-\tilde{\Delta h}}^{\tilde{h}} V'(h) - 1 \, dh,$$

which is strictly positive by the HJB equation with smooth pasting at  $\tilde{h} > \tilde{h} - \tilde{\Delta h}$ . Since financiers' rents also equal

$$\left(\frac{1-\theta}{\theta}\right) (V(\underline{h}) - V_o(\underline{h})),$$

the gap  $V(\underline{h}) - V_o(\underline{h})$  does not vanish, which requires that  $\underline{h} \rightarrow 0$ . But any positive limit for financing threshold  $\underline{h} \rightarrow \varepsilon$  for  $\varepsilon \in (0, \tilde{\Delta h})$ , as a deviation, vanishes the gap when  $\gamma \rightarrow \infty$ , and hence the rents that financiers can extract. Thus, the marginal benefit given  $\varepsilon \rightarrow 0^+$ , in terms of reducing financiers' rents, diverges to infinity as  $\gamma \rightarrow \infty$ .<sup>56</sup> In comparison, the marginal costs include additional flow carry cost  $\varphi \, dt$  and a higher frequency of financing, and both are bounded. Therefore, there is a strictly profitable deviation for sufficiently high  $\gamma$ , a contradiction.

In case (ii), note that  $V(h) - V_o(h)$  converges uniformly to zero on, say,  $[\tilde{h}/2, \tilde{h}]$ . Therefore, the increase in financing rents when  $\underline{h} \rightarrow \tilde{\underline{h}} - \varepsilon$  for  $\varepsilon \in (0, \tilde{h}/2)$  vanishes as  $\gamma \rightarrow \infty$ . But this marginally lower financing threshold when  $\varepsilon \rightarrow 0^+$  has non-vanishing marginal benefits of less flow carry cost  $\varphi \, dt$  and lower frequency of financing. As such, there is a strictly profitable deviation for sufficiently high  $\gamma$ , a contradiction.

<sup>55</sup>It is without loss to assume that the limits exist, since Bolzano-Weierstrass Theorem ensures the existence of convergence subsequences given  $\gamma_n \rightarrow \infty$  as  $n \rightarrow \infty$ .

<sup>56</sup>For clarity, the statement concerns the limit of the derivative  $\frac{d}{dh}(V(0) - V_o(0))$  as  $\gamma \rightarrow \infty$ .

**Part 3: Financing threshold  $\underline{h}$ .** For the first claim on  $\underline{\theta}$ , I first establish an intermediate claim:  $\frac{d\bar{h}}{d\theta} \rightarrow -\infty$  as  $\theta \rightarrow 1^-$ . For this, reformulate Equation (B.6) using (B.5) as

$$\int_{\underline{h}}^{\bar{h}} V'(h) - 1 \, dh = (1 - \theta) \left( \frac{\mu + \lambda\Pi}{\rho + \lambda} - \frac{\varphi}{\rho + \lambda} \Delta h + \frac{r + \lambda}{\rho + \lambda} \underline{h} - V_o(\underline{h}) \right)$$

Totally differentiating both sides with respect to  $\theta$  gives

$$\begin{aligned} \int_{\underline{h}}^{\bar{h}} \frac{d}{d\theta} (V'(h) - 1) \, dh - (V'(\underline{h}) - 1) \frac{d}{d\theta} \underline{h} &= - \left( V(\bar{h}) - \bar{h} - (V_o(\underline{h}) - \underline{h}) \right) \\ &\quad + (1 - \theta) \left( \frac{r + \lambda}{\rho + \lambda} \frac{d}{d\theta} \underline{h} - \frac{\varphi}{\rho + \lambda} \frac{d}{d\theta} \Delta h - \frac{d}{d\theta} V_o(\underline{h}) \right). \end{aligned}$$

Letting  $\theta \rightarrow 1^-$  trivially vanishes the integral term because  $\Delta h \rightarrow 0$  while  $V'(\bar{h}) = 1$ . Rearranging the potentially non-vanishing terms gives

$$\begin{aligned} &\left( V(\bar{h}) - \bar{h} - (V_o(\underline{h}) - \underline{h}) \right) + (1 - \theta) \frac{d}{d\theta} V_o(\underline{h}) \\ &= \left( (V'(\underline{h}) - 1) + (1 - \theta) \frac{r + \lambda}{\rho + \lambda} \right) \frac{d}{d\theta} \underline{h} - (1 - \theta) \frac{\varphi}{\rho + \lambda} \frac{d}{d\theta} \Delta h + o(1 - \theta). \end{aligned}$$

The left-hand side is positive and, especially, the first term does not vanish in the limit since  $\gamma < \infty$ . By Part 2,  $\lim_{\theta \rightarrow 1^-} \frac{d}{d\theta} \underline{h} \in [-\infty, 0]$ . Therefore, it must be that  $\lim_{\theta \rightarrow 1^-} \frac{d}{d\theta} \Delta h = -\infty$ , and thus,

$$\lim_{\theta \rightarrow 1^-} \frac{d}{d\theta} \bar{h} = \lim_{\theta \rightarrow 1^-} \frac{d}{d\theta} \underline{h} + \lim_{\theta \rightarrow 1^-} \frac{d}{d\theta} \Delta h = -\infty.$$

The main claim is straightforward. Note that by Part 1,  $G(0)$  from Equation (B.4) goes to zero as  $\theta \rightarrow 1^-$ . Since the argument only concerns the sign of  $G(0)$  for  $\theta$  near 1, divide  $G(0)$  by  $\theta$  and then substitute Equation (B.5). Differentiate the resulting expression with respect to  $\theta$  to obtain

$$\frac{d}{d\theta} \frac{G(0)}{\theta} = -\gamma \frac{\mu + \lambda\Pi}{\rho + \lambda} - \varphi \left( 1 + \frac{(1 - \theta)\gamma}{\rho + \lambda} \right) \frac{d}{d\theta} \bar{h}.$$

By the intermediate claim, therefore,  $\frac{d}{d\theta} \frac{G(0)}{\theta} \rightarrow \infty$  as  $\theta \rightarrow 1^-$ , such that  $G(0)$  is strictly negative for  $\theta < 1$  sufficiently close. The claim then follows from Appendix B.2.

Next, since (i)  $\underline{h}(\underline{\gamma}) = 0$ , (ii)  $\underline{h} > 0$  for  $\gamma > \underline{\gamma}$ , and yet (iii)  $\underline{h} \rightarrow 0$  as  $\gamma \rightarrow \infty$ , non-monotonicity in  $\gamma \in (\underline{\gamma}, \infty)$  is proven sufficiently if  $\underline{h}$  is continuous at  $\gamma = \underline{\gamma}$ . Suppose by way of contradiction that  $\lim_{\gamma \rightarrow \underline{\gamma}^+} \underline{h} = \tilde{\underline{h}} > 0$ .<sup>57</sup> As  $\gamma \rightarrow \underline{\gamma}^+$ ,  $V = F$  on  $[0, \tilde{\underline{h}}]$  and so  $V = F$  fails to satisfy its own HJB on the interval. But at exactly  $\gamma = \underline{\gamma}$ ,  $V$  must satisfy its own HJB on the interval, and hence  $F$  must also satisfy that same HJB.

<sup>57</sup>Again, Bolzano-Weierstrass Theorem ensures the existence of a convergent subsequence.

Since  $V(0) = F(0)$  always, it follows that  $V = F$  on  $[0, \tilde{h}]$  when  $\gamma = \underline{\gamma}$ . But then  $\underline{h}(\underline{\gamma}) = \tilde{h} > 0$ , contradicting Proposition 2.

Lastly, strict monotonicity of  $\underline{h} = \bar{h} - \Delta h > 0$  in  $\theta$  given  $\sigma = 0$  is immediate from decreasing  $\bar{h}$  in Part 1 and nondecreasing  $\Delta h$  in Part 4.

**Part 4: Financing amount  $\Delta h$ .** Suppose that  $\underline{h} > 0$ . Rearranging (11) which equals  $V(\bar{h}) - V(\underline{h}) - \Delta h$  gives

$$1 + \frac{\varphi}{\gamma} = \frac{V(\bar{h}) - V(\underline{h})}{\Delta h} = \frac{1}{\Delta h} \int_{\bar{h}-\Delta h}^{\bar{h}} V'(h) dh. \quad (\text{B.7})$$

That is, early financing specifies the *average* of the first derivative of  $V$  over  $[\bar{h} - \Delta h, \bar{h}]$ . Recall that dividend optimality stipulates that  $V'(\bar{h}) = 1$ . Therefore, given  $\bar{h} > 0$ , one can equivalently think of  $V$  as satisfying the HJB equation globally below  $\bar{h}$  and  $\Delta h$  as simply choosing when the first derivative on the interval meets the desired average  $1 + \frac{\varphi}{\gamma}$ .

Since the first derivative is concerned,  $V(\bar{h})$  is irrelevant. Given that  $\sigma = 0$  restricts to the startup case by Assumption 1, differentiate the HJB using the startup notation in Section 4.1:

$$V''(h) = -\frac{1}{\kappa - rh} [(\varphi + \lambda)V'(h) - \lambda].$$

If  $r = 0$ , the above equation that determines the evolution of  $V'$  below  $\bar{h}$  is independent of  $(\theta, \gamma)$ . Since the desired average is  $1 + \frac{\varphi}{\gamma}$ , the claims when  $r = 0$  are established.

Suppose  $r > 0$ . Because  $V'(h) > 1$  for  $h < \bar{h}$  and  $\kappa > r\bar{h}$  by Assumption 1, a higher  $\bar{h}$  means that  $V''(\bar{h} - \delta h)$  is pointwise lower. Therefore,

$$V'(\bar{h} - \delta h) = 1 - \int_{\bar{h}-\delta h}^{\bar{h}} V''(h) dh$$

is pointwise higher, and so a higher  $\bar{h}$ , ceteris paribus, reduces  $\Delta h$  in equilibrium. The claim then follows from Part 1.

**Part 5: Financing rent  $(1 - \theta)[N(\underline{h}) + E(\underline{h})]$ .** The claims when  $\underline{h} > 0$  follow from Part 4 and Lemma 3. The asymptotic convergence to zero with respect to  $\theta$  is because  $V(\bar{h}) - \bar{h}$  is bounded above by the frictionless net present value; that with respect to  $\gamma$  is due to Part 2 and Lemma 3. Lastly, suppose by way of contradiction that given  $1 > \theta' > \theta \geq \underline{\theta}$ , financing rent is higher for  $\theta'$ . Since  $\bar{h} = \Delta h$  is lower with  $\theta'$ , insiders with  $\theta'$  incur greater financing rents more often. Thus, their net value  $V(\bar{h}) - \bar{h}$  must be lower than given  $\theta$ . But this contradicts Equation (B.5).  $\square$

# Supplemental Appendix A Other Discussions

## SA.1 Robustness to within-period bargaining

Here, I discuss how  $\theta$  can accommodate various static bargaining environments. I illustrate it using the stylized setup in Section 3, but the implications extend to the main model. Let  $c_t$  continuation value and  $v_t^o$  the firm's reservation value on date  $t$ .

**Multiple financiers on each date.** Suppose that  $n \in \mathbb{N}$  financiers visit the firm on each date. I assume that the firm bargains with one financier at a time. If bargaining were to fail, the firm can bargain with another financier on the same date. Each financier commits to refusing to bargain with the firm conditional on his previous bargaining failure with it. Let  $v_t^m$  firm value from bargaining with the  $m^{\text{th}}$  financier on date  $t$ ,  $m \leq n$ . Then,

$$\begin{aligned} v_t^n &= c_t - (1 - \theta)(c_t - v_t^o) = \theta c_t + (1 - \theta)v_t^o, \\ v_t^{n-1} &= \theta c_t + (1 - \theta)v_t^n = (1 - (1 - \theta)^2)c_t + (1 - \theta)^2 v_t^o, \\ v_t^{n-2} &= \theta c_t + (1 - \theta)v_t^{n-1} = (1 - (1 - \theta)^3)c_t + (1 - \theta)^3 v_t^o \\ &\dots \\ v_t^1 &= \theta c_t + (1 - \theta)v_t^2 = (1 - (1 - \theta)^n)c_t + (1 - \theta)^n v_t^o. \end{aligned}$$

Therefore, it is equivalent to  $\theta_n \equiv (1 - (1 - \theta)^n) \in [\theta, 1]$ ; expectedly,  $\theta_n \rightarrow 1$  as  $n \rightarrow \infty$ .

**Nesting Bertrand competition.** Let  $n = 2$  above, and suppose that each financier can fully commit to refusing to bargain with the firm consecutively, and otherwise can commit to refusing to bargain with an i.i.d. probability  $\chi \in [0, 1]$  in case the firm has failed in bargaining beforehand. Once the firm is denied bargaining by either financier, no more bargaining is allowed on that date. Because bargaining does not fail on the path of play in any subgame, the value of each financier's outside options when he is bargaining with the firm is zero. Letting  $v_t^B$  firm's value from bargaining,

$$\begin{aligned} v_t^B &= \theta c_t + (1 - \theta)(\chi v_t^o + (1 - \chi)v_t^B) \\ \implies v_t^B &= \frac{\theta}{1 - (1 - \theta)(1 - \chi)} c_t + \left(1 - \frac{\theta}{1 - (1 - \theta)(1 - \chi)}\right) v_t^o. \end{aligned}$$

Therefore, this is equivalent to  $\theta_\chi \equiv \frac{\theta}{1 - (1 - \theta)(1 - \chi)} \in [\theta, 1]$ . Note that  $\theta_\chi = 1$  if and only if  $\chi = 0$ , that is, financiers cannot commit at all except upon immediate bargaining failure. This essentially describes Bertrand competition. In sum, when the firm must bargain with each financier bilaterally, perfect competition à la Bertrand between two financiers obtains if and only if financiers lack commitment technology altogether  $\chi = 0$ .

## SA.2 Contraction from two-step recursion: illustration

Take the space  $\mathcal{W}$  and the three self-maps  $\hat{T}, T, T_o : \mathcal{W} \rightarrow \mathcal{W}$  as defined in Appendix B.1. Here, I illustrate the proof that  $\hat{T}$  is a contraction, such that contraction mapping theorem can be applied. Let  $V^* \in \mathcal{W}$  be first-best value function, that is, without financing frictions and  $\tilde{V}_o \in \mathcal{W}$  value function under permanent autarky, that is,  $\gamma = 0$ .

By way of analogy, visualize the infinite-dimensional  $\mathcal{W}$  as a simple  $\mathbb{R}_+$ . A continuous, nondecreasing, and piece-wise differentiable self-map on  $\mathbb{R}_+$  is a contraction if and only if its derivative, wherever defined, is below  $\zeta$  for some  $\zeta \in [0, 1)$ .



Figure SA.1: Two-step recursion as a contraction map

Figure SA.1a describes  $T$  and  $T_o$ . First consider  $T$ . If insiders' outside option is preferred to the first-best  $W \geq V^*$ , they are going to simply take it – the embedded choice (iii) in the definition of  $T$  (see Appendix B.1). Therefore, its slope above  $V^*$  is unity. If  $W < V^*$ , then insiders do not take the option, but this ‘outside’ option is still beneficial to the extent that insiders bargain with financiers. Since better outside option reduces financiers' rents by a fraction  $1 - \theta$  subject to bargaining taking place, the slope of  $T$  below  $V^*$  is (ii)  $1 - \theta$  given bargaining and (i) zero given no bargaining. As shown,  $T$  is not a contraction: any point above  $V^*$  is one of its fixed points.

Next, consider  $T_o$ : insiders are on autarky until the stopping time  $\tau_\gamma$  when the business gives  $W$  as a terminal payoff. A higher terminal payoff benefits insiders but only up to a discounting due to time delay  $\mathbb{E}[e^{-\rho\tau_\gamma}] \in [0, 1)$ , which thus bounds the slope of  $T_o$ . Therefore,  $T_o$  is a contraction and has a unique fixed point.

But  $T_o$ , in itself, is a particularly *uninteresting* contraction. Regardless of  $\gamma \in [0, \infty)$ , its fixed point is always  $\tilde{V}_o$ . If at  $\tau_\gamma$  insiders receive their current value as a ter-

minal payoff, they are exactly compensated for the termination anyway. Accordingly, the graph of  $T_o$  rotates as  $\gamma$  changes but centered on the invariant fixed point  $(\tilde{V}_o, \tilde{V}_o)$ .

Even though  $T_o$  by itself is a trivial contraction, its time discounting due to  $\gamma < \infty$ , as Figure SA.1b illustrates, still makes the concatenated map  $\hat{T} = T \circ T_o$  contractionary, because the slope of  $T$  never exceeds unity. Hence, there exists a unique – and interesting – fixed point  $V = \hat{T}(V)$  by contraction mapping theorem.

### SA.3 Financial slack under investment opportunities

For the present stylized exercise, I slightly modify the cash flow profile in Section 4.1.

**Cash flow and investment opportunities.** The business has ‘normalized’ cash inflow  $\pi dt + \sigma dB_t$ , with  $\pi, \sigma > 0$ . Opportunities to scale up cash inflow to  $\eta(\pi dt + \sigma dB_t)$ ,  $\eta > 1$ , arrive at a Poisson rate  $\lambda > 0$ , but it requires a normalized upfront investment *expense* of  $\xi > 0$ .<sup>58</sup> Firm value  $W(k, h)$  given  $k \in \mathbb{N} \cup \{0\}$  rounds of past investment and funds  $h$  is homogeneous so that  $W(k, h) = \eta^k W\left(1, \frac{h}{\eta^k}\right) \equiv \eta^k V\left(\frac{h}{\eta^k}\right)$ .

To ensure that investment is first-best, I assume the following.

**Assumption SA.3.1.**  $\pi > \lambda\xi$  and  $\frac{(\eta-1)\pi}{\rho} \geq \xi$ .

**Investment choice.** Upon receiving an opportunity to invest, a firm with funds  $h$  can (i) fund the investment internally, with value  $\eta V\left(\frac{h-\xi}{\eta}\right)$ , (ii) forgo the investment, with value  $V(h)$ , or (iii) finance the investment externally, with value given by bargaining as

$$\hat{V}_o(h) + \theta\left(\eta V(\bar{h}) - (\xi + \eta\bar{h} - h) - \hat{V}_o(h)\right),$$

where the firm’s reservation value is  $\hat{V}_o(h) \equiv \max\left\{V_o(h), \eta V_o\left(\frac{h-\xi}{\eta}\right)\right\}$ :

- When  $\hat{V}_o(h) = V_o(h)$ , the firm optimally forgoes investment if financing fails.
- When  $\hat{V}_o(h) = \eta V_o\left(\frac{h-\xi}{\eta}\right)$ , the firm optimally invests even if financing fails. I term this ‘financing the investment with (credible) commitment to execution.’

The term ‘commitment’ is simply a shorthand for when, given that choice (iii) is optimal, the optimality of investment itself does not depend on successful financing.<sup>59</sup>

**Financing access.** I study how access to alternative financing  $\gamma$  affects financial slack and investment, given  $(\rho, r) = (0.07, 0)$ ,  $\theta = 0.5$ ,  $(\pi, \sigma) = (1, 2)$  and  $(\lambda, \xi) = (0.5, 0.7)$ .

Let  $\gamma = 0.3$ . As Figure SA.2 illustrates, firms may sometimes forgo – as shown in black square markers – investment opportunities that increase frictionless net present

<sup>58</sup>In the illustration part from Section 6.2 at Figure 9,  $\xi$  was fund proceeds from divestment.

<sup>59</sup>By subgame perfection, this is equivalent to the firm first funding the investment internally – per choice (i) – and then promptly raising financing – per Section 4.2.



Figure SA.2: Optimal investment policy under dilution from financing

The black curve is a simulated history of normalized internal funds along the left axis. The horizontal lines in blue and red are funding target  $\bar{h}$  and financing threshold  $\underline{h}$ , respectively. Investment opportunities arrive at the vertical dashed lines, each requiring an upfront normalized expense  $\xi$  whose size is indicated on the left axis. The markers in each shaded region indicate the choice made upon each opportunity in accordance with investment policy described in the legend. The magenta bar is normalized financing rent: one at investment financing ( $t \approx 0.44$ ), and one at non-investment financing ( $t \approx 7.64$ ). Access to alternative financing is  $\gamma = 0.3$ .

value. For these firms, financing involves large rent extraction as their access to alternative financing is not so good. Therefore, they optimally forgo investment when the lumpy expense would greatly dilute firm value ex-ante by making the sizable rent extraction by financiers much likelier to be triggered soon. When rent extraction has already become highly likely soon due to low funds, however, firm value is already diluted enough ex-ante so that it is optimal to increase diluted firm value by bargaining with financiers over financing the investment.

Figure SA.3 illustrates optimal financing and investment policies as the strength of financing access varies  $\gamma \in [0, 52]$ . The subplots describe  $\gamma \in \{0, 26\}$ . In Figure SA.3a with  $\gamma = 0$ , firms face substantial dilution from financing because they cannot access alternative financiers. As a result, they delay financing as much as possible and often forgo investment in order to avoid large dilution from financing.

In contrast, Figure SA.3b describes a firm that can expect to find alternative financiers very quickly  $\gamma = 26$ . It finances in small lumps  $\Delta h \approx 0.91$  and thus extremely frequently – 53 times over  $t \in [0, 10]$  in the present simulation – because of the extremely negligible size of financing cost around 0.004 at the financing threshold. Nevertheless, it maintains a sizable funding reserve  $\underline{h} \approx 2.84$ . As discussed in Section 5, this financial slack is what reduces dilution to such a negligible size.

**‘Commitment’ to investment execution and the size of dilution.** Moreover, firms with such good access to financing  $\gamma = 26$  always finance lumpy investments despite keeping funding reserves  $\underline{h} \approx 2.84$  in large excess of the expense  $\xi = 0.7$ . For



Figure SA.3: Financing access and investment policy

In the main plot with comparative statics in  $\gamma$ , colored areas show optimal investment policy given funds  $h$  per the legend. The semi-dashed line in magenta is the size of dilution from financing the investment given the opportunity arrives when  $h_t = \underline{h}$ . The dilution graph kinks downward when ‘commitment’ starts to hold – i.e., where the red curve enters the yellow region. For explanation of the subplots, see Figure SA.2.

any  $h_t \in [\underline{h}, \bar{h}]$ , these firms are willing to invest even if financing fails: even at  $h_t = \underline{h}$ ,

$$\eta V_o \left( \frac{\underline{h} - \xi}{\eta} \right) \approx 23.228 > 22.102 \approx V_o(\underline{h}). \quad (\text{SA.1})$$

This is because both  $\gamma$  and  $\underline{h}$  are high. Because  $\underline{h} \approx 2.84$  is high, firms can fund the investment internally and still have a substantial amount of normalized remaining funds  $\frac{\underline{h} - \xi}{\eta} \approx 1.95$ . Because  $\gamma$  is high, firms’ normalized reservation value falls only slightly as a result  $V_o \left( \frac{\underline{h} - \xi}{\eta} \right) \approx 21.116 < 22.102 \approx V_o(\underline{h})$ . Investment thus raises the actual reservation value, as Inequality (SA.1) shows.

This is how such firms incur negligible dilution even when financing the investment. Even if financing were to fail, investment would still be optimally undertaken. Because investment does not depend on financing, returns on investment do not constitute a surplus from bargaining. Dilution thus becomes quite negligible (approximately 0.054), because this credible ‘commitment’ reduces it by  $(1 - \theta) \left( \eta V_o \left( \frac{\underline{h} - \xi}{\eta} \right) - V_o(h) \right) \approx 0.563$ .

## SA.4 Supporting analysis on Section 6.2

Fully characterizing optimal financing policy with investment choice is analytically intractable. Here, I conduct suggestive analysis in support of the intuitions offered in Section 6.2. Specifically, I focus on a key necessary condition for the optimality of financing marginally earlier before running out of funds and how investment irreversibility differentially affects it depending on productivity.

As in Section 5.2, define  $F$  as the payoff function when insiders immediately raise financing as a one-shot and then revert to the optimal policy. With funds  $h = h_t \rightarrow 0^+$ , financing immediately is preferred to delaying financing by a  $dt$  instant (without considering the option value of averting financing) if and only if, omitting  $h$ ,

$$\rho F - rhF' \geq (A - i_F - \Psi(i_F))F' + \frac{1}{2}\sigma^2 F'' + (i_F - \delta)(F - hF'), \quad (\text{SA.2})$$

where  $i_F$  is optimal investment policy during the instantaneous delay being considered, given by: if investment is reversible  $\mathcal{I} = \mathbb{R}$ ,

$$i_F = \frac{1}{\psi} \left( \frac{F}{F'} - h - 1 \right),$$

and otherwise  $\mathcal{I} = \mathbb{R}_+$ , the greater of the above and zero.

The same derivation as in the proof of Lemma 2 leads to the following equivalent reformulations of Inequality (SA.2) evaluated at  $h \rightarrow 0^+$ . First, in case of reversibility,

$$G_0 \equiv \theta \left( \frac{1}{2\psi} \left( V(\bar{h}) - \bar{h} - 1 - \psi i_F(0) \right) \left( V(\bar{h}) - \bar{h} \right) - \varphi \bar{h} \right) \geq 0. \quad (\text{SA.3})$$

Above, typically  $i_F(0) < 0$ . To see this, observe that

$$i_F(0) = \frac{1}{\psi} \left( \frac{\theta(V(\bar{h}) - \bar{h})}{\theta + (1 - \theta)V'_o(0)} - 1 \right). \quad (\text{SA.4})$$

Here, the fraction inside the parenthesis is mostly less than unity and typically close to zero, because  $V'_o(0)$  is marginal value of funds at the brink of business failure and so tends to be large due to the extreme curvature.<sup>60</sup> Thus, the incentive for early financing given reversible investment, as captured by  $G_0$ , is strengthened by the extent to which firms without access to financing would be incentivized to preserve funds  $V'_o(0)$  through divestment.

Second, when investment is irreversible, the above discussion implies that typically,

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<sup>60</sup>The fraction may still exceed unity if  $\theta$  is close to unity, but then this paper is not relevant.

$i_F(0) = 0$ . Thus, if  $i(\bar{h}) = 0$ , then  $\hat{G}_0 = -\theta\varphi\bar{h}$ , and otherwise,

$$\hat{G}_0 \equiv \theta \left( \frac{1}{2\psi} \left( V(\bar{h}) - \bar{h} - 1 \right)^2 - \varphi\bar{h} \right) \geq 0. \quad (\text{SA.5})$$

Before the main analysis, I first state a handy result.

**Lemma SA.4.1.** *Given that the irreversibility constraint is either absent or slack at funding target, net firm value at funding target is given by*

$$V(\bar{h}) - \bar{h} = 1 + \psi \left( (\rho + \delta) - \sqrt{(\rho + \delta)^2 - \frac{2}{\psi} (A - \rho - \delta - \varphi\bar{h})} \right). \quad (\text{SA.6})$$

*Proof.* Rearrange the main HJB using  $i = \frac{1}{\psi} \left( \frac{V}{V'} - h - 1 \right)$  to obtain

$$\begin{aligned} & \left( \rho + \delta + \frac{1}{\psi} \right) V - \left( A + \frac{1}{2\psi} + \left( r + \delta + \frac{1}{\psi} \right) h \right) V' - \frac{1}{2\psi} \left( \frac{V^2}{V'} - 2hV - h^2V' \right) \\ & - \frac{1}{2}\sigma^2V'' \equiv \tilde{\rho}V - \left( \tilde{\mu} + \tilde{r}h \right) V' - \frac{1}{2\psi} \left( \frac{V^2}{V'} - 2hV - h^2V' \right) - \frac{1}{2}\sigma^2V'' = 0. \end{aligned}$$

Rearrange for  $\bar{V} \equiv V(\bar{h})$  using smooth pasting and super contact:

$$\Omega(\bar{V}|\bar{h}) = 0,$$

where  $\Omega(v|h) \equiv \frac{1}{2\psi}v^2 - \left( \tilde{\rho} + \frac{h}{\psi} \right) v + \left( \tilde{\mu} + \tilde{r}h - \frac{h^2}{2\psi} \right)$ . Then,  $\Omega_v(\bar{V}|\bar{h}) < 0$ , because, letting  $V^*$  first-best value, both roots solving  $\Omega(v^*|0) = 0$  are strictly between  $v^- - \bar{h}$  and  $v^+ - \bar{h}$ , where  $v^\pm$  are the two roots of  $\Omega(v|\bar{h}) = 0$ , even as  $\bar{V} - \bar{h} < V^*$ ; i.e.,  $\bar{V} = v^-$ .<sup>61</sup> A quadratic formula then gives Equation (SA.6).  $\square$

**Early financing when investment is reversible.** Substituting Equation (SA.4) reformulates Inequality (SA.3) as

$$\frac{G_0}{\theta} = \frac{1}{2\psi} \left( V(\bar{h}) - \bar{h} \right)^2 \left( \frac{(1 - \theta)V'_o(0)}{\theta + (1 - \theta)V'_o(0)} \right) - \varphi\bar{h} \geq 0. \quad (\text{SA.7})$$

Under the empirically relevant case where positive investment sometimes occurs in equilibrium, net firm value must strictly exceed unity, since  $i(\bar{h}) = \frac{1}{\psi} (V(\bar{h}) - \bar{h} - 1)$ . Therefore, from Lemma SA.4.1,

$$\varphi\bar{h} < A - \rho - \delta.$$

Given standard parametrization (see Bolton et al., 2011, for example), the right-

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<sup>61</sup>To have a well-defined value of  $V^*$ , assume that  $A \leq (\rho + \delta) \left( \frac{(\rho + \delta)\psi}{2} + 1 \right)$ .

hand side above tends to be quite small relative to  $\frac{1}{2\psi}$ , which is strictly less than  $\frac{1}{2\psi}(V(\bar{h}) - \bar{h})^2$ . Therefore, if  $V'_o(0)$  is sufficiently high and  $\theta$  not too close to unity, Inequality (SA.7) holds broadly, which means that when  $h_t \rightarrow 0^+$ , raising financing marginally earlier is mostly preferred to raising financing marginally later.<sup>62</sup>

**Early financing under irreversibility.** Inequality (SA.5) is, in comparison, more difficult to satisfy because now it is net firm value *minus one* that is being squared. In addition, net value should be lower under irreversibility than without, and so by Lemma SA.4.1, funding target increases, making  $\hat{G}_0$  even lower.

In this context, productivity becomes crucial for the incentives for early financing relative to delayed financing as  $h_t \rightarrow 0^+$ , as the following result suggests.

**Lemma SA.4.2.**  $\hat{G}_0 = -\theta\varphi\bar{h} < 0$  for  $A \leq \rho + \delta$ .  $\frac{d\hat{G}_0}{dA} \geq 0$  if and only if  $\frac{d}{dA}(V(\bar{h}) - \bar{h}) \geq \frac{1}{\rho + \delta}$ .

*Proof.* The second claim is immediate from

$$\begin{aligned} \frac{d}{dA} \frac{\hat{G}_0}{\theta} &= i(\bar{h}) \left( \frac{1 - \varphi \frac{d}{dA} \bar{h}}{\rho + \delta - i(\bar{h})} \right) - \varphi \frac{d}{dA} \bar{h} \geq 0 \iff \frac{i(\bar{h})}{\rho + \delta} \geq \varphi \frac{d}{dA} \bar{h} \\ &\iff \frac{d}{dA} (V(\bar{h}) - \bar{h}) = \frac{1 - \varphi \frac{d}{dA} \bar{h}}{\rho + \delta - i(\bar{h})} \geq \frac{1 - \frac{i(\bar{h})}{\rho + \delta}}{\rho + \delta - i(\bar{h})} = \frac{1}{\rho + \delta}. \end{aligned}$$

The first claim is also immediate since  $A \leq \rho + \delta$  implies that investment is not optimal even on a first-best basis, such that  $i(\bar{h}) = 0$ .  $\square$

Given  $\underline{A} \equiv \inf\{A > 0 \mid \hat{G}_0 \geq 0\}$ , the first claim above implies that  $\frac{d\hat{G}_0}{dA}|_{\underline{A}} \geq 0$  such that

$$\frac{d}{dA} (V(\bar{h}) - \bar{h})|_{\underline{A}} \geq \frac{1}{\rho + \delta}.$$

While the proof remains elusive, it is plausible – and consistent with numerical simulations – that  $\frac{d}{dA} (V(\bar{h}) - \bar{h})|_A > \frac{1}{\rho + \delta}$  for any  $A > \underline{A}$ , such that  $\hat{G}_0$  is strictly increasing in  $A > \underline{A}$ . This is because if  $A$  increases marginally, value should in general rise by a factor between  $\frac{1}{\rho + \delta}$  and  $\frac{1}{\rho + \delta - i(\bar{h})}$ , since investment ranges between  $[0, i(\bar{h})]$ .<sup>63</sup> In fact, consistent with the intuition given that  $i(\bar{h})$  is increasing in  $A$ , numerical simulation

<sup>62</sup>Given the large gap between  $A - \rho - \delta$  and  $\frac{1}{2\psi}$ , the same conclusion holds as long as divestment at funding target  $i(\bar{h}) < 0$  is not severe – that is,  $V(\bar{h}) - \bar{h}$  is not too below unity.

<sup>63</sup>In principle, expected future dilution dampens this factor, as shown by the term  $-\varphi \frac{d\bar{h}}{dA}$  in

$$\frac{d}{dA} (V(\bar{h}) - \bar{h}) = \frac{1 - \varphi \frac{d\bar{h}}{dA}}{\rho + \delta - i(\bar{h})}.$$

But since net value is being evaluated at the funding target  $\bar{h}$ , the magnitude of this effect should be relatively small.

suggests that  $V(\bar{h}) - \bar{h}$  is also convex in  $A$  given  $i(\bar{h}) > 0$ ; basically, greater investment makes marginal increase in capital productivity more valuable.

## Supplemental Appendix B Algorithms

Here, I explain the numerical algorithm. I first give a brief sketch of the algorithm for the main model in Sections 4 and 5, and then provide details in the context of investment choice in Section 6. As for the investment algorithm, Supplemental Appendix SB.1 sets up the main algorithm for the case without irreversibility and fluctuations in parameters. I then introduce Markov chains, both discrete and continuous, in Supplemental Appendix SB.2. Last, Supplemental Appendix SB.3 describes how to solve the model under investment irreversibility.

**Sketch of the algorithm.** The key idea of the algorithm is that the following differential equations must be *globally* satisfied: given sufficiently high  $H > 0$ ,

$$\begin{aligned} &(\rho + \lambda)V(h) - (\mu + rh)V'(h) - \frac{1}{2}\sigma^2V''(h) = \text{NL}(V, V_o, h) \\ &\equiv \max \left\{ (\rho + \lambda)V(h) - (\mu + rh)V'(h), \lambda(\Pi + h), \right. \\ &\quad \left. (\rho + \lambda) \left( \theta(V(H) - H + h)(1 - \theta)V_o(h) \right) - (\mu + rh)V'(h) - \frac{1}{2}\sigma^2V''(h) \right\}, \\ &(\rho + \lambda + \gamma)V_o(h) - (\mu + rh)V'_o(h) - \frac{1}{2}\sigma^2V''_o(h) = \text{NL}_o(V, V_o, h) \\ &\equiv \max \left\{ (\rho + \lambda + \gamma)V_o(h) - (\mu + rh)V'_o(h), \lambda(\Pi + h) + \gamma V(h) \right\} \end{aligned}$$

for all  $h \in (0, H)$ , along with the boundary conditions

$$V(0) = \theta(V(H) - H), \quad V'(H) = 1, \quad V_o(0) = 0, \quad V'_o(H) = 1.$$

The first element in NL is the maximum for  $h \in (\bar{h}, H)$ , the second for  $h \in (0, \bar{h}) \setminus B$ , and the third for  $h \in B$ . Similarly, the first element in  $\text{NL}_o$  is the maximum for  $h \in (\bar{h}_o, H)$  and the second for  $h \in (0, \bar{h}_o)$ . ‘NL’ stands for ‘nonlinear in  $(V, V_o)$ .’

Given initial guess for  $(V^{old}, V_o^{old})$ , compute NL and  $\text{NL}_o$  as vectors with respect  $h$ , use linear solver to get  $(V^{new}, V_o^{new})$  solving the above, and iterate until convergence.

One benefit of this approach is that it does not impose Lemma 2. This is useful in Section 6 because, due to intractability introduced by investment choice, it is elusive to analytically establish monotone financing strategy in the same fashion. Numerical results displayed in Section 6 do have monotone financing strategy when financing is early, not because it is imposed but because it is numerically predicted.

### SB.1 Main algorithm for investment extension

**Formulation.** Start by setting some  $H > 0$ . It should be higher than  $\bar{h}^o$ , the funding target under exclusion, which is higher than  $\bar{h}$  but typically by a slight margin.  $V$  on

$[0, H]$  satisfies:

$$h \geq \bar{h} \implies 0 = V_{hh}(h) \quad (\because V_h = 1 \text{ on } [\bar{h}, \infty)) \quad (\text{SB.1})$$

$$\begin{aligned} h \in [\underline{h}, \bar{h}] \implies \rho V(h) - rhV_h(h) &= \max_i (A - i - \Psi(i))V_h + (i - \delta)(V - hV_h) + \frac{1}{2}\sigma^2 V_{hh} \\ &= \left(A + \frac{1}{2\psi} + \left(\delta + \frac{1}{\psi}\right)h\right)V_h - \left(\delta + \frac{1}{\psi}\right)V + \frac{1}{2\psi} \frac{(V - hV_h)^2}{V_h} + \frac{1}{2}\sigma^2 V_{hh} \end{aligned} \quad (\text{SB.2})$$

$$h \leq \underline{h} \implies V(h) = \theta(V(H) - H + h) + (1 - \theta)V^o(h). \quad (\text{SB.3})$$

Note that (SB.1) implies  $V(H) - H = V(\bar{h}) - \bar{h}$ , which is being substituted into (SB.3). Next,  $V^o$  on  $[0, H]$  satisfies:

$$h \geq \bar{h}^o \implies 0 = V_{hh}^o(h) \quad (\because V_h^o = 1 \text{ on } [\bar{h}^o, \infty)) \quad (\text{SB.4})$$

$$\begin{aligned} h \in [0, \bar{h}^o] \implies \rho V^o(h) - rhV_h^o(h) &= \left(A + \frac{1}{2\psi} + \left(\delta + \frac{1}{\psi}\right)h\right)V_h^o - \left(\delta + \frac{1}{\psi}\right)V^o + \frac{1}{2\psi} \frac{(V^o - hV_h^o)^2}{V_h^o} + \frac{1}{2}\sigma^2 V_{hh}^o \\ &\quad + \gamma(V(h) - V^o(h)). \end{aligned} \quad (\text{SB.5})$$

For ease of notation, define

$$\alpha \equiv \rho + \delta + \frac{1}{\psi}, \quad \beta(h) \equiv A + \frac{1}{2\psi} + \left(r + \delta + \frac{1}{\psi}\right)h, \quad \xi(v, v_h, h) \equiv \frac{1}{2\psi} \frac{(v - hv_h)^2}{v_h}.$$

The five piecewise equalities above – (SB.1) through (SB.5) – switch to strict inequalities when evaluated outside the respective intervals, with left-hand sides being higher. Therefore, these can be summarized as follows: for  $h \in [0, H]$ ,

$$\begin{aligned} \alpha V(h) - \beta(h)V_h(h) - \frac{1}{2}\sigma^2 V_{hh}(h) &= \text{NL}(V, V^o, h) \\ &\equiv \max \left\{ \alpha V(h) - \beta(h)V_h(h), \xi(V(h), V_h(h), h), \right. \\ &\quad \left. \alpha \left( \theta(V(H) - H + h) + (1 - \theta)V^o(h) \right) - \beta(h)V_h(h) - \frac{1}{2}\sigma^2 V_{hh}(h) \right\}, \end{aligned} \quad (\text{SB.6})$$

$$\begin{aligned} (\alpha + \gamma)V^o(h) - \beta(h)V_h^o(h) - \frac{1}{2}\sigma^2 V_{hh}^o(h) &= \text{NL}^o(V, V^o, h) \\ &\equiv \max \left\{ (\alpha + \gamma)V^o(h) - \beta(h)V_h^o(h), \xi(V^o(h), V_h^o(h), h) + \gamma V(h) \right\}. \end{aligned} \quad (\text{SB.7})$$

Both NL and  $\text{NL}^o$  capture the nonlinear components of the pair of differential equations. In the expression for NL, the first element gives the maximum on  $[\bar{h}, H]$ , the

second on  $[\underline{h}, \bar{h}]$  and the last on  $[0, \underline{h}]$ , and similarly in  $\text{NL}^o$  given  $\underline{h}^o = 0$ .<sup>64</sup> Last, the boundary conditions are:

$$V(0) = \theta(V(H) - H), \quad V_h(H) = 1 \quad (\text{SB.8})$$

$$V^o(0) = 0, \quad V_h^o(H) = 1. \quad (\text{SB.9})$$

**Discretization and linearization.** Let us discretize the fund space  $[0, H]$  into  $N_h$  evenly spaced grids, and let  $\Delta H \equiv \frac{H}{N_h-1}$  the grid size. For now, let  $i \in \{1, 2, \dots, N_h\}$  index  $[0, H]$  increasingly, and denote  $h \in [0, H]^{N_h}$  as the column vector discretizing  $[0, H]$ , such that  $h(0) = 0$ ,  $h(N_h) = H$ . Posit  $V_0, V_0^o \in \mathbb{R}^{N_h}$  as column vectors representing conjectured approximate value functions under inclusion and exclusion, respectively. Let  $W_0 \in \mathbb{R}^{2N_h}$  with

$$W_0 \equiv \begin{pmatrix} V_0 \\ V_0^o \end{pmatrix},$$

represent the stacked value functions.  $i \in \{N_h + 1, \dots, 2N_h\}$  represents funds under exclusion.

The core of the algorithm is to summarize the left-hand sides of the combined HJB equations (SB.6), (SB.7) as well as the boundary conditions (SB.8), (SB.9) into a single  $2N_h \times 2N_h$  sparse matrix  $M(W)$ , which depends on the true stacked value function  $W$ , such that  $M(W) \cdot W = \mathbf{NL}(W)$ , where  $\mathbf{NL}$  is essentially a stack of  $\text{NL}$  and  $\text{NL}^o$  but with some additional adjustments at the respective end rows, to be specified soon.  $M$  should depend on  $W$  exclusively due to the upwinding method described shortly. I will start with some initial  $W_0$  and use a linear solver to obtain  $W_1$  such that

$$\frac{W_1 - W_0}{\Delta t} + M(W_0) \cdot W_1 = \mathbf{NL}(W_0).$$

Above, the first term on the left-hand side is the pseudo-time derivative to enable efficient convergence, with  $\Delta t \in \mathbb{R}_{++}$ . Until  $W_1$  is sufficiently close to  $W_0$ , update  $W_0$  and repeat.

In principle,  $M(\cdot)$  is a mapping from  $\mathbb{R}^{2N_h}$  to  $\mathbb{R}^{2N_h \times 2N_h}$ , but the sparsity substantially reduces its rank. This mapping should capture everything linear in both the combined HJB equations, including the linear differential terms, and the boundary conditions.

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<sup>64</sup>The first elements of  $\text{NL}$  and  $\text{NL}^o$  make use of the fact that  $V_{hh}, V_{hh}^o \leq 0$  with the equality if and only if  $h \geq \bar{h}$ ,  $h \geq \bar{h}^o$ , respectively. The second elements are the standard HJB equations. The third element in  $\text{NL}$  derives from  $V \geq \theta(V(H) - H + h) + (1 - \theta)V^o$ , with the equality if and only if  $h \leq \underline{h}$ .

**Upwinding the derivatives.** I will approximate the combined HJB equations (SB.6), (SB.7) for the interior rows of  $h$  only, i.e.,  $i = 2, 3, \dots, N_h - 1$ . The endpoints  $i \in \{1, N_h\}$  will be reserved for the boundary conditions (SB.8), (SB.9). For each of the interior rows  $i \in \{2, 3, \dots, N_h - 1\}$ , I approximate the first and second derivatives of a given approximated function  $V$  as follows: define

$$\begin{aligned}\Delta_h^f &\equiv \left(0, -\frac{1}{\Delta H}, \frac{1}{\Delta H}\right), & \Delta_h^b &\equiv \left(-\frac{1}{\Delta H}, \frac{1}{\Delta H}, 0\right), \\ \Delta_h^c &\equiv \left(-\frac{1}{2\Delta H}, 0, \frac{1}{2\Delta H}\right), & \Delta_h^2 &\equiv \left(\frac{1}{\Delta H^2}, -\frac{2}{\Delta H^2}, \frac{1}{\Delta H^2}\right).\end{aligned}$$

Then, letting  $\widehat{V}(i) \equiv (V(i-1), V(i), V(i+1))'$ ,  $\Delta_h^f \cdot \widehat{V}(i)$  denotes the forward first difference,  $\Delta_h^b \cdot \widehat{V}(i)$  the backward first difference,  $\Delta_h^c \cdot \widehat{V}(i)$  the centered first difference, and  $\Delta_h^2 \cdot \widehat{V}(i)$  the second difference, all in  $h$ .

I follow the standard numerical method of ‘upwinding’ where the forward/backward difference is used in approximating the first derivative with a positive/negative drift. Determining the sign of the drift in cash flow, however, is somewhat tricky since it depends on which region  $-[0, \underline{h}]$ ,  $(\underline{h}, \bar{h}]$  or  $(\bar{h}, H]$  under inclusion and  $[0, \bar{h}^o]$  or  $(\bar{h}^o, H]$  under exclusion – the current internal funds level belongs to. Determining the region is equivalent to determining which element is the maximum in NL and NL<sup>o</sup> in Equations (SB.6) and (SB.7), respectively, whose elements all involve first derivatives.

Therefore, I employ the centered first difference to determine the regions and then use them to implement the upwinding. Given  $V_0$  and  $V_0^o$ , define ten row indicator vectors –  $f_m, b_m \in \{0, 1\}^{N_h}$  for  $m = 1, 2, 3$  and  $f_m^o, b_m^o \in \{0, 1\}^{N_h}$  for  $m = 1, 2$  – such that  $i \in \{1, N_h\} \implies \forall m, f_m(i) = b_m(i) = f_m^o(i) = b_m^o(i) = 0$  and for  $i \in \{2, 3, \dots, N_h - 1\}$ ,  $f_1(i) = f_1^o(i) = 1$ ,

$$\begin{aligned}f_2(i) &\equiv \mathbb{1}\left(A - E\left(V_0(i), \Delta_h^c \cdot \widehat{V}_0(i), h(i)\right) + \left(r + \delta - I\left(V_0(i), \Delta_h^c \cdot \widehat{V}_0(i), h(i)\right)\right)h(i) > 0\right), \\ f_2^o(i) &\equiv \mathbb{1}\left(A - E\left(V_0^o(i), \Delta_h^c \cdot \widehat{V}_0^o(i), h(i)\right) + \left(r + \delta - I\left(V_0^o(i), \Delta_h^c \cdot \widehat{V}_0^o(i), h(i)\right)\right)h(i) > 0\right), \\ f_3(i) &\equiv \mathbb{1}\left(A - E\left(V_0^o(i) - \theta(V_0(N_h) - H + h(i)), \Delta_h^c \cdot \widehat{V}_0(i) - \theta, h(i)\right) \right. \\ &\quad \left. + \left(r + \delta - I\left(V_0^o(i) - \theta(V_0(N_h) - H + h(i)), \Delta_h^c \cdot \widehat{V}_0(i) - \theta, h(i)\right)h(i)\right) > 0\right),\end{aligned}$$

and  $b_m(i) = 1 - f_m(i)$ ,  $b_m^o(i) = 1 - f_m^o(i)$  for all  $m$ . Above, I and E denote optimized values of gross investment  $i$ <sup>65</sup> and total investment expense  $i + \Psi(i)$ , respectively, given

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<sup>65</sup>I and  $\delta$  enter the drift in cash flow because  $h$  is internal funds per capital.

as

$$I(v, v_h, h) = \frac{1}{\psi} \left( \frac{v}{v_h} - h - 1 \right), \quad E(v, v_h, h) = \frac{1}{2\psi} \left( \left( \frac{v - hv_h}{v_h} \right)^2 - 1 \right).$$

As can be inferred from  $f_1 = f_1^o = 1$ , I use forward differences on  $(\bar{h}, H]$  and  $(\bar{h}^o, H]$ .

Let

$$\Delta_{hm}(i) \equiv f_m(i)\Delta_h^f + b_m(i)\Delta_h^b, \quad \Delta_{hmo}(i) \equiv f_m^o(i)\Delta_h^f + b_m^o(i)\Delta_h^b.$$

Let  $m^*(i) \in \{1, 2, 3\}$  and  $m^{o*}(i) \in \{1, 2\}$  index the maximum, respectively in the sets

$$\left\{ \alpha V_0(i) - \beta(h(i))\Delta_{h1}(i) \cdot \widehat{V}_0(i), \xi(V_0(i), \Delta_{h2}(i) \cdot \widehat{V}_0(i), h(i)), \alpha(\theta(V_0(N_h) - H + h(i)) \right. \\ \left. + (1 - \theta)V_0^o(i)) - \beta(h(i))\Delta_{h3}(i) \cdot \widehat{V}_0(i) - \frac{1}{2}\sigma^2\Delta_h^2 \cdot \widehat{V}_0(i) \right\}, \quad (\text{SB.10})$$

$$\left\{ (\alpha + \gamma)V_0^o(i) - \beta(h(i))\Delta_{h1o}(i) \cdot \widehat{V}_0^o(i), \xi(V_0^o(i), \Delta_{h2o}(i) \cdot \widehat{V}_0^o(i), h(i)) + \gamma V_0(i) \right\}. \quad (\text{SB.11})$$

Last, let  $f(i) \equiv f_{m^*(i)}(i)$ ,  $f^o(i) \equiv f_{m^{o*}(i)}^o(i)$ ,  $b(i) \equiv 1 - f(i)$ ,  $b^o(i) \equiv 1 - f^o(i)$ , and

$$\Delta_h(i) \equiv f(i)\Delta_h^f + b(i)\Delta_h^b, \quad \Delta_h^o(i) \equiv f^o(i)\Delta_h^f + b^o(i)\Delta_h^b.$$

These three-dimensional row vectors  $\Delta_h(i)$ ,  $\Delta_h^o(i) \in \mathbb{R}^3$  implement the upwinding for first differences in the construction of  $M(\cdot)$ , to which I now transition.

**Constructing the matrix.** As a reminder,  $M(W_0)$  is a  $2N_h \times 2N_h$  sparse matrix, because  $W_0$  is a stacked vector of  $V_0, V_0^o$ . For each of  $i = 2, 3, \dots, N_h - 1$  rows, let

$$M(i, i - 1 : i + 1 \mid W_0) \equiv (0, \alpha, 0) - \beta(h(i))\Delta_h(i) - \frac{1}{2}\sigma^2\Delta_h^2,$$

where  $M(i, i - 1 : i + 1)$  denotes the  $i^{\text{th}}$  row from the  $(i - 1)^{\text{th}}$  column to the  $(i + 1)^{\text{th}}$  column, to account for the left-hand side of Equation (SB.6). Similarly, for each of  $(N_h + i)^{\text{th}}$  rows with  $i = 2, \dots, N_h - 1$ , implement the left-hand side of (SB.7) by

$$M(N_h + i, N_h + i - 1 : N_h + i + 1 \mid W_0) \equiv (0, \alpha + \gamma, 0) - \beta(h(i))\Delta_h^o(i) - \frac{1}{2}\sigma^2\Delta_h^2.$$

Next, construct the ‘nonlinear’ column vector  $\mathbf{NL}(W_0) \in \mathbb{R}^{2N_h}$  as follows: for each of  $i = 2, 3, \dots, N_h - 1$  rows,  $\mathbf{NL}(i \mid W_0)$  is the maximum in the set (SB.10) and  $\mathbf{NL}(N_h + i \mid W_0)$  in the set (SB.11). The rows  $i = 1, N_h, N_h + 1, 2N_h$  will be separately specified right below.

Last, the boundary conditions (SB.8) and (SB.9) are implemented as:

$$\begin{aligned}
M(1, 1 \mid W_0) &\equiv \alpha, \quad M(1, N_h \mid W_0) \equiv -\theta \left( \frac{1}{\Delta t} + \alpha \right), \quad \mathbf{NL}(1 \mid W_0) \equiv -\theta \left( \frac{1}{\Delta t} + \alpha \right) H, \\
M(N_h, N_h - 1 \mid W_0) &\equiv -\left( \frac{1}{\Delta t} + \alpha \right), \quad M(N_h, N_h \mid W_0) \equiv \alpha, \\
\mathbf{NL}(N_h \mid W_0) &\equiv \left( \frac{1}{\Delta t} + \alpha \right) \Delta H, \\
M(N_h + 1, N_h + 1 \mid W_0) &\equiv 0, \quad \mathbf{NL}(N_h + 1 \mid W_0) \equiv 0, \\
M(2N_h, 2N_h - 1 \mid W_0) &\equiv -\left( \frac{1}{\Delta t} + \alpha + \gamma \right), \quad M(2N_h, 2N_h \mid W_0) \equiv \alpha + \gamma, \\
\mathbf{NL}(2N_h \mid W_0) &\equiv \left( \frac{1}{\Delta t} + \alpha + \gamma \right) \Delta H.
\end{aligned}$$

Any unspecified element of  $M(W_0)$  is set to zero, making it highly sparse.

**Iteration to solution.** Posit  $H$ . Start with some initial guess  $V_0$ ,  $V_0^o$ , and stack them into  $W_0$ . Obtain  $W_1$  that solves

$$\left( \frac{1}{\Delta t} \mathbb{I}_{2N_h} + M(W_0) \right) \cdot W_1 = \frac{1}{\Delta t} W_0 + \mathbf{NL}(W_0),$$

where  $\mathbb{I}_{2N_h}$  is the  $2N_h \times 2N_h$  identity matrix, also highly sparse. If  $W_1$  is close enough to  $W_0$ , stop;  $V \equiv W_1(1 : N_h)$  and  $V^o \equiv W_1(N_h + 1 : 2N_h)$ . Otherwise, update  $W_0 \equiv aW_1 + (1 - a)W_0$  for some weight  $a \in (0, 1]$  and repeat.

If  $H$  is set too high, the algorithm might converge too slowly and the solution becomes unnecessarily coarse on  $[0, \bar{h}]$ . On the other hand, if  $H$  is too low, then possibly  $H < \bar{h}^o$ , in which case the algorithm fails. Therefore, I run the algorithm twice, an initializer and a verifier. During initialization, I use an adequate fraction (say 0.2) of the first-best value  $V^*$  as the initial  $H$  and use a high error tolerance. If convergence fails, I raise the initial  $H$ . If it succeeds, I choose a new  $H$  to be only slightly higher than  $\bar{h}^o$  and run the verifier with a low error tolerance.

## SB.2 Fluctuations in parameters

**Stacked value.** Let  $N_s$  denote the number of Markov states. For a continuous Markov chain,  $N_s$  is the number of grids in state-space discretization. I use  $s \in \{1, 2, \dots, N_s\}$  to index the Markov states. Given  $(V_0(i_h, s), V_0^o(i_h, s))$  for  $(i_h, s) \in \{1, \dots, N_h\} \times \{1, \dots, N_s\}$ , define

$$W_0 \equiv \begin{pmatrix} V_0(:, 1)' & \dots & V_0(:, N_s)' & V_0^o(:, 1)' & \dots & V_0^o(:, N_s)' \end{pmatrix}' \in \mathbb{R}^{2N_h N_s}$$

as their stacked column vector.  $i \in \{1, \dots, 2N_h N_s\}$  now jointly indexes  $(i_h, s)$  and inclusion/exclusion. The mapping  $M : \mathbb{R}^{2N_h N_s} \rightarrow \mathbb{R}^{2N_h N_s \times 2N_h N_s}$  will be defined in a fashion overall identical to Section [SB.1](#) for each of the  $N_h \times N_h$  blocks corresponding to  $s \in \{1, 2, \dots, N_s\}$  along the main diagonal  $M((s-1)N_h + 1 : sN_h, (s-1)N_h + 1 : sN_h \mid W_0)$ . There will be, however, an additional sparse matrix for the Markov chain and some changes to NL,  $NL^o$ .

**Discrete Markov chain.** Consider a Markov chain in Poisson arrival rates of transition given as

$$\begin{pmatrix} -\lambda^1 & \lambda_2^1 & \dots & \lambda_{N_s}^1 \\ \lambda_1^2 & -\lambda^2 & \dots & \lambda_{N_s}^2 \\ \dots & \dots & \dots & \dots \\ \lambda_1^{N_s} & \lambda_2^{N_s} & \dots & -\lambda^{N_s} \end{pmatrix},$$

where  $\lambda_{s'}^s \geq 0$  is the Poisson rate of transition at  $s$  to  $s' \neq s$ , and  $\lambda^s \equiv \sum_{s' \neq s} \lambda_{s'}^s$ .

Modify the combined HJB equations ([SB.6](#)) and ([SB.7](#)): for  $s \in \{1, \dots, N_s\}$ ,

$$\begin{aligned} & (\alpha(s) + \lambda^s)V(h, s) - \beta(h, s)V_h(h, s) - \frac{1}{2}\sigma^2 V_{hh}(h, s) - \sum_{s' \neq s} \lambda_{s'}^s V(h, s') = \text{NL}(V, V^o, h, s) \\ & \equiv \max \left\{ (\alpha(s) + \lambda^s)V(h, s) - \beta(h, s)V_h(h, s) - \sum_{s' \neq s} \lambda_{s'}^s V(h, s'), \xi(V(h, s), V_h(h, s), h, s), \right. \\ & \quad (\alpha(s) + \lambda^s) \left( \theta(s)(V(H, s) - H + h) + (1 - \theta(s))V^o(h, s) \right) - \beta(h, s)V_h(h, s) \\ & \quad \left. - \frac{1}{2}\sigma^2 V_{hh}(h, s) - \sum_{s' \neq s} \lambda_{s'}^s V(h, s') \right\}, \quad (\text{SB.12}) \end{aligned}$$

$$\begin{aligned} & (\alpha(s) + \lambda^s + \gamma(s))V^o(h, s) - \beta(h, s)V_h^o(h, s) - \frac{1}{2}\sigma^2 V_{hh}^o(h) - \sum_{s' \neq s} \lambda_{s'}^s V^o(h, s') \\ & = \text{NL}^o(V, V^o, h, s) \equiv \max \left\{ (\alpha(s) + \lambda^s + \gamma(s))V^o(h, s) - \beta(h, s)V_h^o(h, s) \right. \\ & \quad \left. - \sum_{s' \neq s} \lambda_{s'}^s V^o(h, s'), \xi(V^o(h, s), V_h^o(h, s), h, s) + \gamma(s)V(h) \right\}. \quad (\text{SB.13}) \end{aligned}$$

The dependence of  $\alpha, \beta, \theta, \gamma, \xi$  on  $s$  captures the fluctuations in parameters.

Define a Markov chain matrix for the entire  $(i_h, s)$  space by

$$\Lambda \equiv \begin{pmatrix} -\lambda^1 \widetilde{\mathbb{I}}_{N_h} & \lambda_2^1 \widetilde{\mathbb{I}}_{N_h} & \dots & \lambda_{N_s}^1 \widetilde{\mathbb{I}}_{N_h} \\ \lambda_1^2 \widetilde{\mathbb{I}}_{N_h} & -\lambda^2 \widetilde{\mathbb{I}}_{N_h} & \dots & \lambda_{N_s}^2 \widetilde{\mathbb{I}}_{N_h} \\ \dots & \dots & \dots & \dots \\ \lambda_1^{N_s} \widetilde{\mathbb{I}}_{N_h} & \lambda_2^{N_s} \widetilde{\mathbb{I}}_{N_h} & \dots & -\lambda^{N_s} \widetilde{\mathbb{I}}_{N_h} \end{pmatrix} \in \mathbb{R}^{N_h N_s \times N_h N_s},$$

where  $\widetilde{\mathbb{I}}_{N_h}$  is the  $N_h \times N_h$  identity matrix but with the first and last main diagonal

elements replaced with zero; the first and the last rows in each block are preserved for the boundary conditions. Proceed to extend  $\Lambda$  to both inclusion and exclusion by defining

$$\mathbf{\Lambda} \equiv \begin{pmatrix} \Lambda & \mathbf{0} \\ \mathbf{0} & \Lambda \end{pmatrix} \in \mathbb{R}^{2N_h N_s \times 2N_h N_s}.$$

The zero off-diagonal blocks indicate that  $\Lambda$  is orthogonal to re-inclusion upon  $\gamma$ .

The construction of each main diagonal block of  $M(W_0)$  – i.e.,  $M((s-1)N_h + 1 : sN_h, (s-1)N_h + 1 : sN_h \mid W_0)$  – is unchanged, both for the interior rows  $i \in \{(s-1)N_h + 2, \dots, sN_h - 1\}$  that implement the combined HJB (SB.12), (SB.13)<sup>66</sup> and for the boundaries  $i \in \{(s-1)N_h + 1, sN_h\}$  that implement the same boundary conditions.  $\mathbf{NL}(W_0)$  is adjusted slightly for the interior, following the modified definition of  $\mathbf{NL}, \mathbf{NL}^o$  in (SB.12), (SB.13). Once  $M(\cdot)$ ,  $\mathbf{NL}(\cdot)$  are constructed, iteratively solve

$$\left( \frac{1}{\Delta t} \mathbb{I}_{2N_h N_s} + M(W_0) - \mathbf{\Lambda} \right) \cdot W_1 = \frac{1}{\Delta t} W_0 + \mathbf{NL}(W_0).$$

**Continuous Markov chain.** Let  $s_t$  follow  $ds_t = \mu_s(s_t) dt + \sigma_s(s_t) dZ_t$ . Discretize the state space into  $N_s$  grids with size  $\Delta S$ . Let  $s \in \{1, 2, \dots, N_s\}$  index the state space increasingly. The above law of motion is ‘discretized’ as a Markov chain:

$$\begin{aligned} \mu_s(s) \geq 0 &\implies \lambda_{s-1}^s = \frac{\sigma_s(s)^2}{2\Delta S^2}, \quad \lambda_{s+1}^s = \frac{\mu_s(s)}{\Delta S} + \frac{\sigma_s(s)^2}{2\Delta S^2}, \\ \mu_s(s) < 0 &\implies \lambda_{s-1}^s = -\frac{\mu_s(s)}{\Delta S} + \frac{\sigma_s(s)^2}{2\Delta S^2}, \quad \lambda_{s+1}^s = \frac{\sigma_s(s)^2}{2\Delta S^2}. \end{aligned}$$

As for the endpoints  $s \in \{1, N_s\}$ , mean reversion will generally allow upwinding of the first-order terms  $\pm \frac{\mu_s(s)}{\Delta S}$ . The second-order terms, however, cannot be correctly computed, as they go outside the grid. I therefore use  $W_0$  to compute second as well as third finite differences at  $s \in \{2, N_s - 1\}$  and use them to linearly approximate the endpoint second derivatives.<sup>67</sup>

Once the discretized Markov chain is set up, follow the same procedure as above.

### SB.3 Investment irreversibility

The algorithm remains mostly the same and is modified only as follows. First, whenever  $\frac{1}{\psi}$  appears in the above algorithm, multiply it by  $1 + (\phi - 1) \cdot \mathbb{1}(\frac{V_0}{\Delta_h V_0} - h < 1)$ ; use  $V_0^o, \Delta_h^o$  instead of  $V_0, \Delta_h$  when appropriate. The indicator function tracks whether the firm divests. Second, move all terms involving this modified expression to the inside of  $\mathbf{NL}$  instead of  $M$ , as investment versus divestment can make the system volatile.

<sup>66</sup>The maximizing indices  $m^*, m^{o*}$  are based on the modified  $\mathbf{NL}$  and  $\mathbf{NL}^o$  in (SB.12), (SB.13).

<sup>67</sup>At the endpoints, the ‘discrete jump’ interpretation might not hold: the algorithm still works.

## Supplemental Appendix C Additional Analysis

This Appendix provides additional analysis for a particularly tractable case of a startup (see Section 4.1) with zero internal yield  $r = 0$ . Zero internal yield allows analytic solutions for the HJB equations in Section 4.3, and zero volatility  $\sigma = 0$  as specified for a startup eliminates the option effect discussed in Section 5.4, streamlining the algebra.

A key goal of this exercise is to demonstrate that a firm's *future* prospect – that is,  $\Pi$  – increases financial slack.<sup>68</sup>

### SC.1 Closed-form solution procedure

1. In all cases,  $V'(\bar{h}) = 1$ .
2. Solve the model with  $\gamma = 0$ . By Proposition 2,  $\underline{h} = 0$ , and  $\bar{h}$  is implicitly defined by  $V(0) = \theta(V(\bar{h}) - \bar{h})$ .
3. For  $\gamma > 0$ , first determine whether  $\underline{h} = 0$  or  $\underline{h} > 0$ . This can be done as follows: (i) posit the value of  $\bar{h}$  obtained in Step 2, and (ii) evaluate Inequality (10) for  $h = 0$ . If it fails, then set  $\underline{h} = 0$  and assign to  $\bar{h}$  the value obtained in Step 2.
4. If the inequality holds, then it must be that  $\underline{h} > 0$ .<sup>69</sup> Use the following conditions to implicitly determine  $(\underline{h}, \bar{h})$ :
  - (a) Stationary Recursion:  $V(\bar{h}) - V(\underline{h}) = \left(1 + \frac{\rho}{\gamma}\right) \Delta h$ , and
  - (b) Smooth Pasting:  $V'(\underline{h}) = \theta + (1 - \theta)V'_o(\underline{h})$ .

### SC.2 Analytic equilibrium solution

The business incurs a fixed flow expense  $\kappa dt$ ,  $\kappa > 0$ , until success arrives at Poisson rate  $\lambda > 0$  upon which the business terminates with one-time payoff  $\Pi > 0$ . As discussed, assume  $r = 0$ . Let us reiterate the first part of Assumption 1 as a reference.

**Assumption SC.2.1** (Positive net present value).  $\lambda\Pi > \kappa$ .

Then,  $V$  on  $(\underline{h}, \bar{h})$  satisfies the following ODE:

$$\begin{aligned} \rho V(h) &= \lambda(\Pi + h - V(h)) - \kappa V'(h) \\ \implies V(h) &= -ce^{-\frac{\rho+\lambda}{\kappa}h} + \frac{\lambda}{\rho+\lambda} \left( \Pi + h - \frac{\kappa}{\rho+\lambda} \right), \end{aligned}$$

<sup>68</sup>The result is consistent with Section 3 in that the two conditions for financial slack – Inequalities (1) and (2) – are easier to satisfy when terminal payoff  $v$  is higher.

<sup>69</sup>This is not necessarily the case if  $\sigma > 0$  due to the option effect.

for some  $c \in \mathbb{R}$ . In addition, since  $V'(\bar{h}) = 1$ , we have

$$V(h) = \frac{1}{\rho + \lambda} \left( \lambda(\Pi + h) - \frac{\kappa}{\rho + \lambda} \left( \lambda + \rho e^{\frac{\rho + \lambda}{\kappa}(\bar{h} - h)} \right) \right), \quad (\text{SC.1})$$

$$V(\bar{h}) = \frac{1}{\rho + \lambda} \left( \lambda(\Pi + \bar{h}) - \kappa \right). \quad (\text{SC.2})$$

### SC.2.1 Baseline: no re-inclusion

First consider  $\gamma = 0$ . Since  $\underline{h} = 0$  by Proposition 2, the equilibrium – just  $\bar{h}$  in this case – is implicitly defined by the stationary recursion as follows:

$$\begin{aligned} V(0) &= x(0)V(\bar{h}) = \theta(V(\bar{h}) - \bar{h}) \\ \iff \theta \rho((\rho + \lambda)\bar{h} + \kappa) + (1 - \theta)\lambda((\rho + \lambda)\Pi - \kappa) &= \rho \kappa \exp\left(\frac{\rho + \lambda}{\kappa}\bar{h}\right). \end{aligned} \quad (\text{SC.3})$$

In the first line,  $V_o(0) = 0$  is used. Note that the solution to Equation SC.3 is positive if and only if Assumption SC.2.1 holds.

With Equation (SC.3), comparative statics is straightforward.

**Proposition SC.2.1** (Comparative statics – Startups without re-inclusion).

$$\frac{\partial \bar{h}}{\partial \theta} = -\frac{1}{\rho} \frac{\lambda \Pi - \kappa - \rho \bar{h}}{\exp\left(\frac{\rho + \lambda}{\kappa}\bar{h}\right) - \theta} < 0, \quad \lim_{\theta \rightarrow 0} \bar{h} < \frac{\lambda \Pi - \kappa}{\rho}, \quad \lim_{\theta \rightarrow 1} \bar{h} = 0, \quad \text{and} \quad \lim_{\theta \rightarrow 1} \frac{\partial \bar{h}}{\partial \theta} = -\infty, \quad (\text{SC.4})$$

$$\frac{\partial \bar{h}}{\partial \Pi} = \frac{\lambda}{\rho} \frac{1 - \theta}{\exp\left(\frac{\rho + \lambda}{\kappa}\bar{h}\right) - \theta} > 0, \quad (\text{SC.5})$$

$$\lim_{\lambda \rightarrow \kappa/\Pi} \bar{h} = \lim_{\lambda \rightarrow \infty} \bar{h} = 0. \quad (\text{SC.6})$$

### SC.2.2 General comparative statics

Consider the general case of  $\gamma \geq 0$ . Since  $\sigma = 0$ , Inequality (10) for  $h = 0$  implies that  $\underline{h} > 0$  if and only if

$$\rho \bar{h} < \frac{(1 - \theta)\gamma}{\rho + \lambda + (1 - \theta)\gamma} (\lambda \Pi - \kappa). \quad (\text{SC.7})$$

By Section SC.1, the following result is obtained.

**Lemma SC.2.1** (Startup financing). *Denote*

$$\eta \equiv \frac{(1 - \theta)\gamma}{\rho + \lambda + (1 - \theta)\gamma}, \quad \xi \equiv \frac{\lambda \Pi - \kappa}{\rho}.$$

The equilibrium is characterized by  $\underline{h} = 0$  and  $\bar{h}$  implicitly defined by Equation (SC.3) if

$$\theta\rho((\rho + \lambda)\eta\xi + \kappa) + (1 - \theta)\lambda((\rho + \lambda)\Pi - \kappa) \geq \rho\kappa \exp\left(\frac{\rho + \lambda}{\kappa}\eta\xi\right). \quad (\text{SC.8})$$

If the inequality is strictly reversed, then  $\underline{h} = \bar{h} - \Delta h > 0$  and  $\bar{h} > \Delta h$  is implicitly defined by

$$\begin{aligned} & \frac{1 - \theta}{\rho + \lambda} \left( (\rho + \lambda + \theta\gamma)\lambda\Pi - \theta\rho\gamma \left( \bar{h} + \frac{\kappa}{\rho + \lambda + \theta\gamma} \right) - (\lambda + \theta\gamma)\kappa \right) \\ &= \rho \left( \frac{\rho + \lambda + \gamma}{\gamma} \Delta h + (1 - \theta) \frac{\kappa}{\rho + \lambda + \theta\gamma} \right) \exp\left(\frac{\rho + \lambda + \theta\gamma}{\kappa}(\bar{h} - \Delta h)\right), \end{aligned} \quad (\text{SC.9})$$

where  $\Delta h = \bar{h} - \underline{h} > 0$  is given by

$$1 + \frac{\rho + \lambda}{\kappa} \left( 1 + \frac{\rho + \lambda}{\gamma} \right) \Delta h = \exp\left(\frac{\rho + \lambda}{\kappa} \Delta h\right). \quad (\text{SC.10})$$

**Proposition SC.2.2** (Comparative statics – startups).  $\bar{h}$  strictly increases in  $\Pi$ . When  $\underline{h} > 0$ ,  $\underline{h}$  strictly increases in  $\Pi$  and  $\Delta h$  is constant in  $\Pi$  and strictly decreasing in  $\lambda$ . Last,  $\bar{h}$ ,  $\underline{h}$  and  $\Delta h$  converge to zero as either (i)  $\Pi$  goes to  $\kappa/\lambda$  or (ii)  $\lambda$  goes to either  $\kappa/\Pi$  or  $\infty$ .

**Proposition SC.2.3** (Breakeven re-inclusion – Startups).  $\underline{\gamma}$  strictly decreases in  $\Pi$ , and converges to zero as  $\Pi$  goes to  $\infty$ . It goes to  $\infty$  as  $\Pi$  goes down to  $\kappa/\lambda$ , the lower bound in Assumption SC.2.1.

### SC.3 Proofs

**Proposition SC.2.1** (Comparative statics – Startups without re-inclusion).

*Proof.* Most results above are straightforward. For the first inequality in (SC.4), it is sufficient to show that  $\lambda\Pi - \kappa - \rho\bar{h}(0) > 0$ , where  $\bar{h}(\theta)$  is  $\bar{h}$  expressed as a function of  $\theta \in [0, 1]$ .

To prove the claim, note that

$$\begin{aligned}
& \lambda\Pi - \kappa > \rho\bar{h}(0) \\
\iff \exp\left(\frac{\rho + \lambda}{\kappa} \frac{\lambda\Pi - \kappa}{\rho}\right) &> \exp\left(\frac{\rho + \lambda}{\kappa} \bar{h}(0)\right) \\
&= \frac{\lambda}{\rho} \left( (\rho + \lambda) \frac{\Pi}{\kappa} - 1 \right) \quad \because (\text{SC.3}) \text{ with } \theta \equiv 0 \\
&= 1 + \frac{\rho + \lambda}{\kappa} \frac{\lambda\Pi - \kappa}{\rho} \\
\iff \frac{\rho + \lambda}{\kappa} \frac{\lambda\Pi - \kappa}{\rho} &> 0,
\end{aligned}$$

which is equivalent to Assumption SC.2.1.  $\square$

**Lemma SC.2.1** (Startup financing).

*Proof.* If (SC.7) holds, then Inequality (SC.7) also holds with  $\bar{h}$  defined by (SC.3). Therefore,  $\underline{h} = 0$ . Now, consider the case where (SC.7) fails. Let us use Smooth Pasting and Stationary Recursion to determine  $(\underline{h}, \bar{h})$ .

**Smooth Pasting.** For ease of notation, denote  $\underline{V}_o \equiv V_o(\underline{h})$  and  $\underline{V} \equiv \theta(V(\bar{h}) - \bar{h} + \underline{h}) + (1 - \theta)\underline{V}_o$ . The insiders' non-excluded and excluded value functions for  $h \in [\underline{h}, \bar{h}]$  are given by

$$\begin{aligned}
V(h) &= \int_0^{(h-\underline{h})/\kappa} \lambda e^{-(\rho+\lambda)t} (\Pi + h - \kappa t) dt + e^{-\frac{\rho+\lambda}{\kappa}(h-\underline{h})} \underline{V} \\
&= \frac{\lambda}{\rho + \lambda} \left( \Pi + h - \frac{\kappa}{\rho + \lambda} \right) - \left( \frac{\lambda}{\rho + \lambda} \left( \Pi + \underline{h} - \frac{\kappa}{\rho + \lambda} \right) - \underline{V} \right) e^{-\frac{\rho+\lambda}{\kappa}(h-\underline{h})}, \\
V_o(h) &= V(h) - e^{-\frac{\rho+\lambda+\gamma}{\kappa}(h-\underline{h})} (\underline{V} - \underline{V}_o).
\end{aligned}$$

$V_o(h)$  is derived based on the observation that, given the strategy of waiting on  $(\underline{h}, h)$  regardless of market access, the only difference that exclusion creates is that one has  $\underline{V}_o$  instead of  $\underline{V}$  at  $h = \underline{h}$  if neither success nor re-inclusion occurs while the internal funds  $h$  run down to  $\underline{h}$ .

Note that

$$\begin{aligned}
V(\bar{h}) &= \frac{1}{\rho + \lambda} (\lambda(\Pi + \bar{h}) - \kappa) \\
\implies \frac{\rho\kappa}{(\rho + \lambda)^2} &= \left( \frac{\lambda}{\rho + \lambda} \left( \Pi + \underline{h} - \frac{\kappa}{\rho + \lambda} \right) - \underline{V} \right) \exp\left(-\frac{\rho + \lambda}{\kappa}(\bar{h} - \underline{h})\right) \\
\implies V(h) &= \frac{\lambda}{\rho + \lambda} \left( \Pi + h - \frac{\kappa}{\rho + \lambda} \right) - \frac{\rho\kappa}{(\rho + \lambda)^2} \exp\left(\frac{\rho + \lambda}{\kappa}(\bar{h} - h)\right), \text{ and} \\
\underline{V} - \underline{V}_o &= \frac{\theta}{1 - \theta} \left( -\frac{\rho}{\rho + \lambda} \left( \bar{h} - \underline{h} + \frac{1}{\rho + \lambda} \kappa \right) + \frac{\rho\kappa}{(\rho + \lambda)^2} \exp\left(\frac{\rho + \lambda}{\kappa}(\bar{h} - \underline{h})\right) \right).
\end{aligned}$$

Next, denote by  $V_d$  a payoff function on  $(\underline{h}, \bar{h}]$  for the deviation strategy of immediate financing. That is, for  $h \in (\underline{h}, \bar{h}]$ ,

$$V_d(h) \equiv \theta(V(\bar{h}) - \bar{h} + h) + (1 - \theta)V_o(h).$$

Smooth pasting condition is

$$V'(\underline{h}) = V'_d(\underline{h}) = \theta + (1 - \theta) \left( V'(\underline{h}) + \frac{\rho + \lambda + \gamma}{\kappa} (V - V_o) \right),$$

which, after some algebra, is equivalent to Equation (SC.10).

**Stationary Recursion.** This time, start by deriving  $V_o$  on  $[0, \underline{h}]$ , which satisfies:

$$\begin{aligned} \rho V_o(h) &= \gamma \left( \theta(V(\bar{h}) - \bar{h} + h) + (1 - \theta)V_o(h) - V_o(h) \right) + \lambda(\Pi + h - V_o(h)) - \kappa V'_o(h), \\ V_o(0) &= 0 \\ \implies V_o(h) &= \frac{1}{\rho + \lambda + \theta\gamma} \left[ \lambda\Pi + \theta\gamma(V(\bar{h}) - \bar{h}) - \frac{\lambda + \theta\gamma}{\rho + \lambda + \theta\gamma} \kappa + (\lambda + \theta\gamma)h \right. \\ &\quad \left. - \left( \lambda\Pi + \theta\gamma(V(\bar{h}) - \bar{h}) - \frac{\lambda + \theta\gamma}{\rho + \lambda + \theta\gamma} \kappa \right) \exp \left( -\frac{\rho + \lambda + \theta\gamma}{\kappa} h \right) \right]. \end{aligned}$$

Since  $\Delta h \equiv \bar{h} - \underline{h} > 0$  has been determined by Equation (SC.10),  $\bar{h}$  is obtained by the recursion:

$$V(\bar{h} - \Delta h) = \theta(V(\bar{h}) - \Delta h) + (1 - \theta)V_o(\bar{h} - \Delta h).$$

Simplifying and substituting (SC.10) give Equation (SC.9).  $\square$

**Proposition SC.2.2** (Comparative statics – startups)

*Proof.* Immediate from Lemma SC.2.1 via implicit differentiation.  $\square$

**Proposition SC.2.3** (Breakeven re-inclusion – Startups).

*Proof.*  $\underline{\gamma}$  is defined by  $\underline{\eta}\xi = \bar{h}^*$ , where  $(\underline{h}^*, \bar{h}^*)$  is the equilibrium associated with  $\gamma = \underline{\gamma}$  and  $(\underline{\eta}, \xi)$  given by Proposition SC.2.1. Since  $\gamma = \underline{\gamma}$  implies  $\underline{h}^* = 0$ , Proposition SC.2.1, in particular Equation (SC.5), holds with  $\bar{h}$  replaced with  $\underline{\eta}\xi$ . Note that

$$\begin{aligned} \frac{\lambda}{\rho} \frac{1 - \theta}{\exp \left( \frac{\rho + \lambda}{\kappa} \underline{\eta}\xi \right) - \theta} &= \frac{\partial(\underline{\eta}\xi)}{\partial\Pi} = \underline{\eta} \frac{\partial\xi}{\partial\Pi} + \xi \frac{\partial\underline{\eta}}{\partial\Pi}, \quad \frac{\partial\xi}{\partial\Pi} = \frac{\lambda}{\rho} \\ \implies \frac{\partial\underline{\eta}}{\partial\Pi} &= \frac{\lambda}{\lambda\Pi - \kappa} \left( \frac{1 - \theta}{\exp \left( \frac{\rho + \lambda}{\kappa} \underline{\eta}\xi \right) - \theta} - \frac{1 - \theta}{1 + \frac{\rho + \lambda}{\underline{\gamma}} - \theta} \right). \end{aligned}$$

Since  $\bar{h}^* = \underline{\eta}\xi$ , smooth pasting holds at  $\underline{h}^* = 0$ . Therefore, from Equation (SC.10),

$$\exp\left(\frac{\rho + \lambda}{\kappa}\underline{\eta}\xi\right) = 1 + \frac{\rho + \lambda}{\gamma}\frac{\rho + \lambda + \gamma}{\kappa}\underline{\eta}\xi.$$

Next, assume for now that  $\underline{\eta}\xi = \bar{h}^* > \frac{\kappa}{\rho + \lambda + \underline{\gamma}}$ , which will be established at the end. Then,

$$\exp\left(\frac{\rho + \lambda}{\kappa}\underline{\eta}\xi\right) > 1 + \frac{\rho + \lambda}{\underline{\gamma}},$$

Thus,  $\partial\underline{\eta}/\partial\Pi < 0$ . Since  $\underline{\eta} = \frac{(1-\theta)\underline{\gamma}}{\rho + \lambda + (1-\theta)\underline{\gamma}}$ , we have  $\partial\underline{\gamma}/\partial\Pi < 0$ .

Next is the convergence claim. Since  $\bar{h}^*$  satisfies Equation (SC.3), it goes to  $\infty$  as  $\Pi$  does. Note that  $\Delta h^* \equiv \bar{h}^* - \underline{h}^* = \bar{h}^*$  satisfies Equation (SC.10) with  $\gamma = \underline{\gamma}$ . Since  $\rho, \lambda, \kappa$  are fixed, the only way for the solution of Equation (SC.10) to be satisfied by a  $\Delta h$  that diverges to infinity is by having the linear coefficient on the LHS also diverge to infinity. This can only be achieved if  $\underline{\gamma}$  goes to zero, as claimed.

The divergence claim is straightforward from  $\underline{\eta}\xi > \frac{\kappa}{\rho + \lambda + \underline{\gamma}}$ . Since  $\xi = (\lambda\Pi - \kappa)/\rho$  and  $\underline{\eta}$  is in the unit interval, the LHS vanishes as  $\Pi$  goes down to  $\kappa/\lambda$ . Therefore, the RHS also vanishes, i.e.,  $\underline{\gamma} \rightarrow \infty$ .

Finally, as for the intermediate claim on the strict lower bound on  $\underline{\eta}\xi$ , first rearrange the Smooth Pasting condition – i.e., Equation (SC.10) – into the following:

$$\frac{\gamma}{\rho + \lambda} \left( \exp\left(\frac{\rho + \lambda}{\kappa}\Delta h\right) - 1 \right) = \frac{\rho + \lambda + \gamma}{\kappa}\Delta h.$$

Denote the LHS and RHS above as functions of  $\Delta h$ . Note that  $\text{LHS}(0) = \text{RHS}(0)$  and  $\text{LHS}'(0) < \text{RHS}'(0)$ . Therefore, the LHS crosses the RHS only once and from below on  $\mathbb{R}_{++}$ . Note that

$$\text{LHS}\left(\frac{\kappa}{\rho + \lambda + \gamma}\right) = \frac{\gamma}{\rho + \lambda} \left( \exp\left(\frac{\rho + \lambda}{\rho + \lambda + \gamma}\right) - 1 \right) < 1 = \text{RHS}\left(\frac{\kappa}{\rho + \lambda + \gamma}\right).$$

This holds for any set of parameters because, letting  $f(x) \equiv x \left( \exp\left(\frac{1}{1+x}\right) - 1 \right)$ , we have

$$\forall x > 0, f'(x) > 0, \text{ and } \lim_{x \rightarrow \infty} f(x) = 1.$$

Therefore,  $\text{LHS}(\Delta h) < \text{RHS}(\Delta h)$  for any  $\Delta h \in (0, \frac{\kappa}{\rho + \lambda + \gamma}]$ . That is, if Smooth Pasting holds at  $\underline{h}$ , then it must be that  $\Delta h = \bar{h} - \underline{h} > \frac{\kappa}{\rho + \lambda + \underline{\gamma}}$ . Since  $\gamma = \underline{\gamma}$  means that Smooth Pasting holds at  $\underline{h}^* = 0$ , it must be that  $\underline{\gamma}\xi = \bar{h}^* = \Delta h^* > \frac{\kappa}{\rho + \lambda + \underline{\gamma}}$ , as claimed.  $\square$